

Quantum Electronic Phases in Twisted Trilayer Graphene

Aaron Sharpe

UCSD

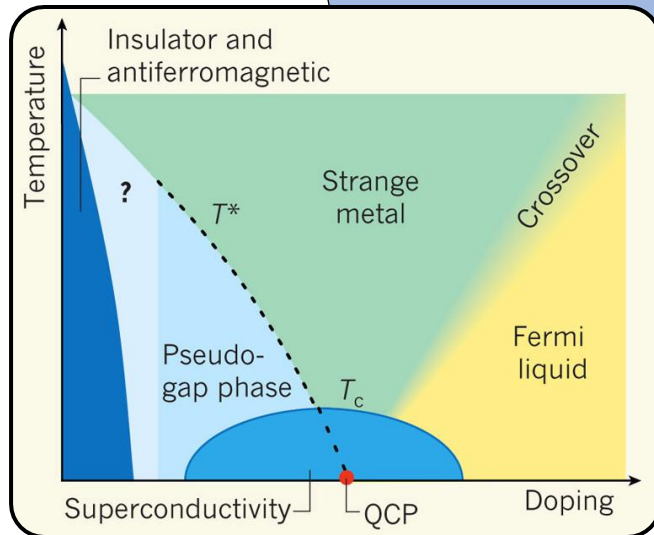
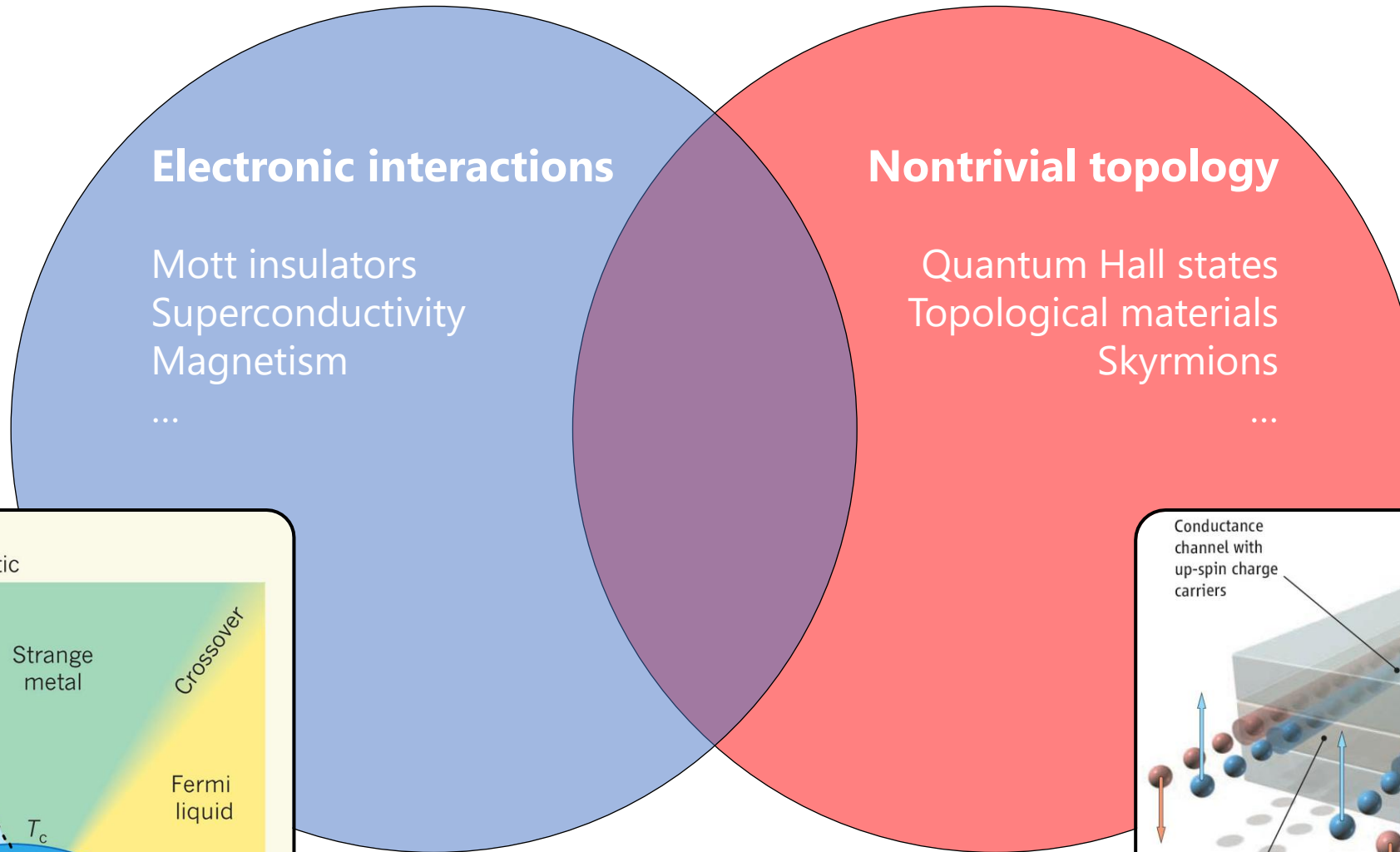
October 29th, 2025

arXiv: 2509.07977

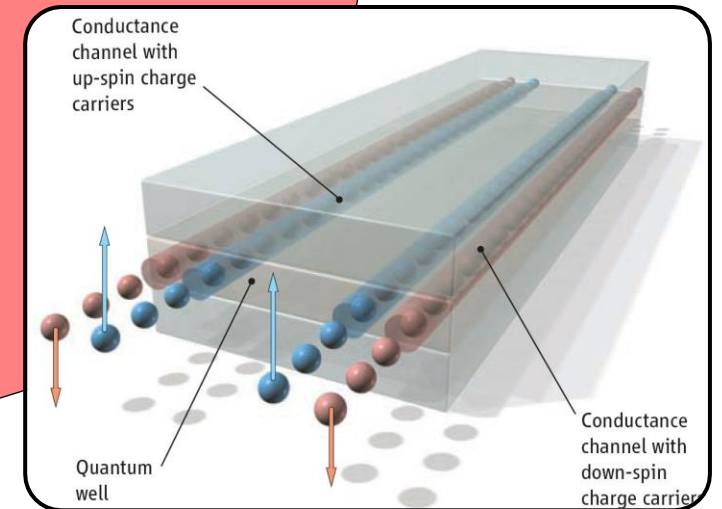
arXiv: 2509.03583

Slides available @ aaronsharpe.science

When interactions meet topology

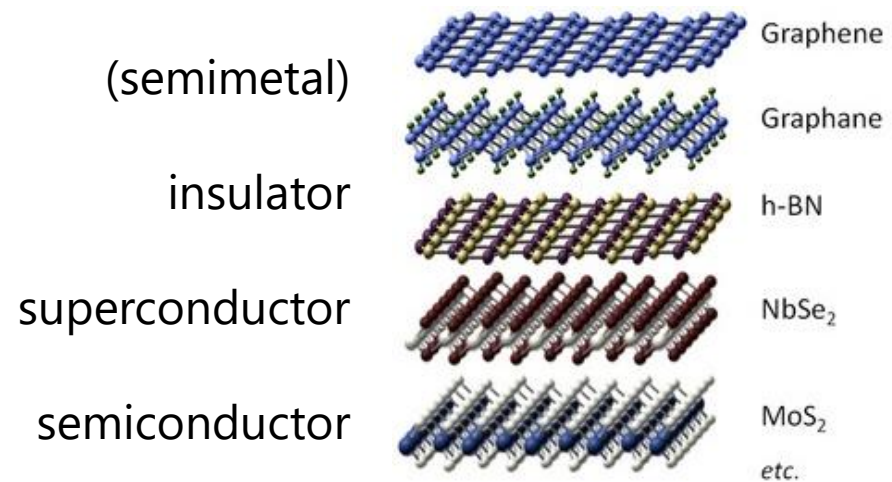


Varma, *Nature* (2010)

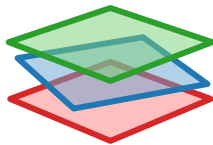
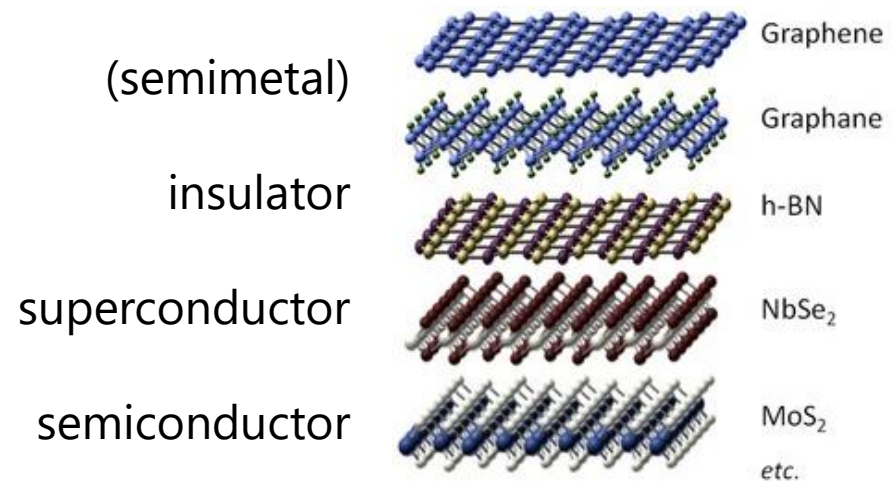


König, et al., *Science* (2007)

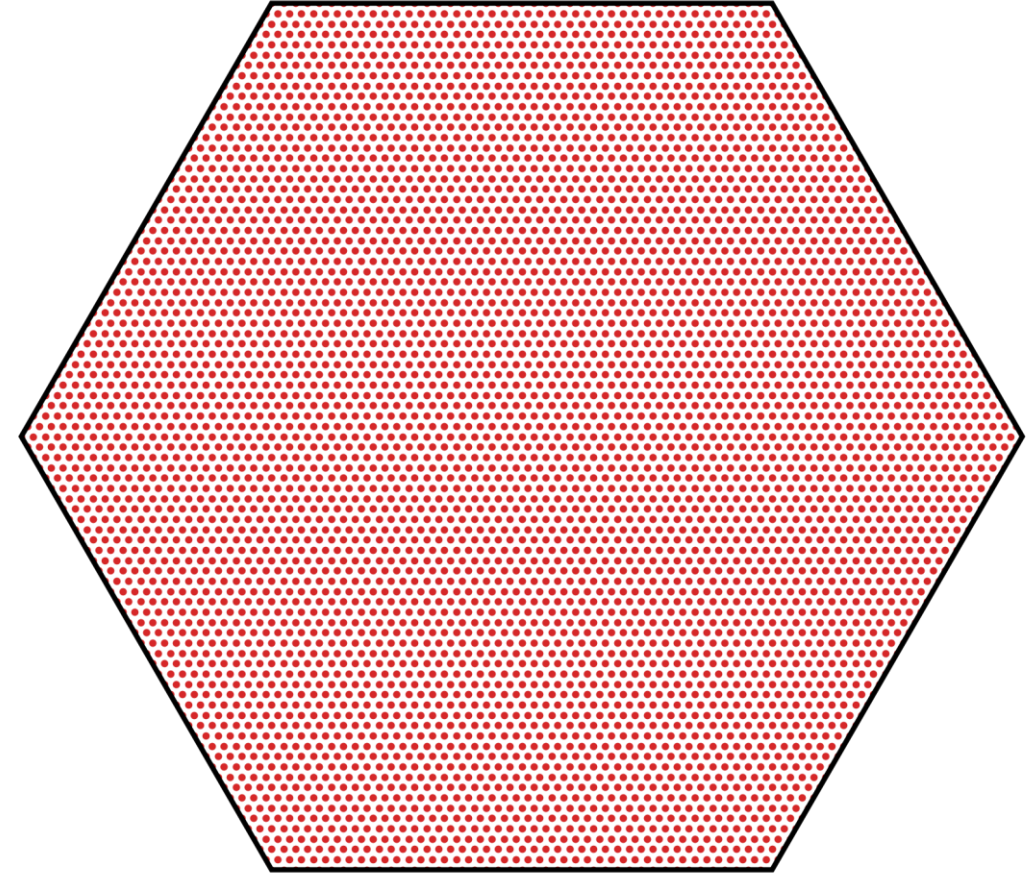
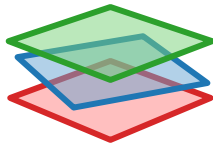
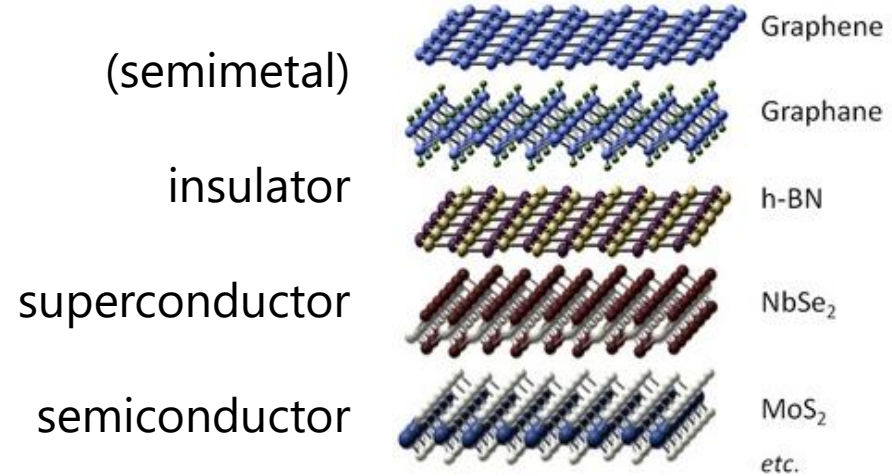
Van der Waals materials with a twist



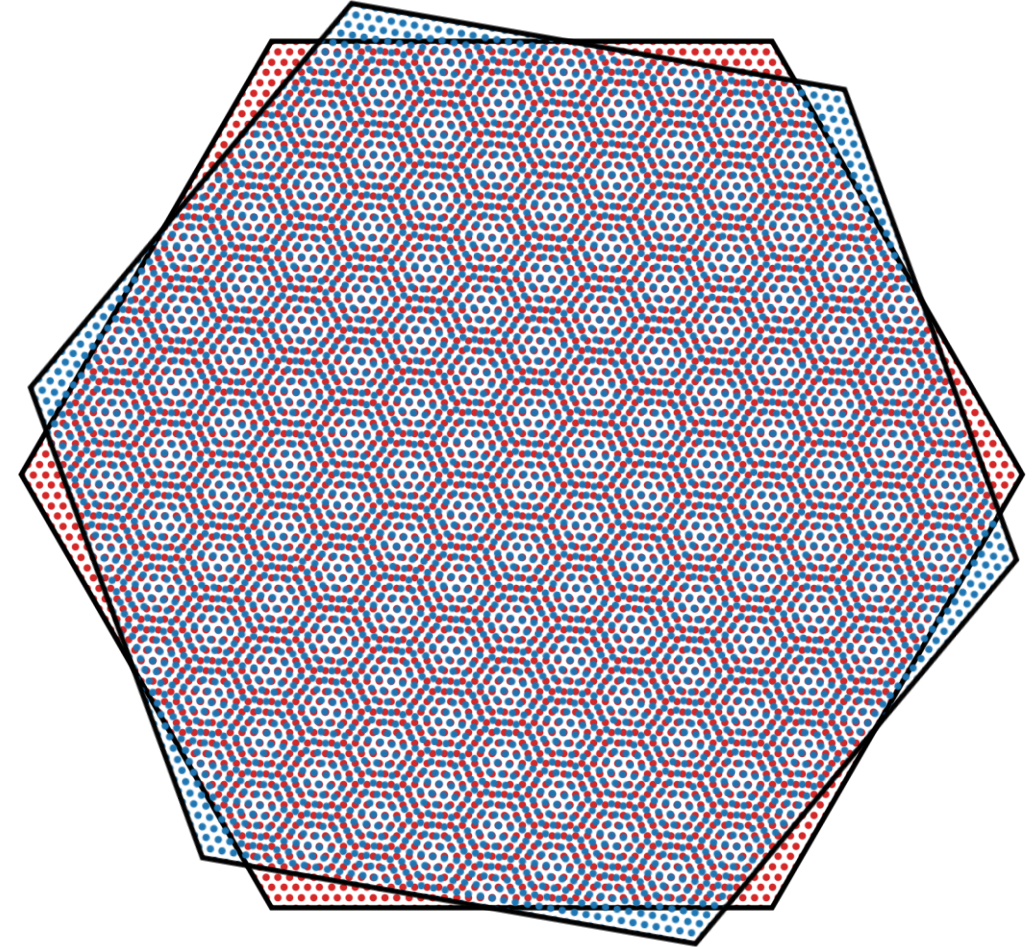
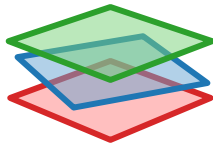
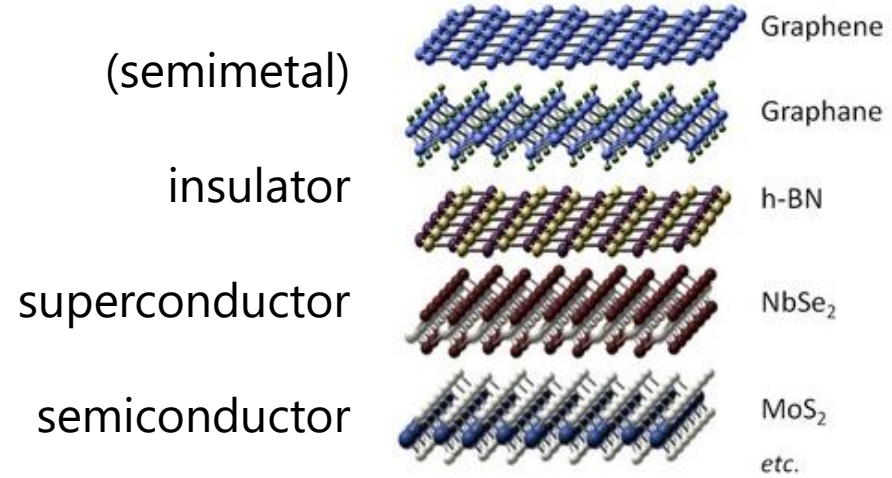
Van der Waals materials with a twist



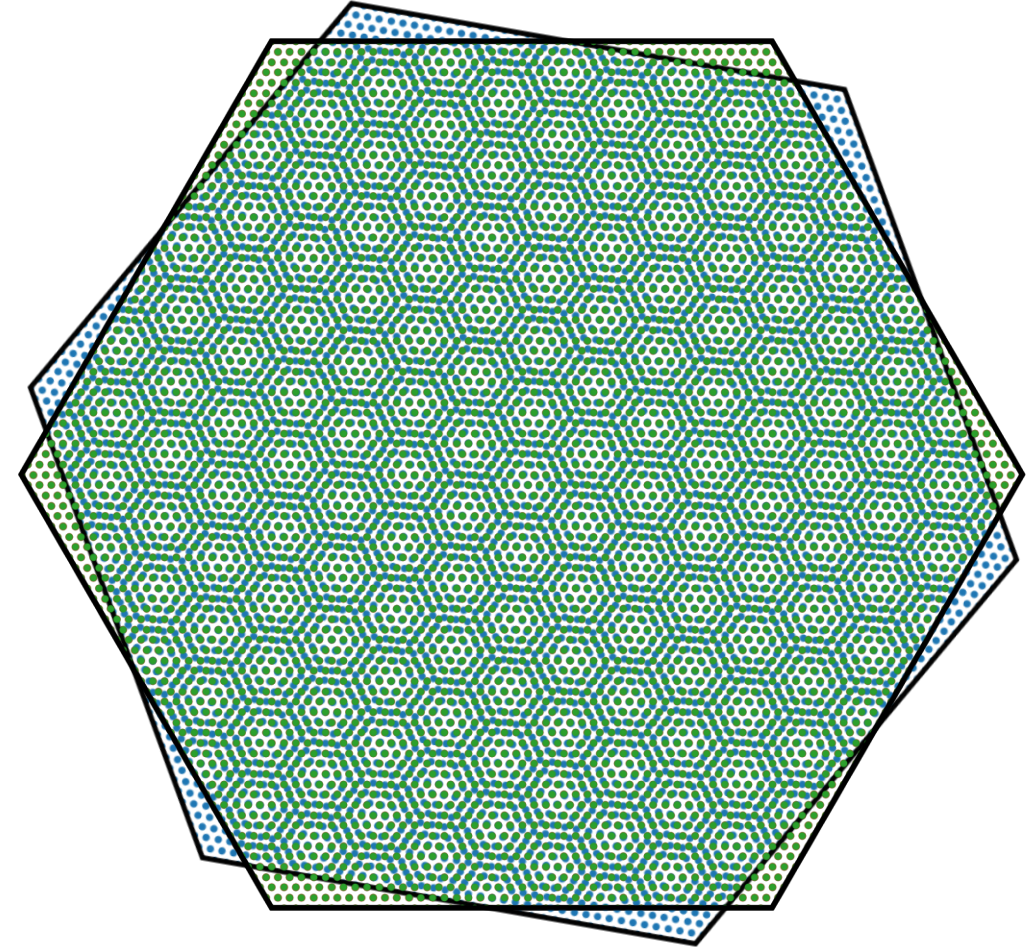
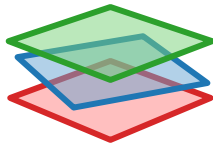
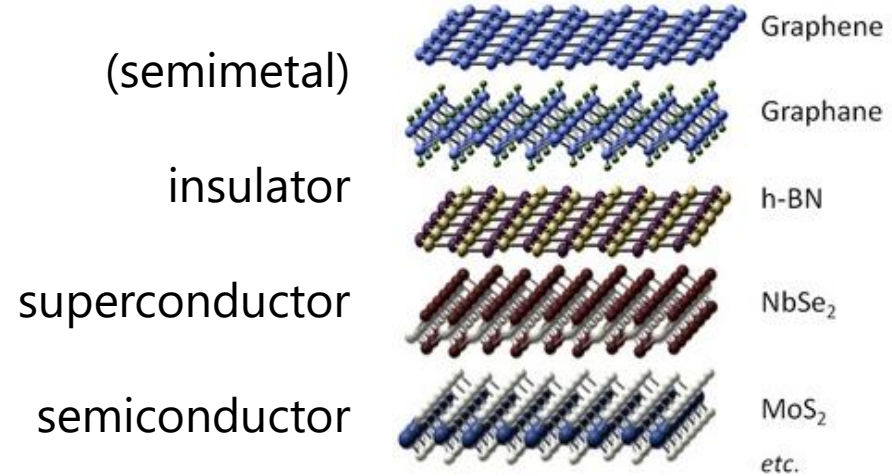
Van der Waals materials with a twist



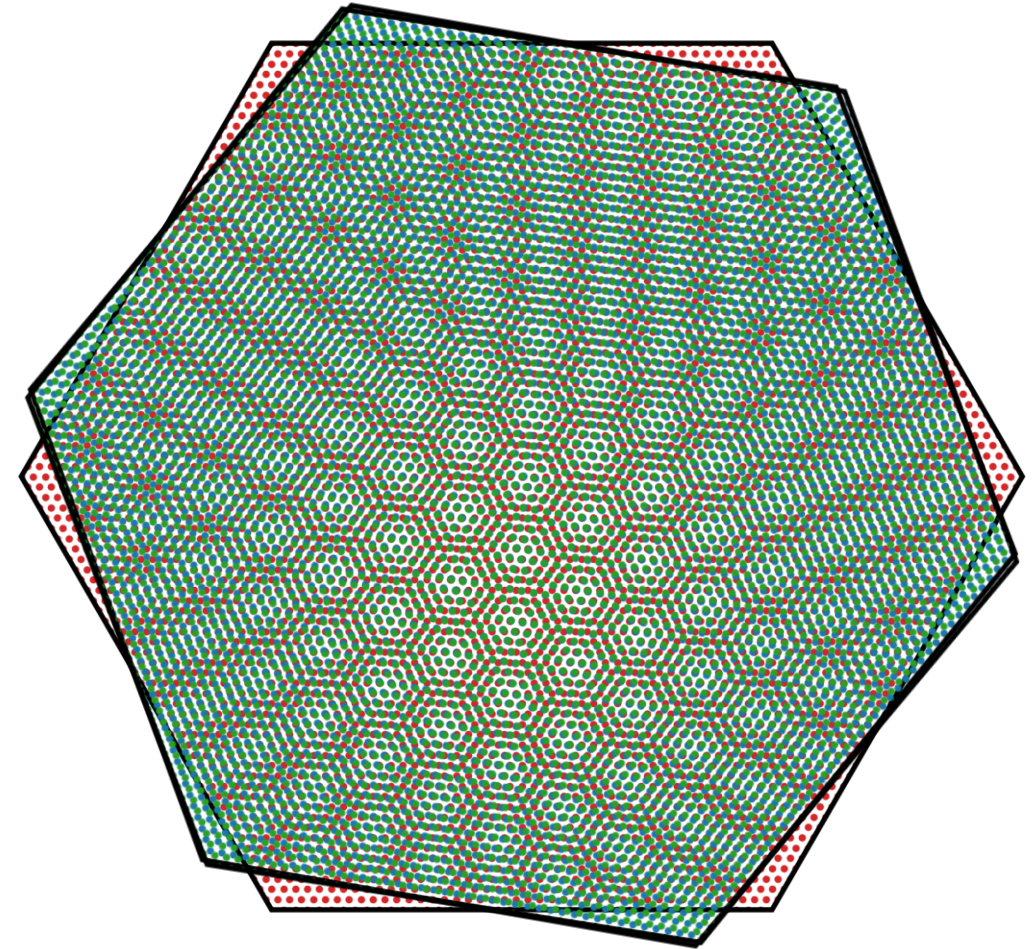
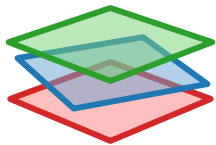
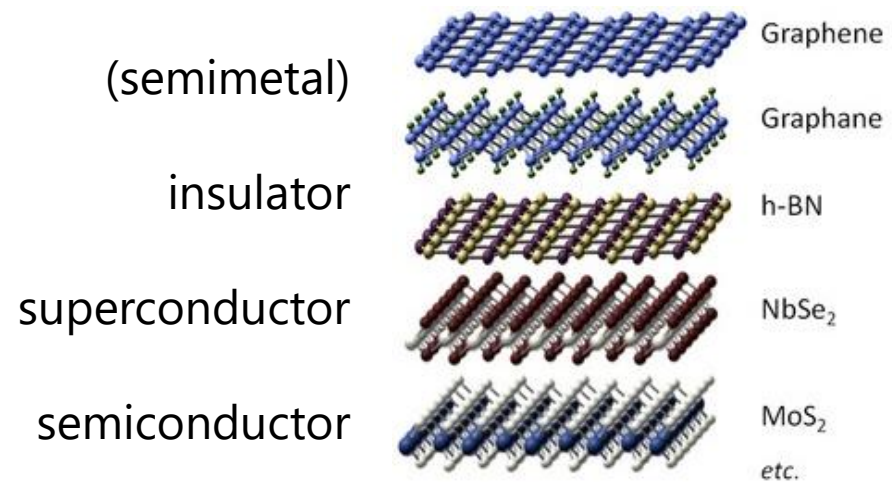
Van der Waals materials with a twist



Van der Waals materials with a twist

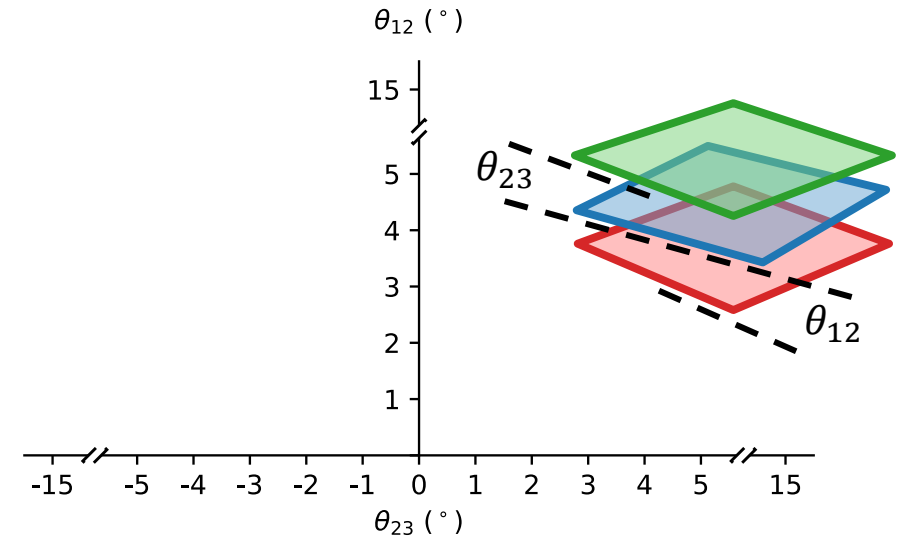


Van der Waals materials with a twist



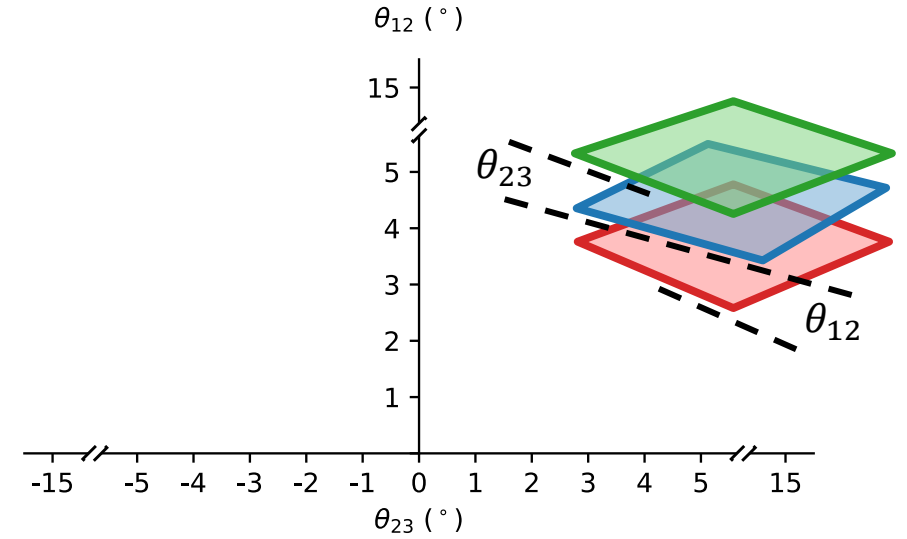
Problem statement:

1. A **third twisted layer** introduces *supermoiré* length scale. Do we also find **strong electronic correlations** in trilayer structures?



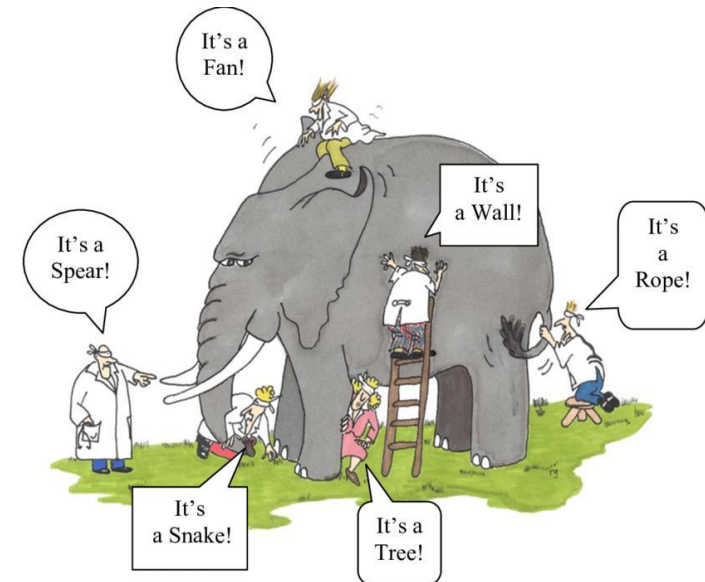
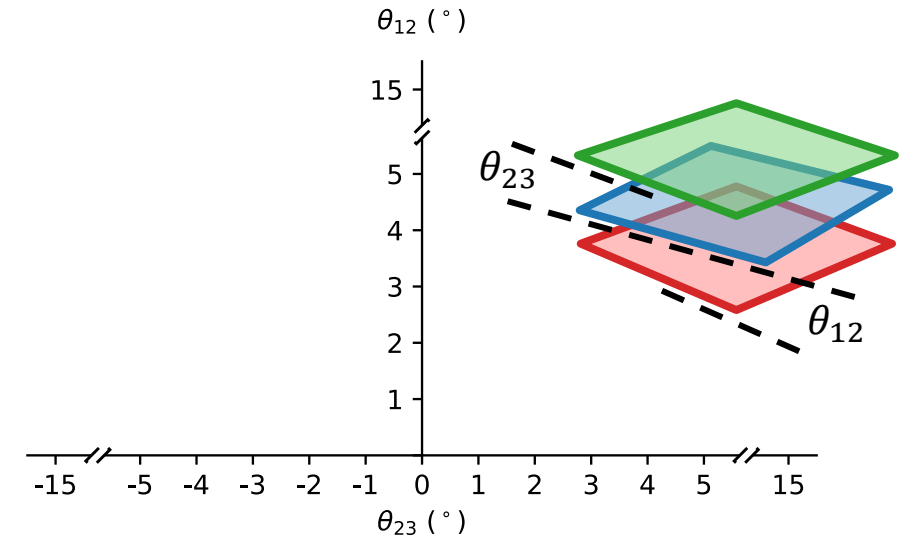
Problem statement:

1. A **third twisted layer** introduces *supermoiré* length scale. Do we also find **strong electronic correlations** in trilayer structures?
2. The **evolution with twist**? Distinct regions in angle-angle space? What **drives** the resultant phenomenology?



Problem statement:

1. A **third twisted layer** introduces *supermoiré* length scale. Do we also find **strong electronic correlations** in trilayer structures?
2. The **evolution with twist**? Distinct regions in angle-angle space? What **drives** the resultant phenomenology?
3. Comparing **different** devices is challenging. How will we systematically explore this space?



Outline

Introduction:

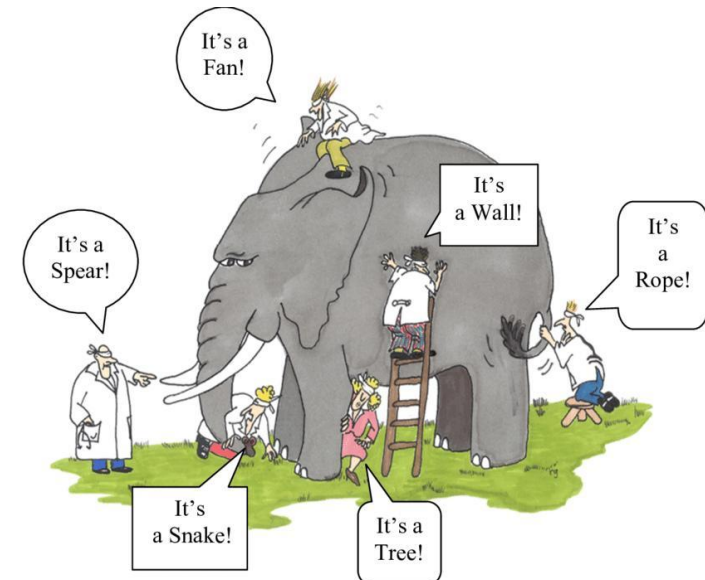
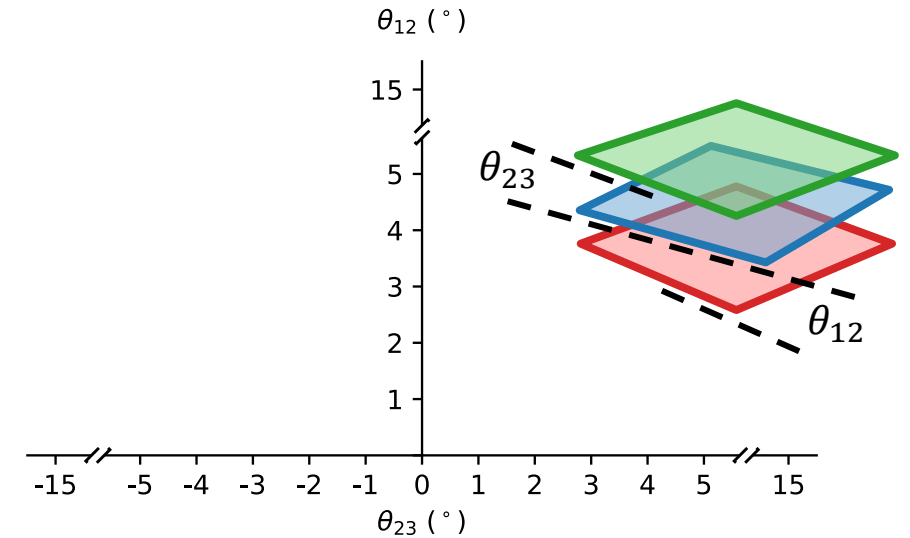
- From moiré to supermoiré
- Helical vs. alternating stacking
- Lattice relaxation

Linking thermodynamic probes and transport:

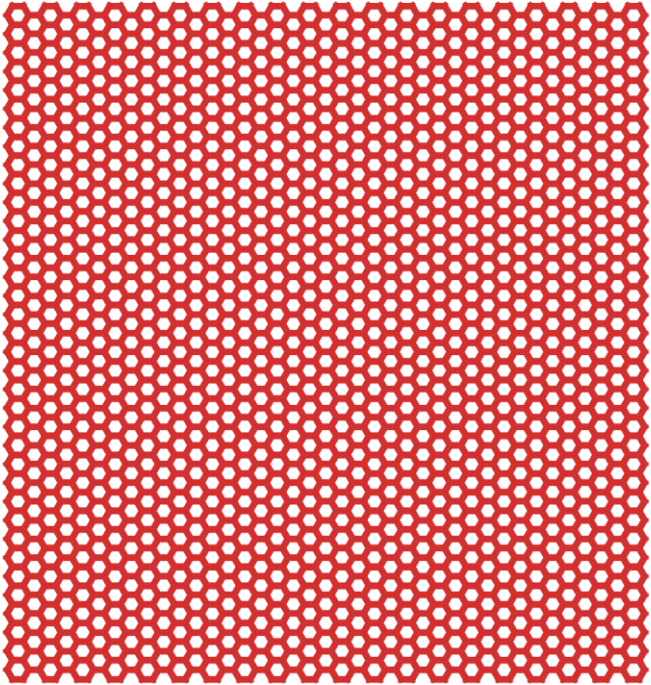
- Fine-grained evolution of correlations
- Cross comparison of distinct experimental probes

Exploring the “magic continuum”:

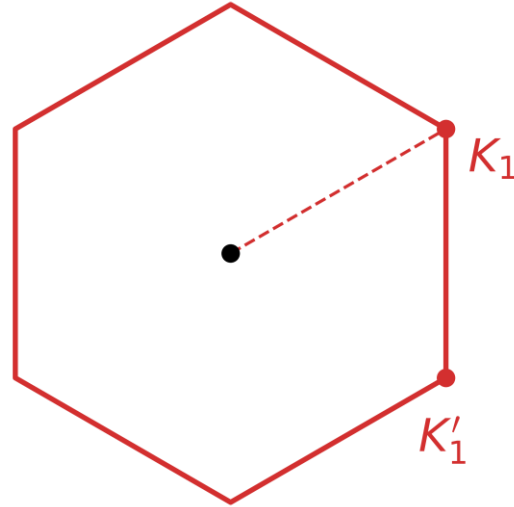
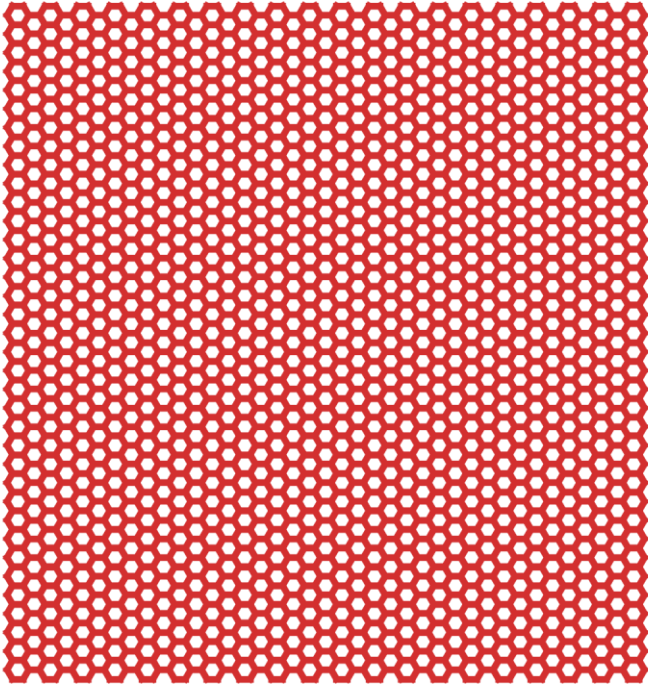
- Broad comparison across many devices in angle-angle space



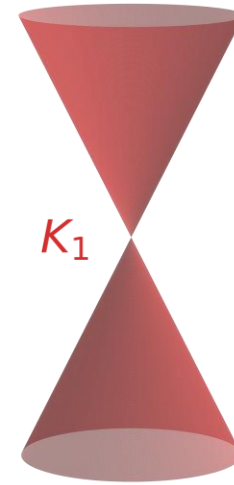
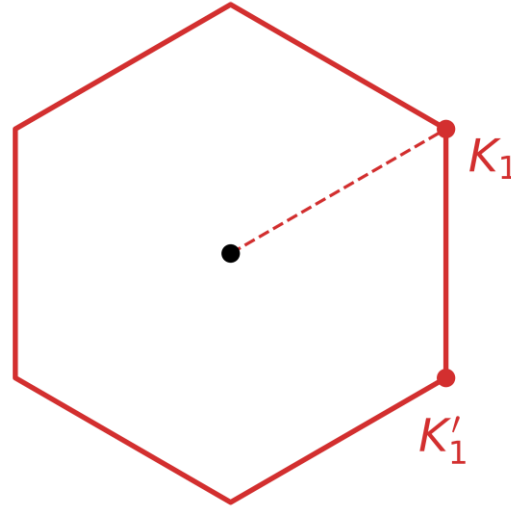
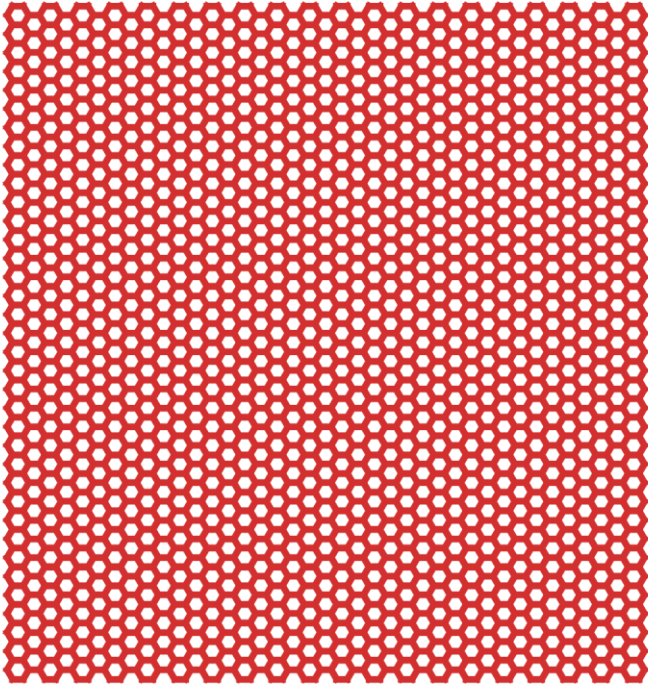
Bandstructure engineering with moirés



Bandstructure engineering with moirés

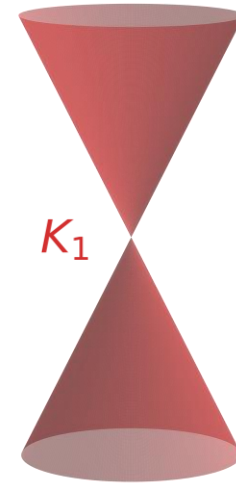
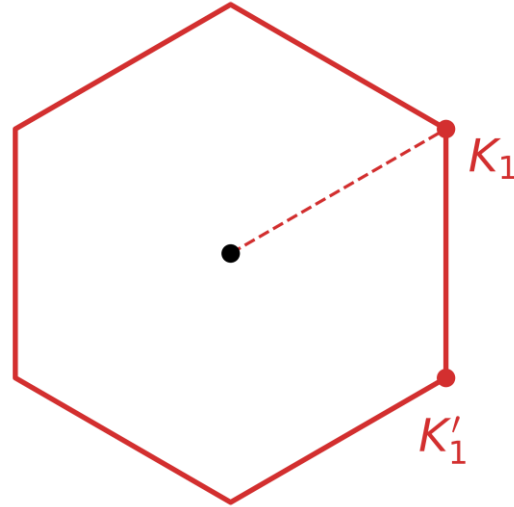
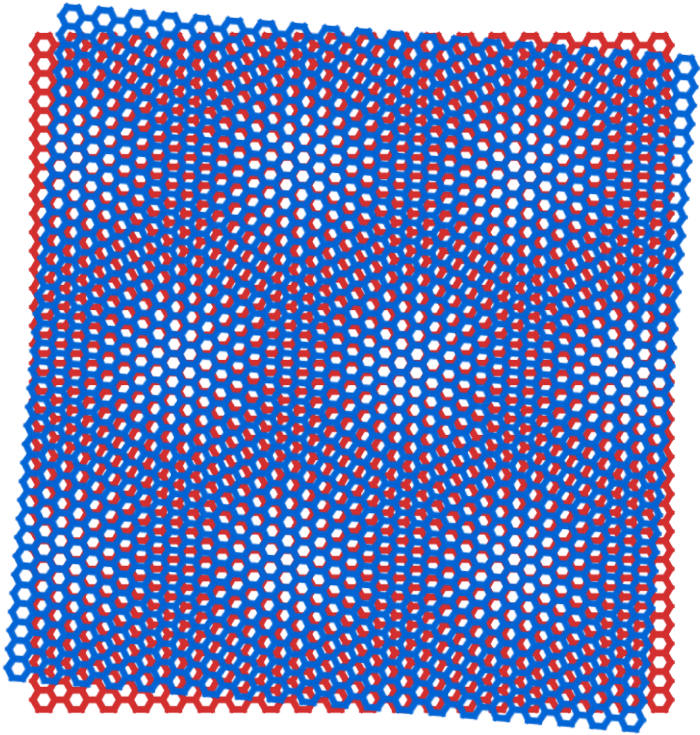


Bandstructure engineering with moirés



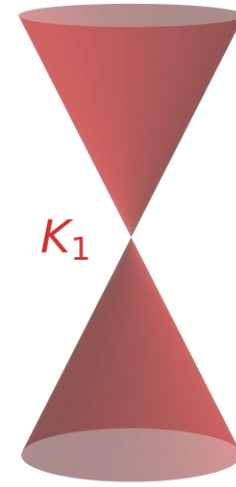
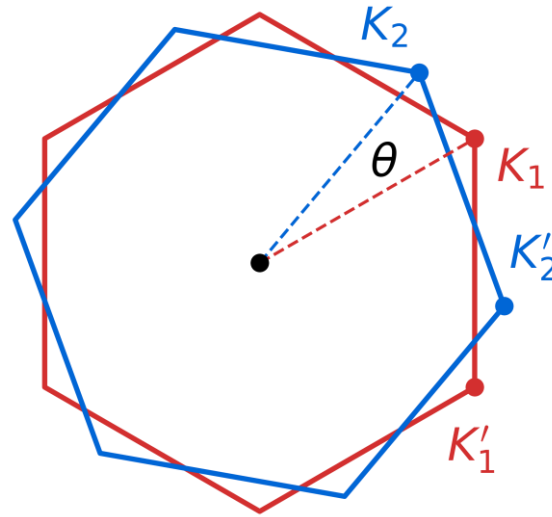
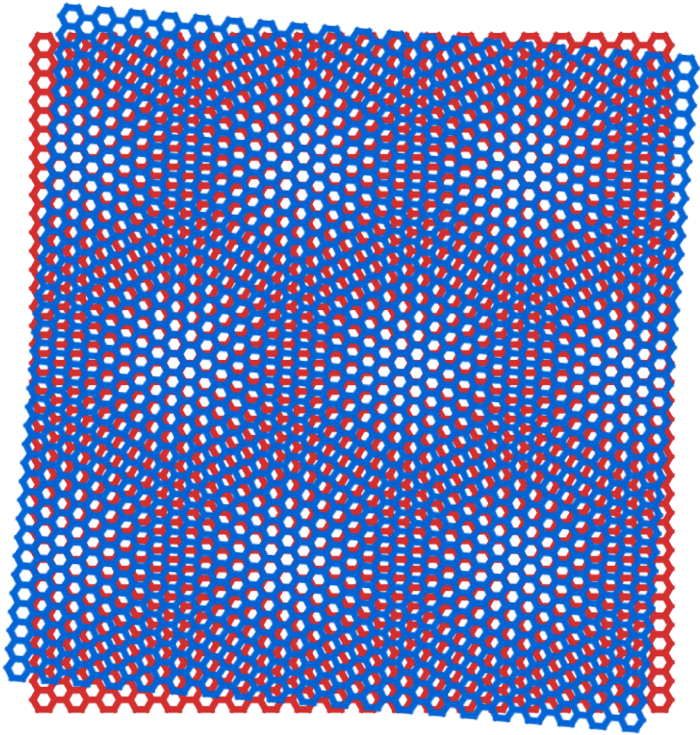
4-fold degenerate
spin x valley

Bandstructure engineering with moirés



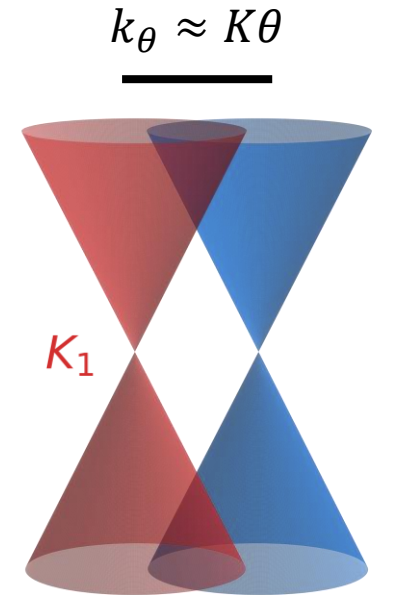
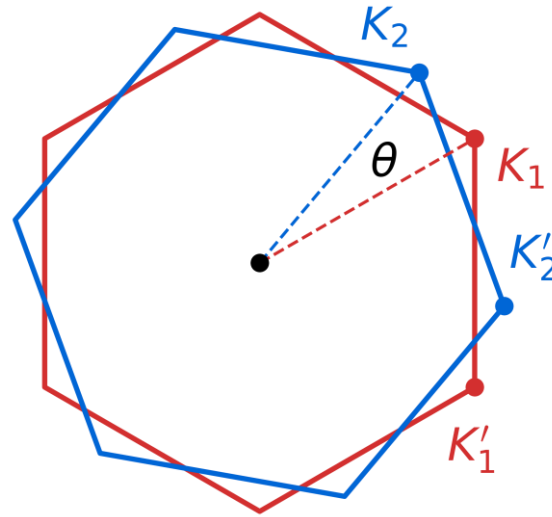
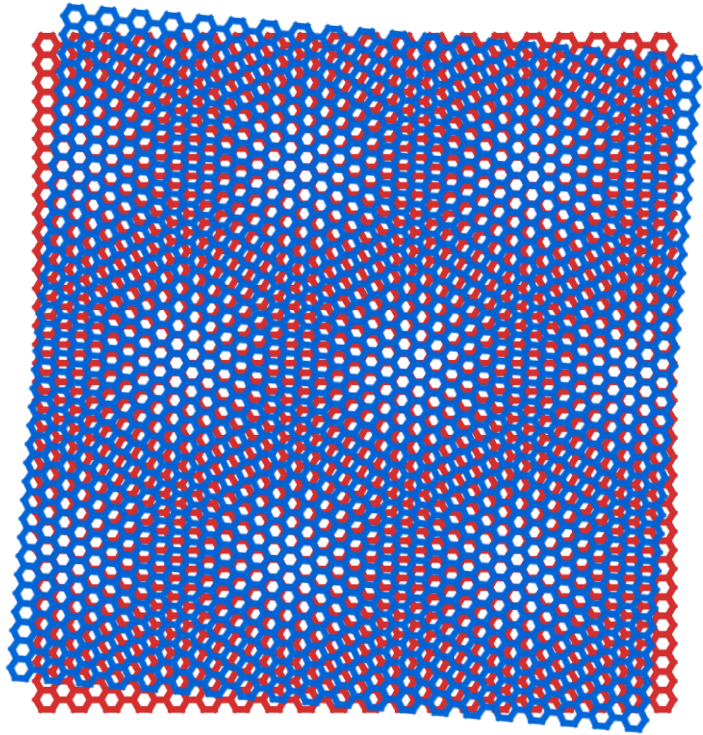
4-fold degenerate
spin x valley

Bandstructure engineering with moirés



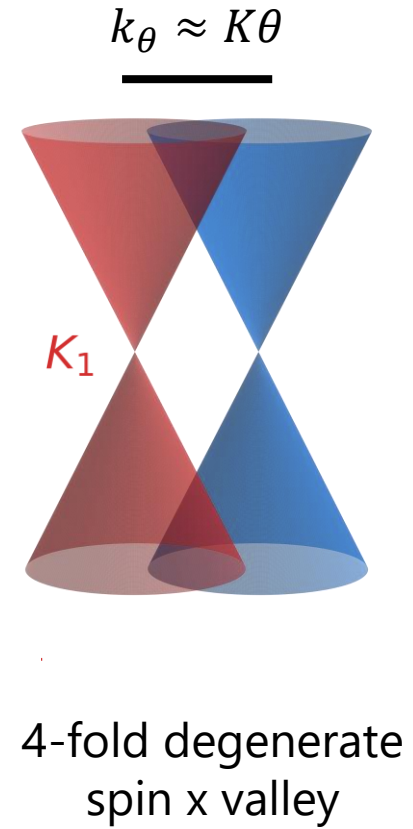
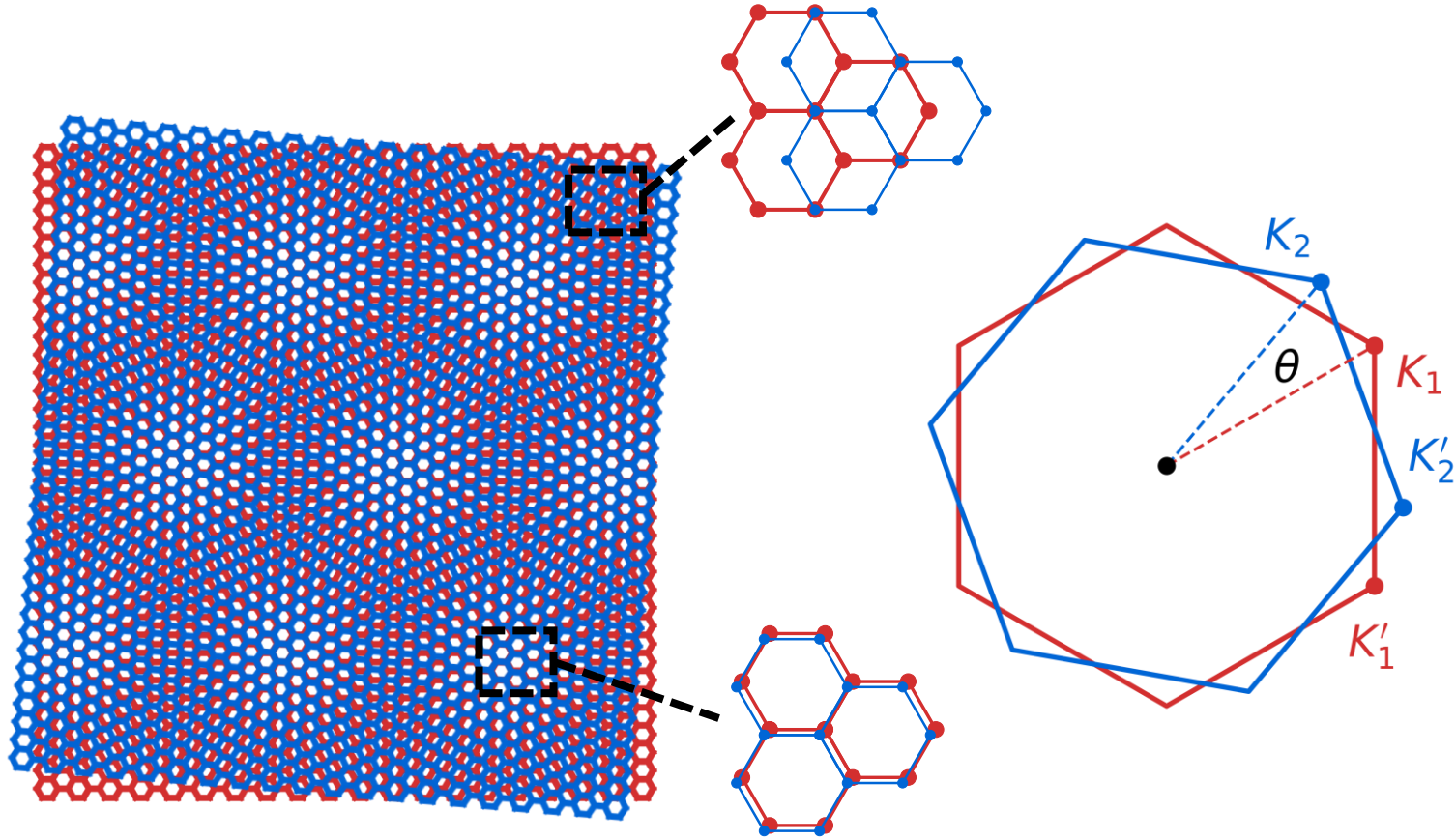
4-fold degenerate
spin x valley

Bandstructure engineering with moirés

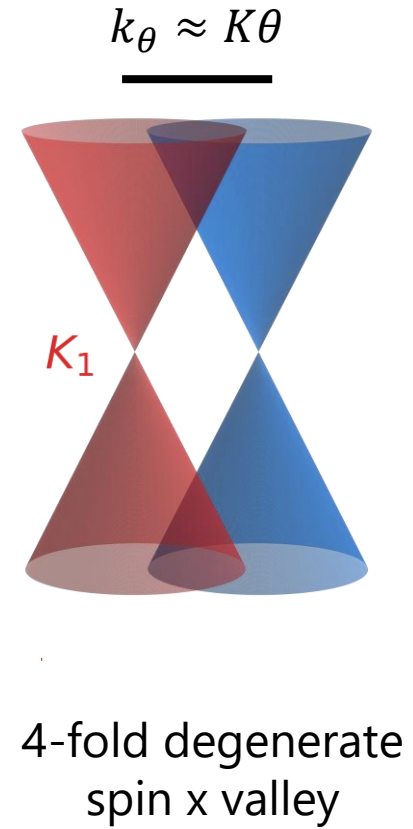
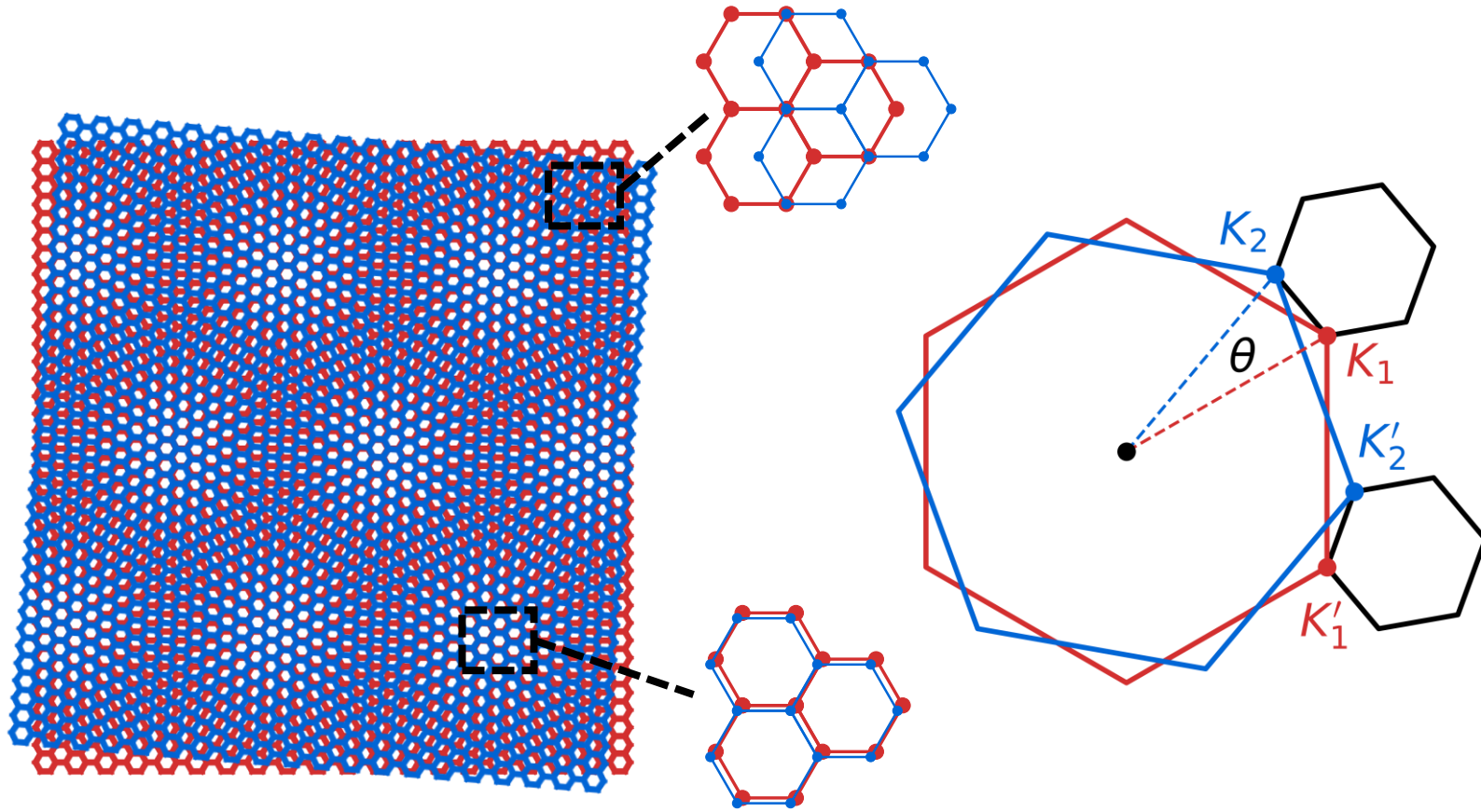


4-fold degenerate
spin x valley

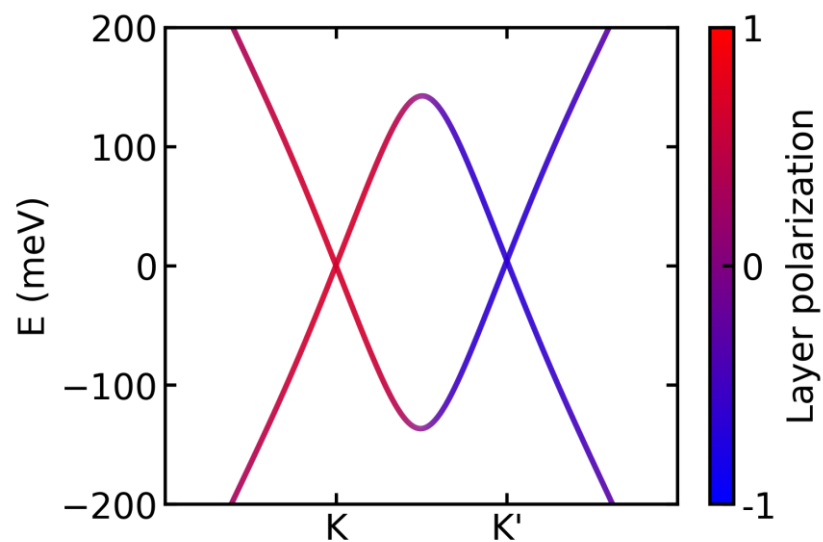
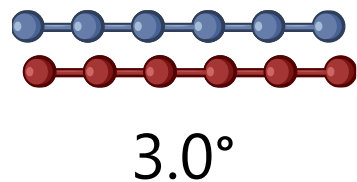
Bandstructure engineering with moirés



Bandstructure engineering with moirés

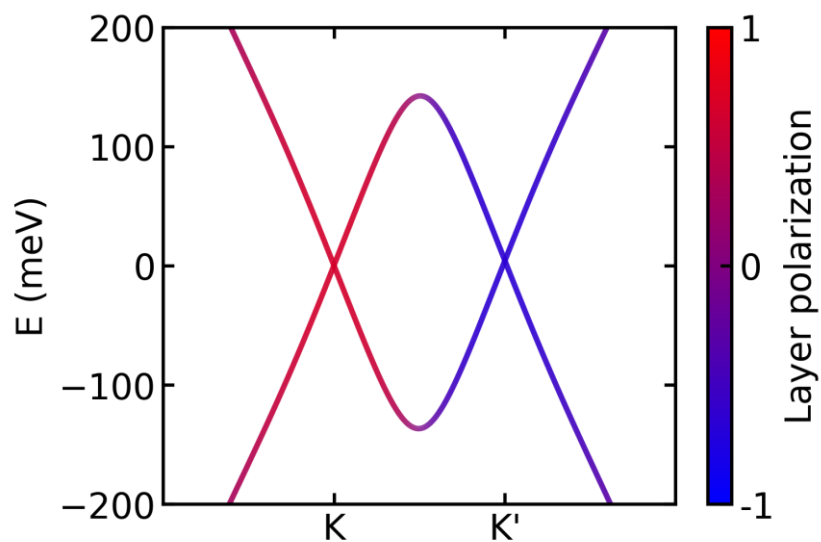
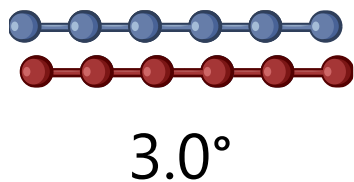


Bandstructure engineering with moirés



~Electronically decoupled
from momentum mismatch

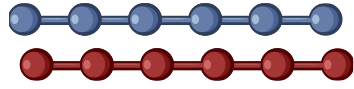
Bandstructure engineering with moirés



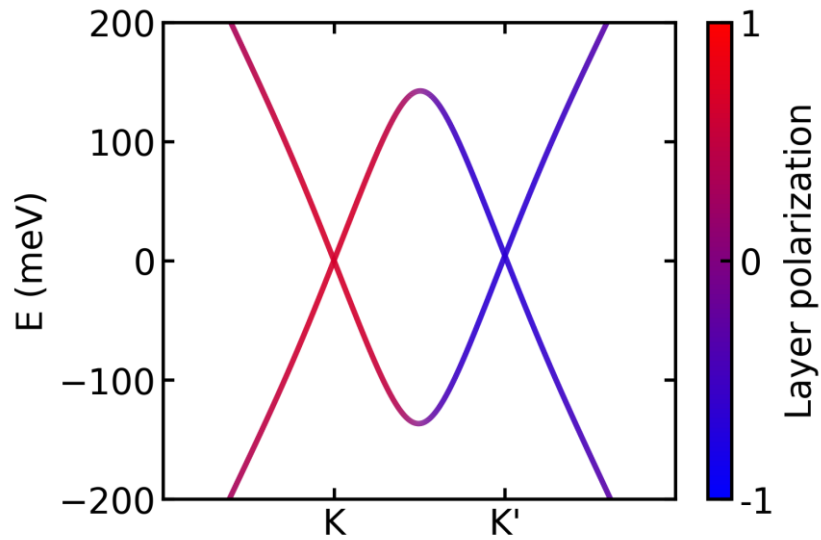
~Electronically decoupled
from momentum mismatch

Decreasing twist angle

Bandstructure engineering with moirés

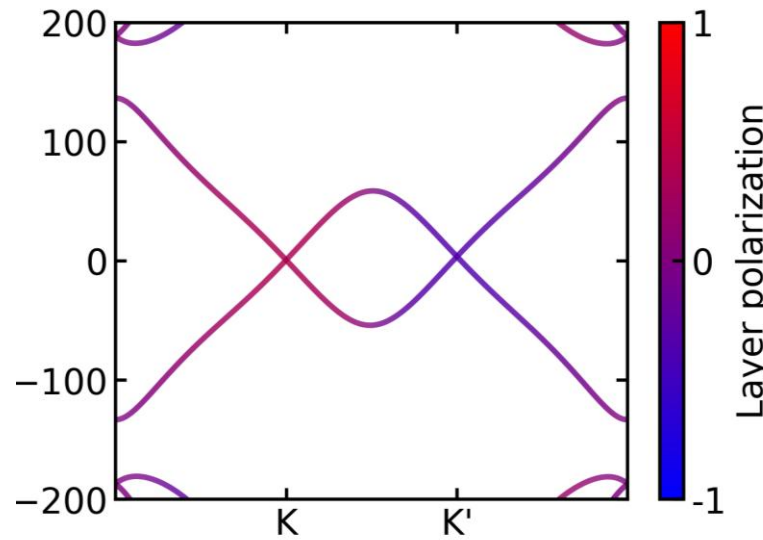


3.0°



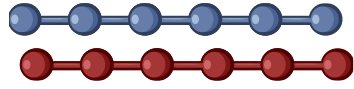
~Electronically decoupled
from momentum mismatch

2.0°

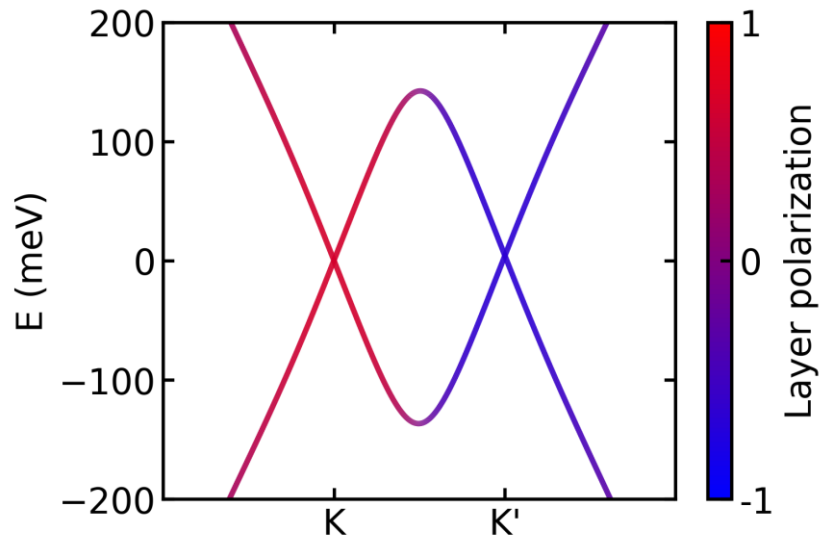


Decreasing twist angle →

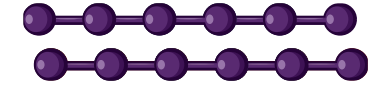
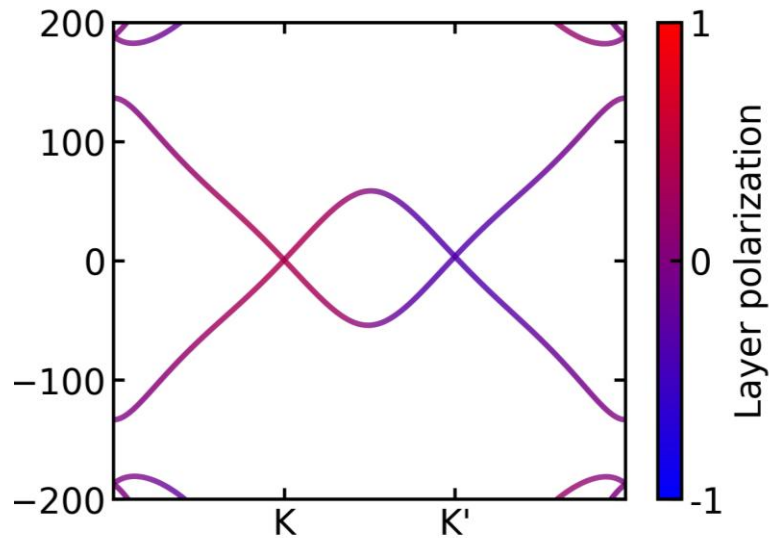
Bandstructure engineering with moirés



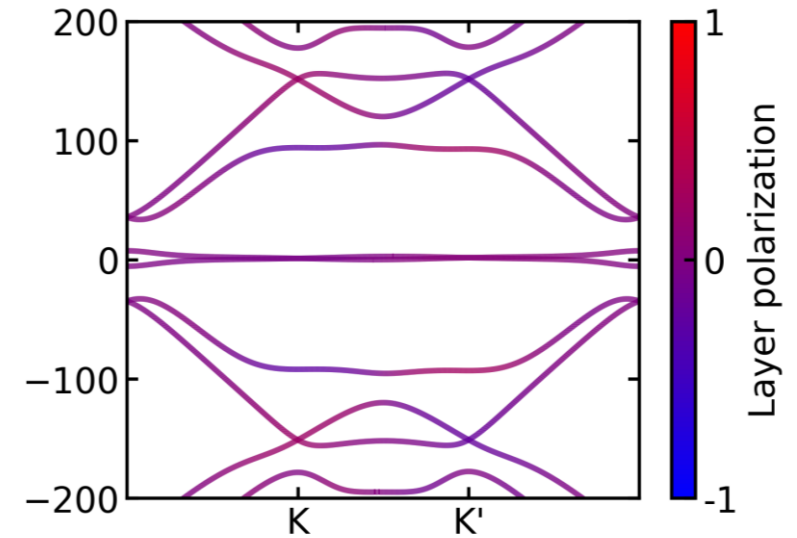
3.0°



2.0°

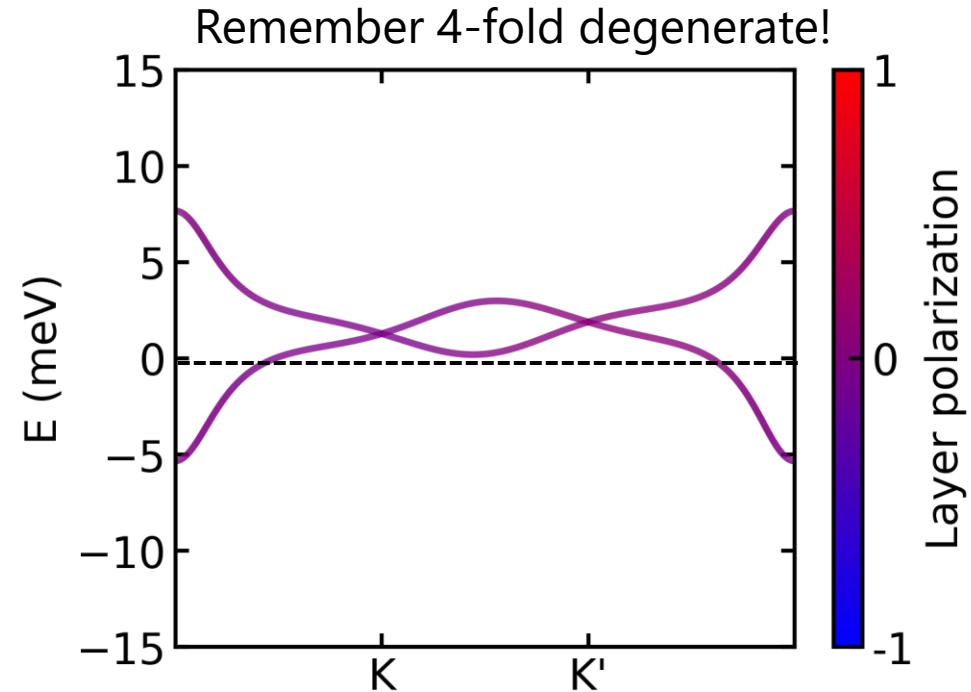
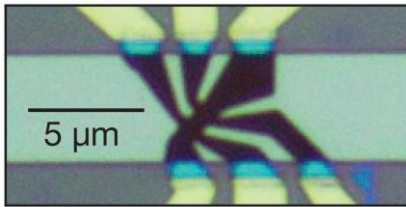
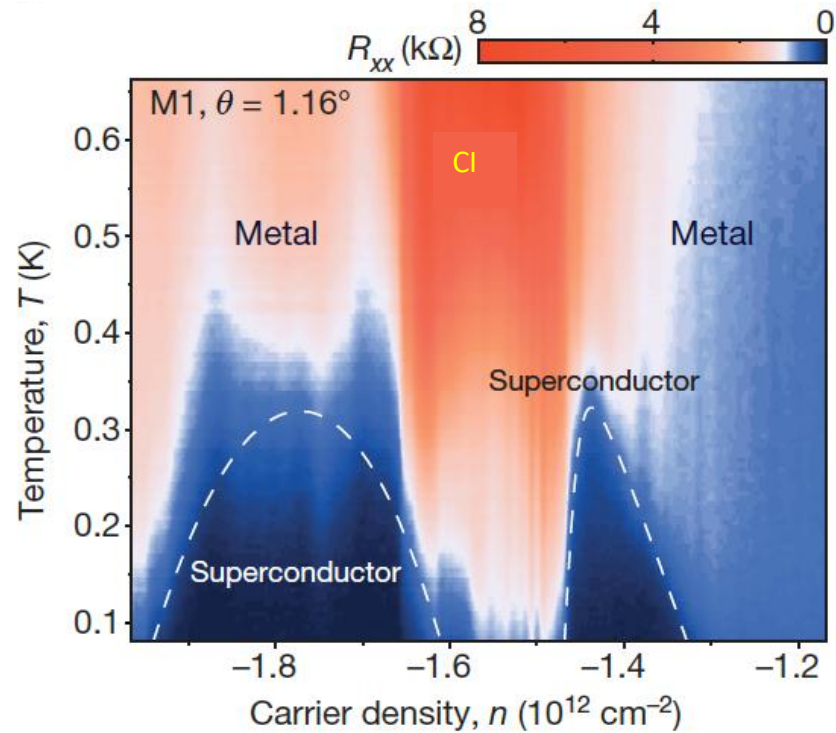


1.1°



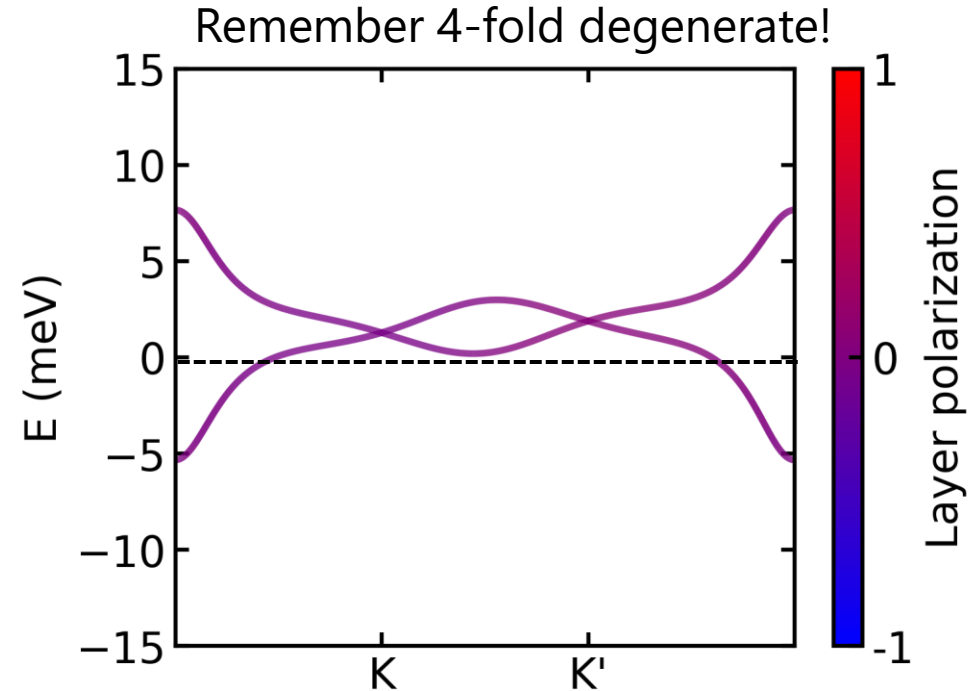
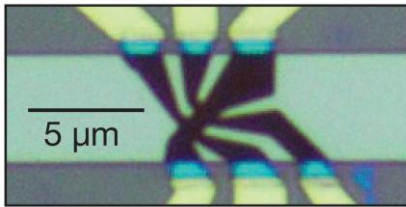
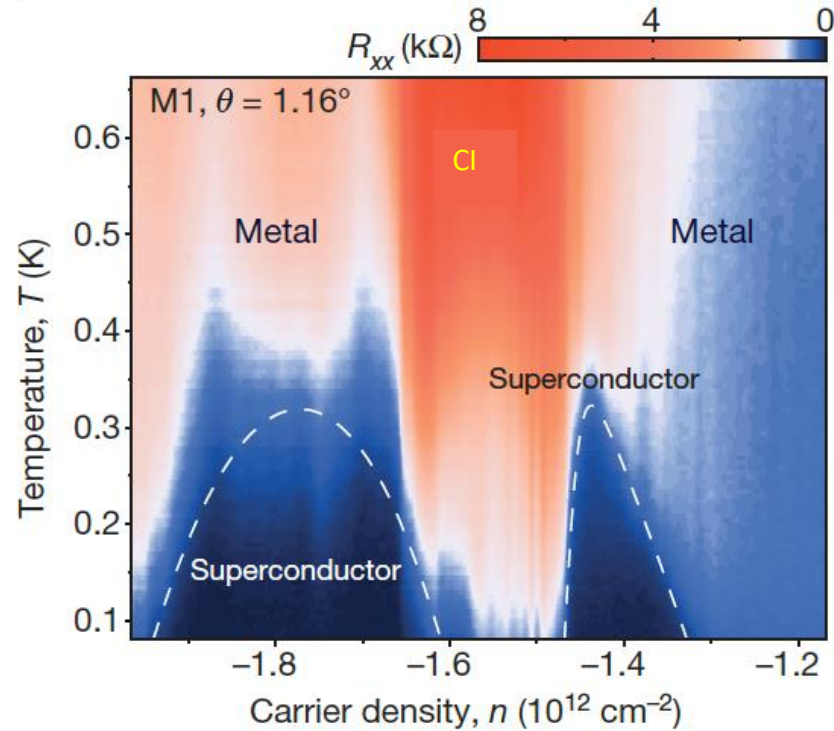
Decreasing twist angle

Magic angle twisted bilayer graphene (MATBG)



Cao, et al. Nature (2018)
Yoo et al. Nat. Nano. (2019)
Bistritzer and MacDonald, PNAS (2011)

Magic angle twisted bilayer graphene (MATBG)



Let's add a third layer of graphene!

Factors we expect to be important:

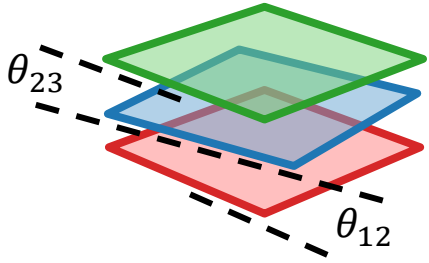
1. Two interlayer twist angles
2. Intermoiré shift (*a la* atomic shifts)
3. Atomic reconstruction

Cao, et al. Nature (2018)

Yoo et al. Nat. Nano. (2019)

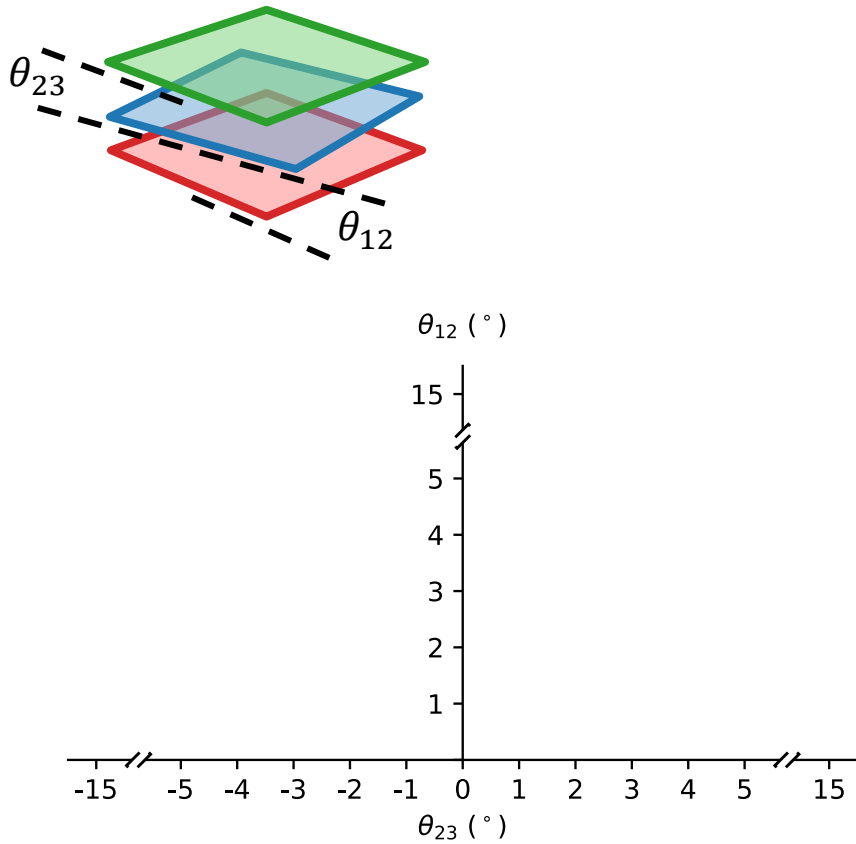
Bistritzer and MacDonald, PNAS (2011)

Where to begin? “Magic Angle Twisted Trilayer”



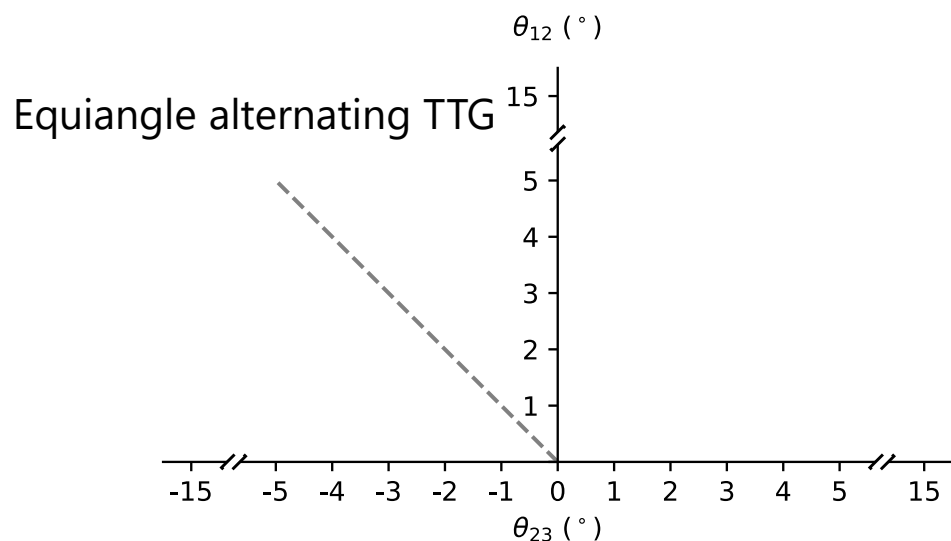
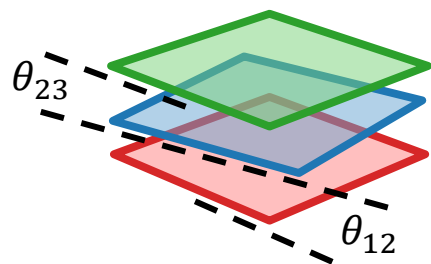
Khalaf, et al. PRB (2019)
Park, et al. Nature (2021)
Hao, et al. Science (2021)
Kim, et al. Nature (2022)

Where to begin? “Magic Angle Twisted Trilayer”



Khalaf, et al. PRB (2019)
Park, et al. Nature (2021)
Hao, et al. Science (2021)
Kim, et al. Nature (2022)

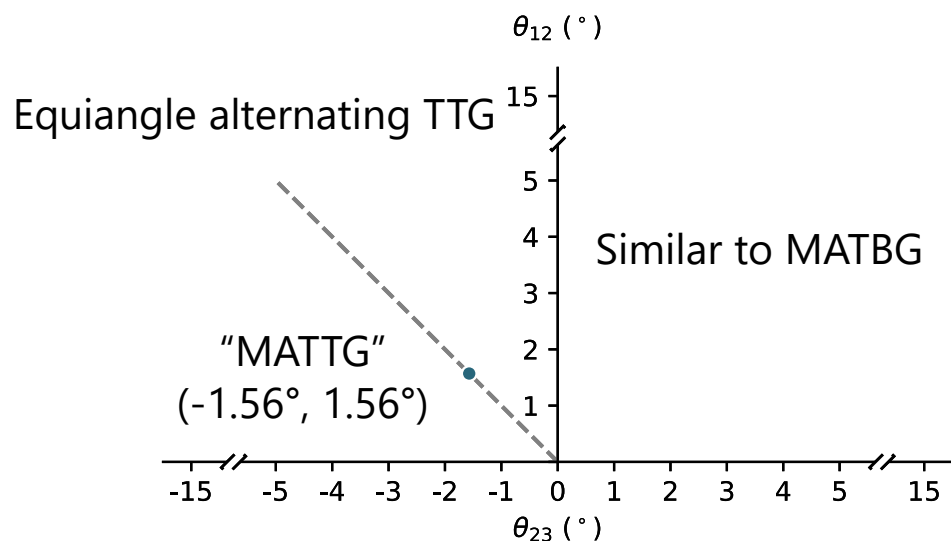
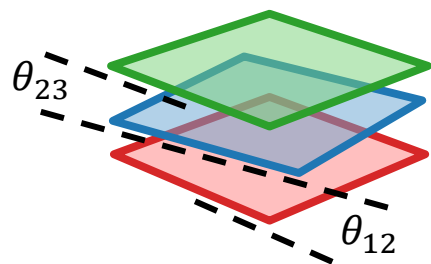
Where to begin? “Magic Angle Twisted Trilayer”



Khalaf, et al. PRB (2019)
Park, et al. Nature (2021)
Hao, et al. Science (2021)
Kim, et al. Nature (2022)

For equiangle alternating TTG
moirés patterns are *exactly* commensurate

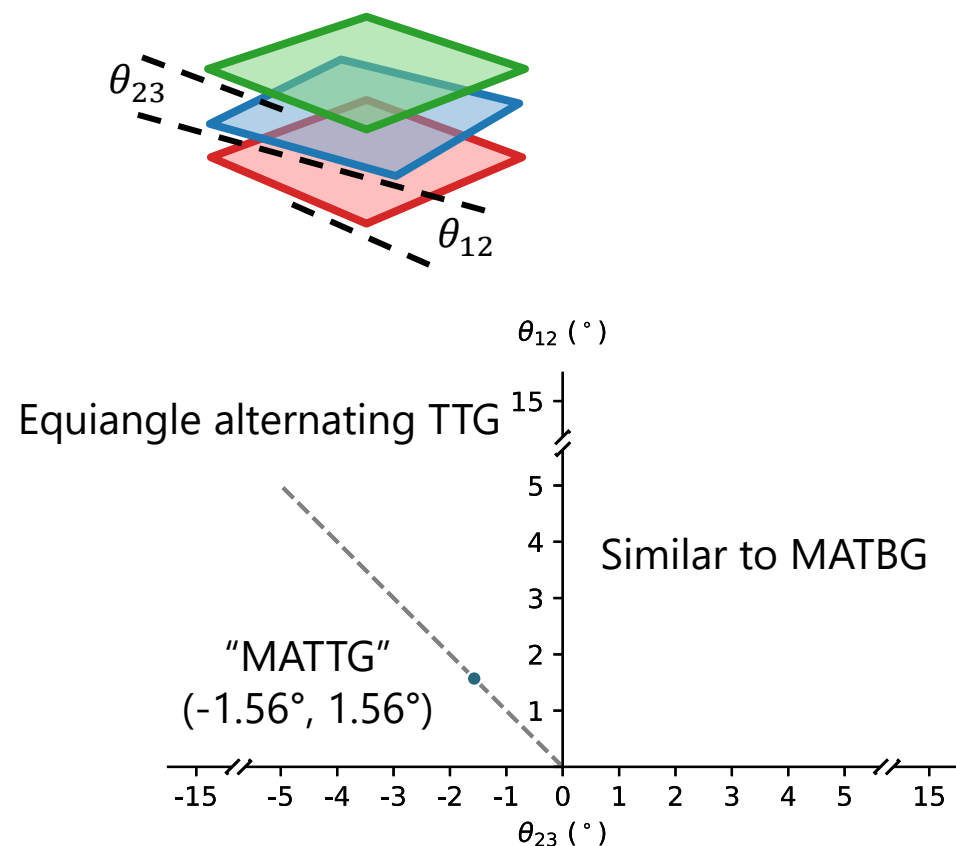
Where to begin? “Magic Angle Twisted Trilayer”



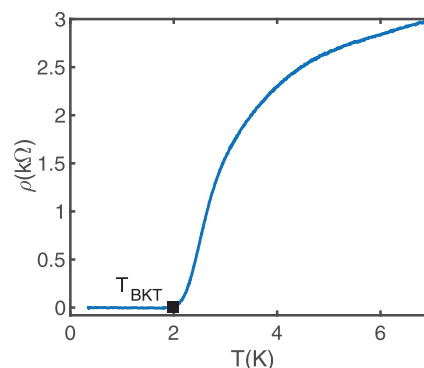
Khalaf, et al. PRB (2019)
Park, et al. Nature (2021)
Hao, et al. Science (2021)
Kim, et al. Nature (2022)

For equiangle alternating TTG
moirés patterns are *exactly* commensurate

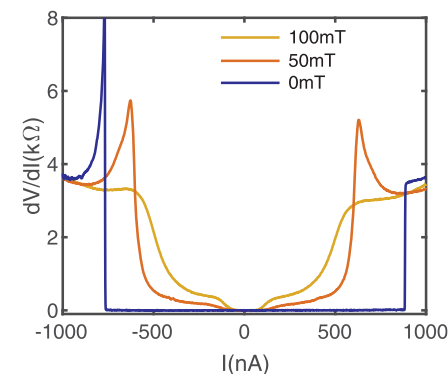
Where to begin? “Magic Angle Twisted Trilayer”



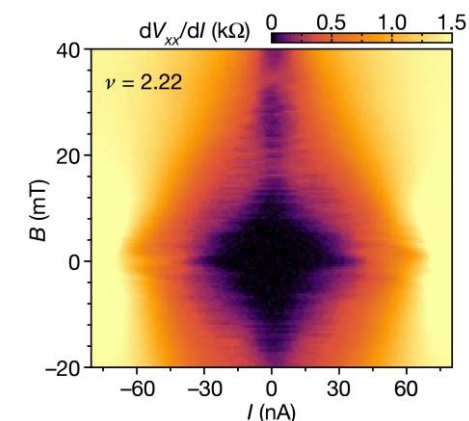
Dissipationless



Non-linear



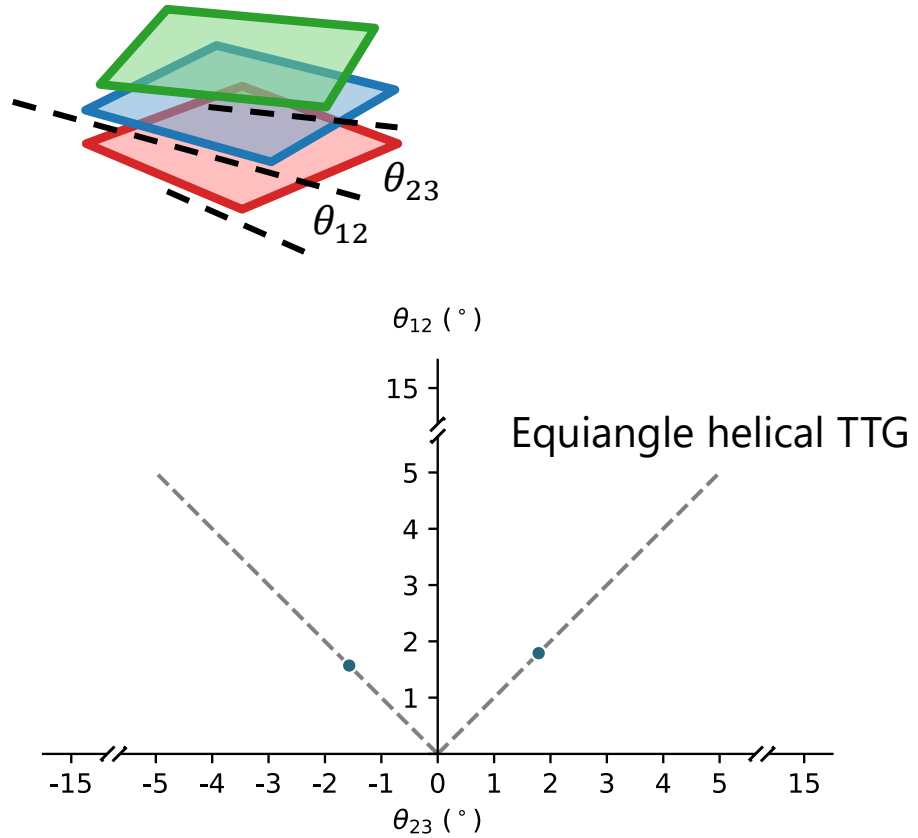
Phase coherent



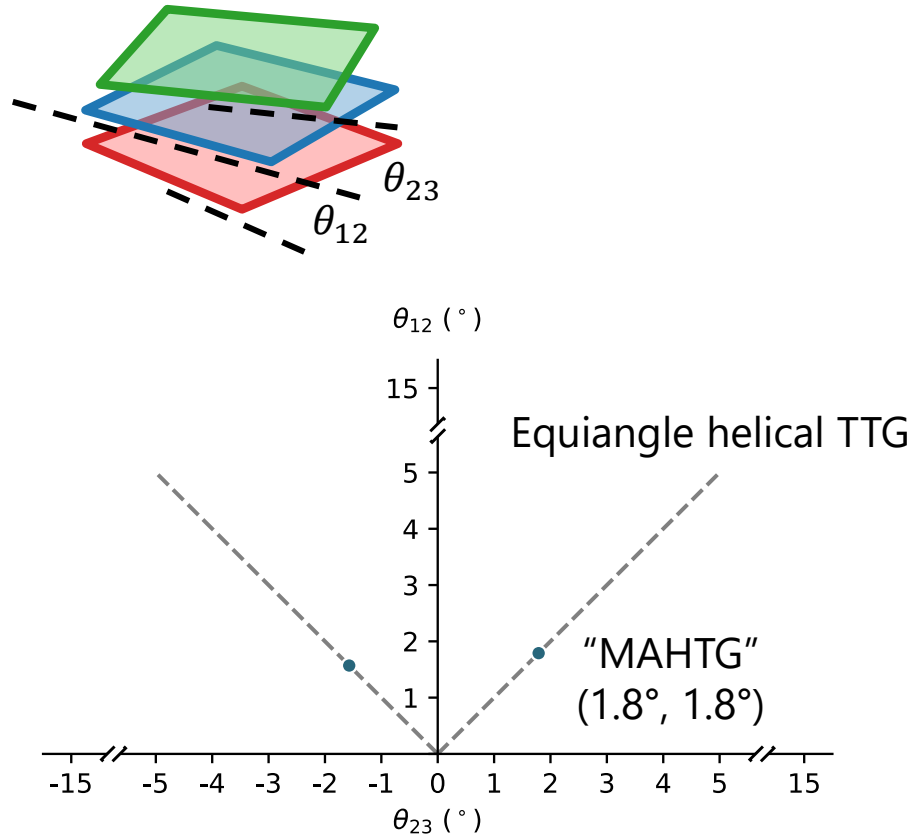
Khalaf, et al. PRB (2019)
Park, et al. Nature (2021)
Hao, et al. Science (2021)
Kim, et al. Nature (2022)

For equiangle alternating TTG
moirés patterns are *exactly* commensurate

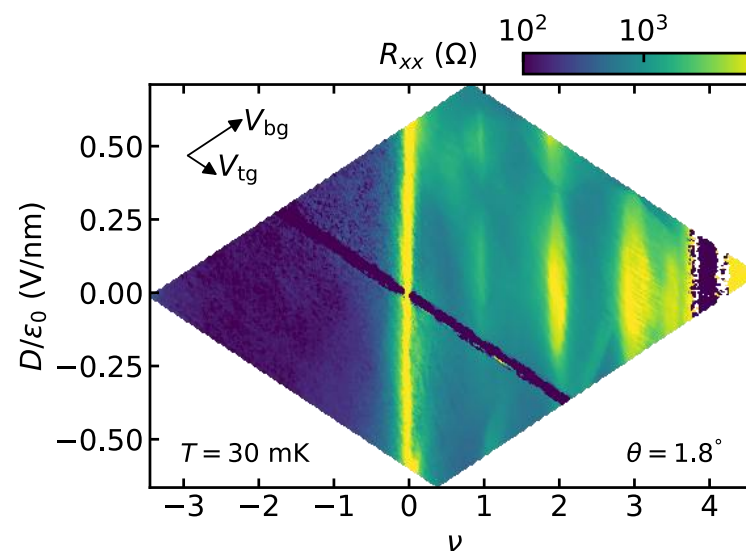
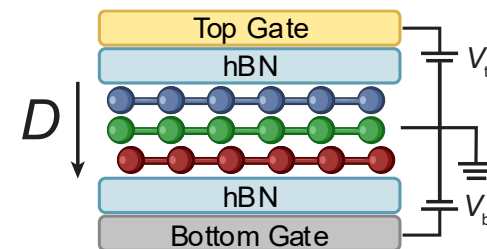
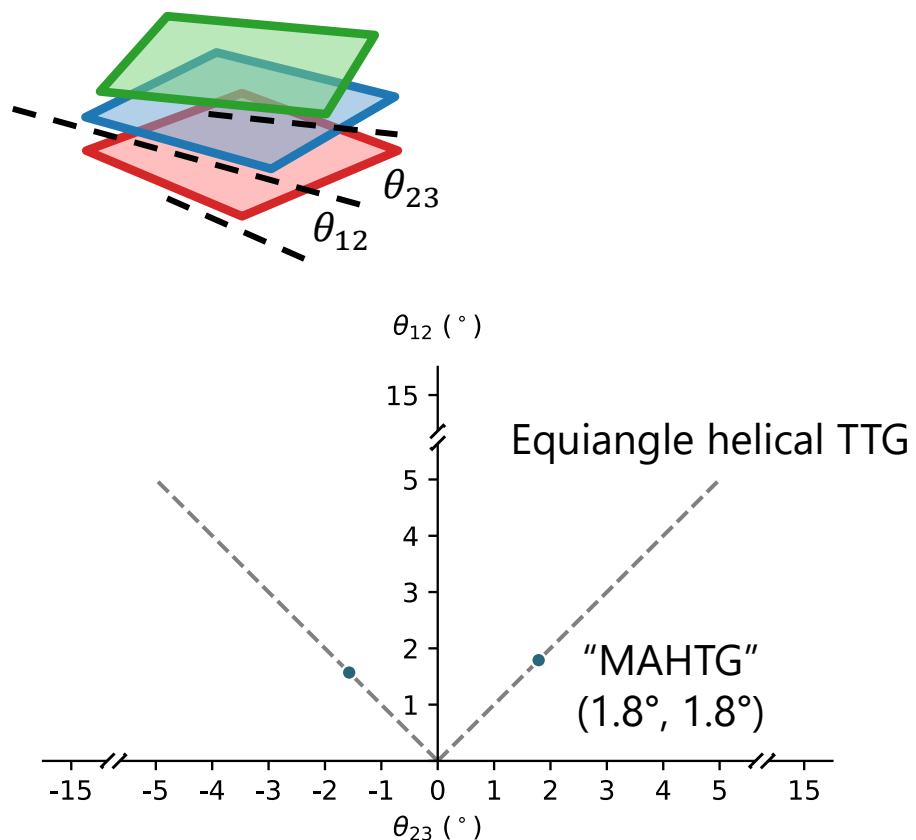
Equiangle Helical Trilayer



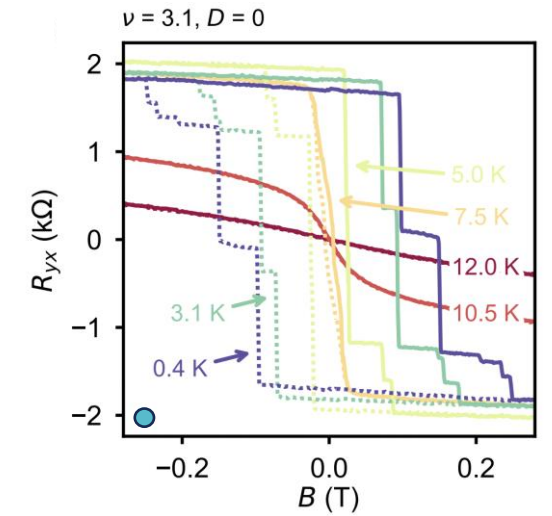
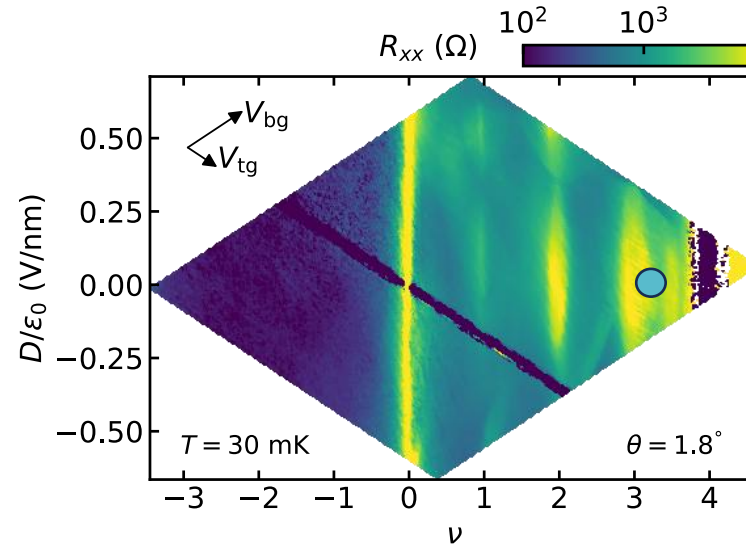
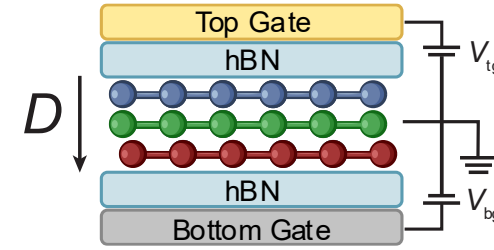
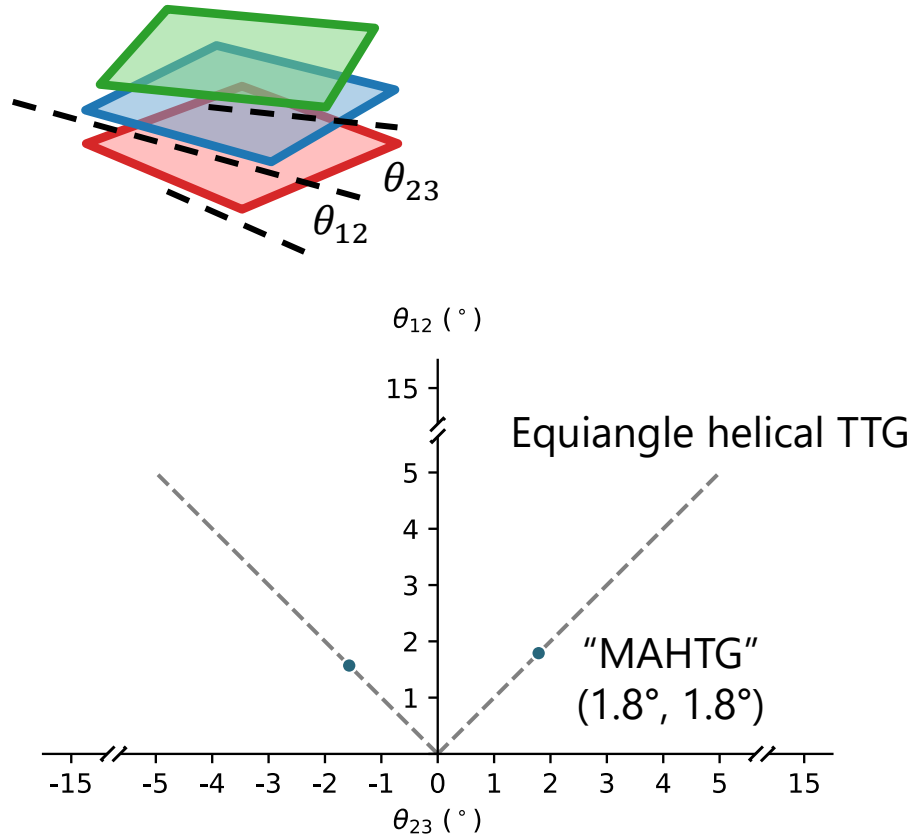
Equiangle Helical Trilayer



Equiangle Helical Trilayer

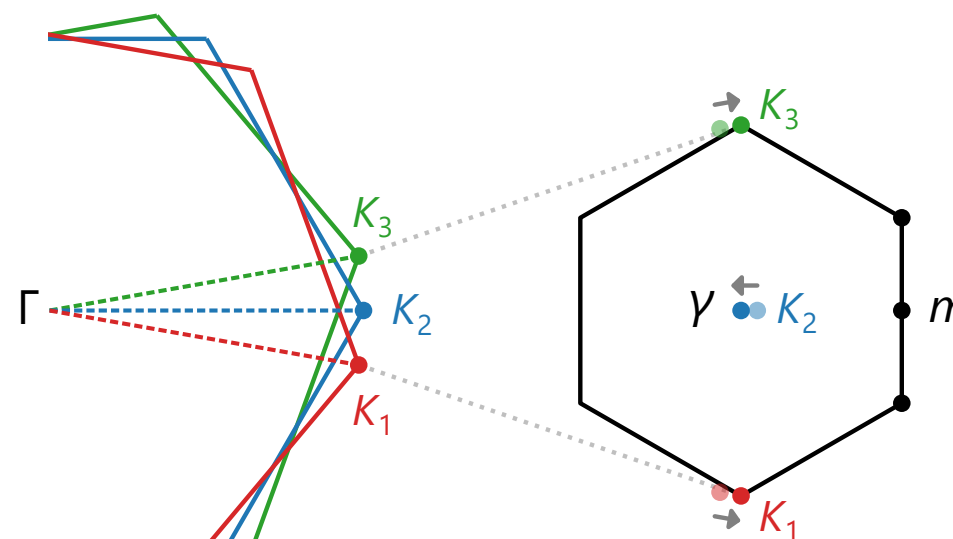
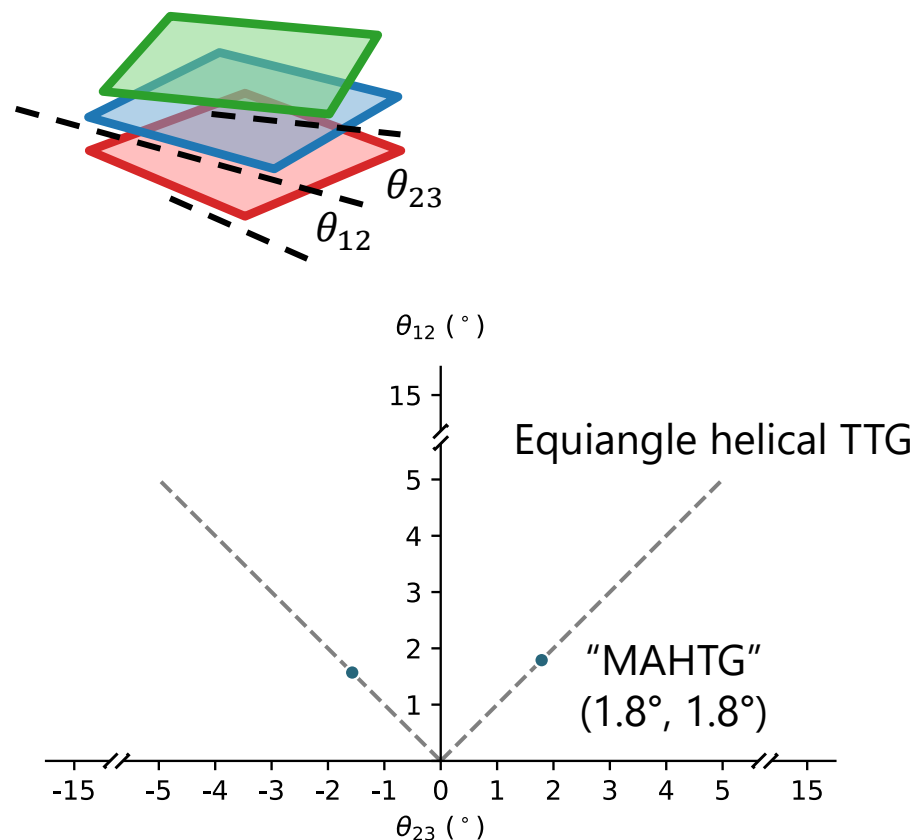


Equiangle Helical Trilayer



No signs of superconductivity so far.
Unquantized anomalous Hall. Domain physics?
What is driving this change in the resulting physics?

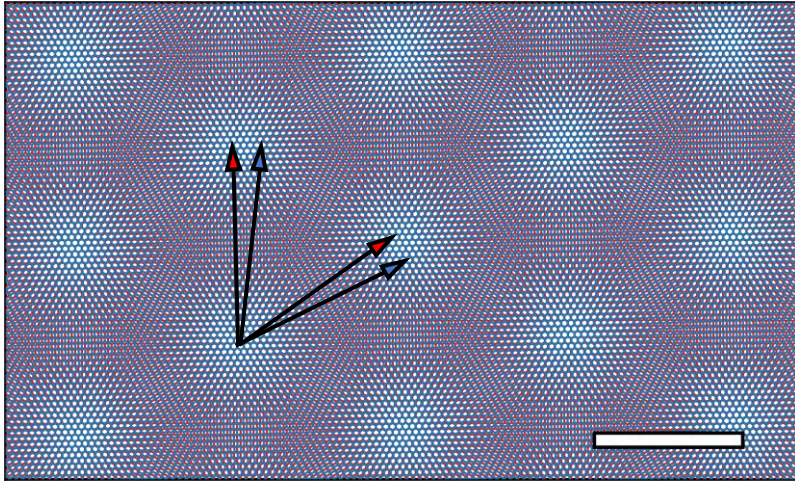
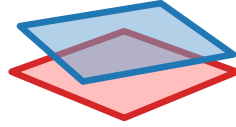
Generically have incommensurate moirés



In HTG, the two moirés are rotated!
We have a moiré of moiré

From moiré to multimoiré

Layers **1** and **2**



Scale bars: 5 nm

Yang, et al. PRB (2024)

Nakatsuji, et al. PRX (2023)

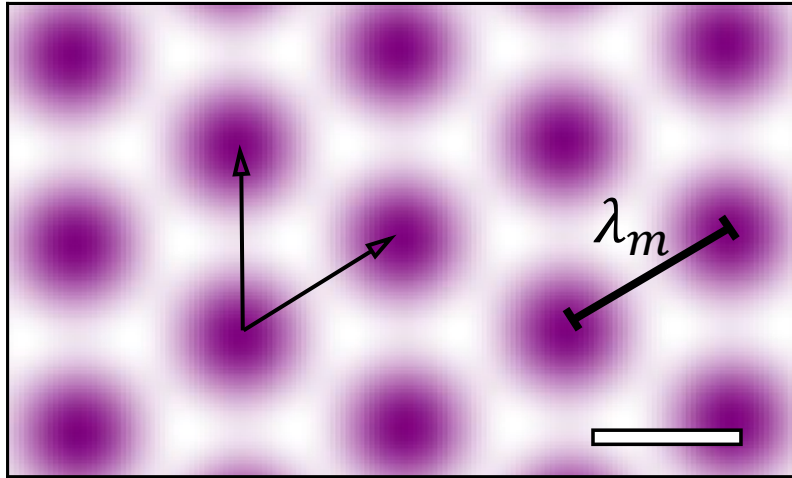
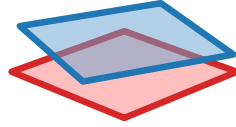
Foo, et al. PRR (2024)

Bistritzer and MacDonald, PNAS (2011)

Atomic lattices that make up moirés generally incommensurate $\rightarrow$ continuum approximation

From moiré to multimoiré

Layers **1** and **2**



Scale bars: 5 nm

Yang, et al. PRB (2024)

Nakatsuji, et al. PRX (2023)

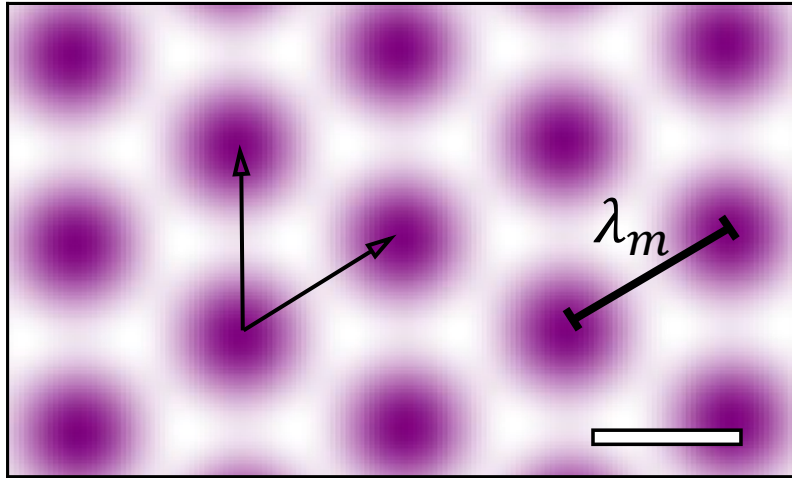
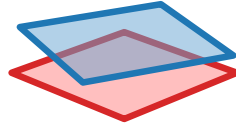
Foo, et al. PRR (2024)

Bistritzer and MacDonald, PNAS (2011)

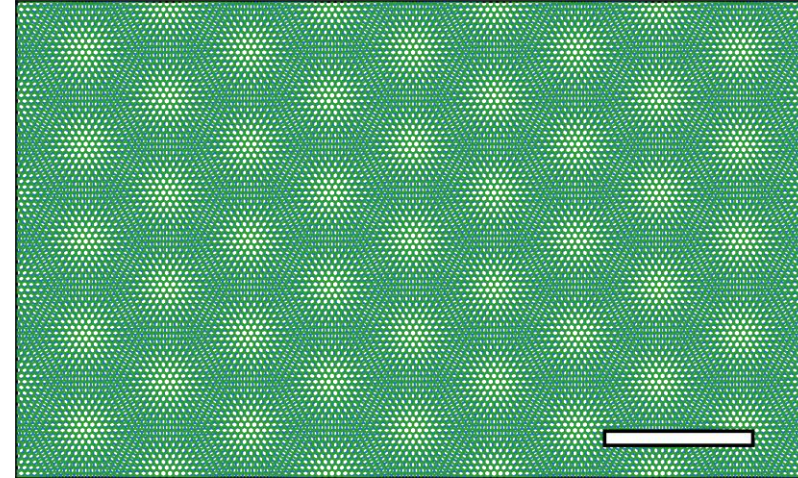
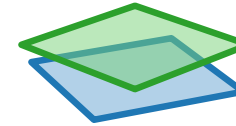
Atomic lattices that make up moirés generally incommensurate $\rightarrow$ continuum approximation

From moiré to multimoiré

Layers **1** and **2**



Layers **2** and **3**



Scale bars: 5 nm

Yang, et al. PRB (2024)

Nakatsuji, et al. PRX (2023)

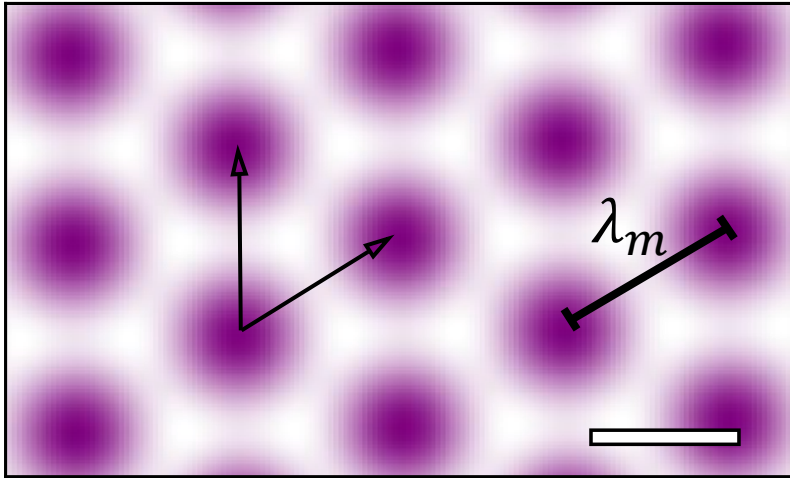
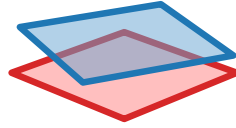
Foo, et al. PRR (2024)

Bistritzer and MacDonald, PNAS (2011)

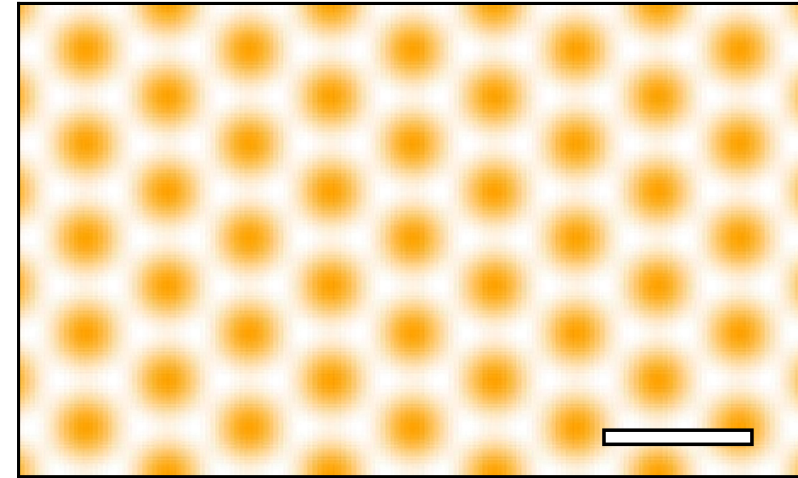
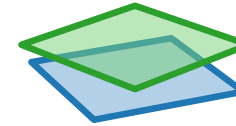
Atomic lattices that make up moirés generally incommensurate → continuum approximation

From moiré to multimoiré

Layers **1** and **2**



Layers **2** and **3**



Scale bars: 5 nm

Yang, et al. PRB (2024)

Nakatsuji, et al. PRX (2023)

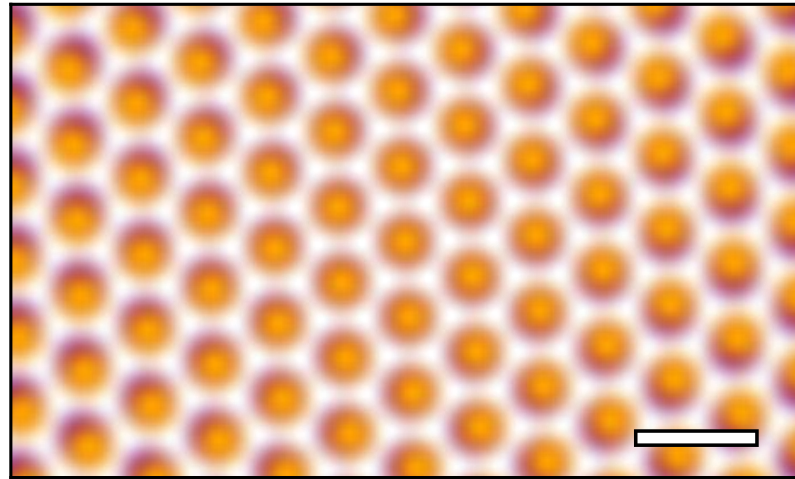
Foo, et al. PRR (2024)

Bistritzer and MacDonald, PNAS (2011)

Atomic lattices that make up moirés generally incommensurate → continuum approximation

Incommensurability and the supermoiré length scale

$$(\theta_{12}, \theta_{23}) = (2^\circ, 2^\circ)$$



● AA₁₂ ● AA₂₃

Scale bars: 10 nm

Yang, et al. PRB (2024)

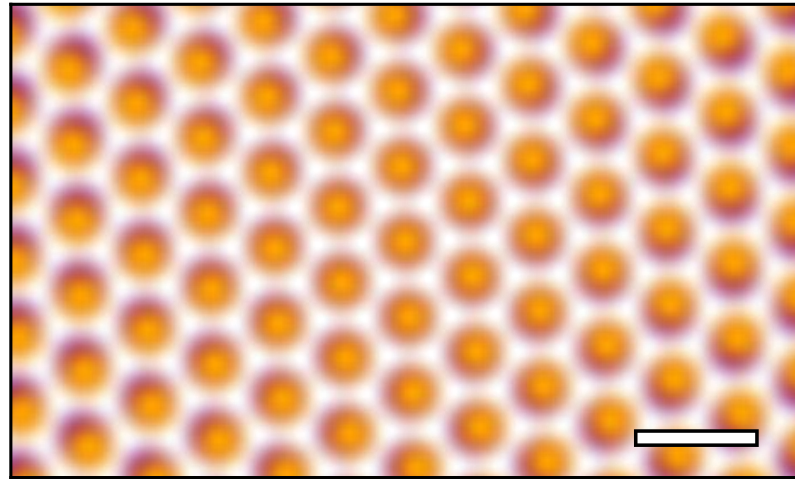
Nakatsuji, et al. PRX (2023)

Foo, et al. PRR (2024)

Bistritzer and MacDonald, PNAS (2011)

Incommensurability and the supermoiré length scale

$$(\theta_{12}, \theta_{23}) = (2^\circ, 2^\circ)$$



● AA₁₂ ● AA₂₃

Scale bars: 10 nm

Translate window



Yang, et al. PRB (2024)

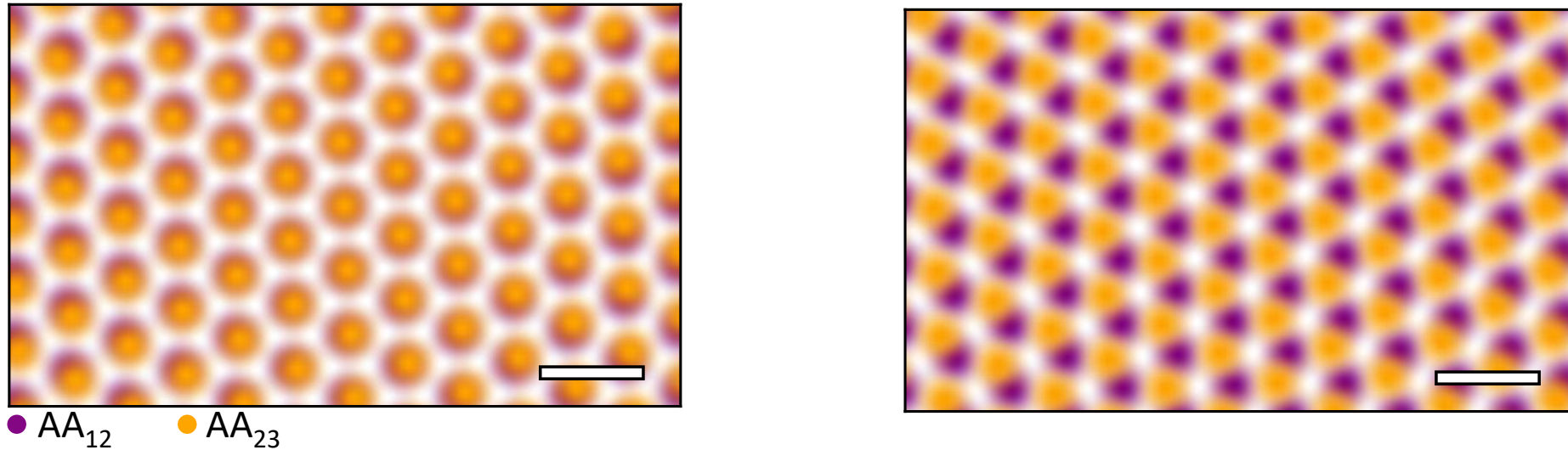
Nakatsuji, et al. PRX (2023)

Foo, et al. PRR (2024)

Bistritzer and MacDonald, PNAS (2011)

Incommensurability and the supermoiré length scale

$$(\theta_{12}, \theta_{23}) = (2^\circ, 2^\circ)$$



Scale bars: 10 nm

Yang, et al. PRB (2024)

Nakatsuji, et al. PRX (2023)

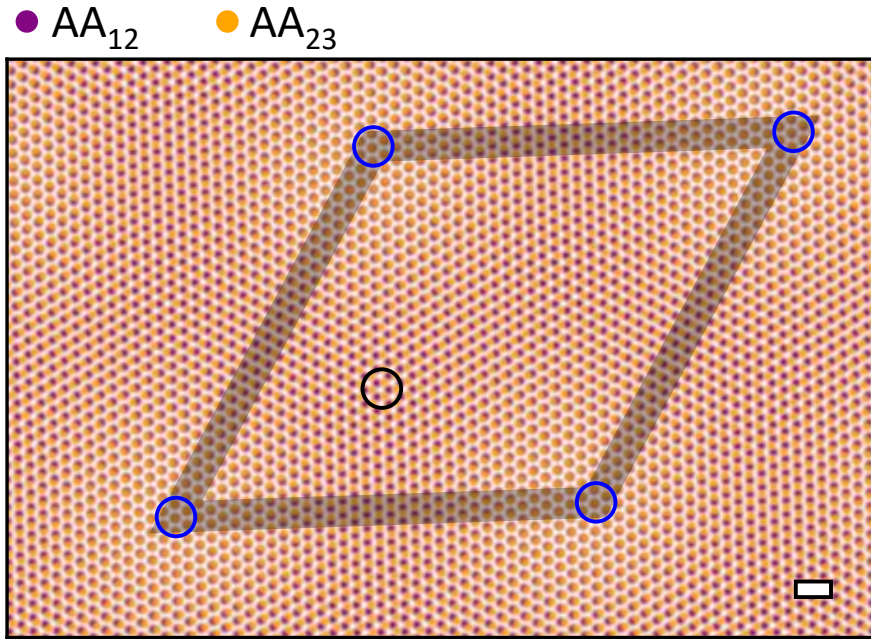
Foo, et al. PRR (2024)

Bistritzer and MacDonald, PNAS (2011)

Translate window



Shift dependent electronic bands



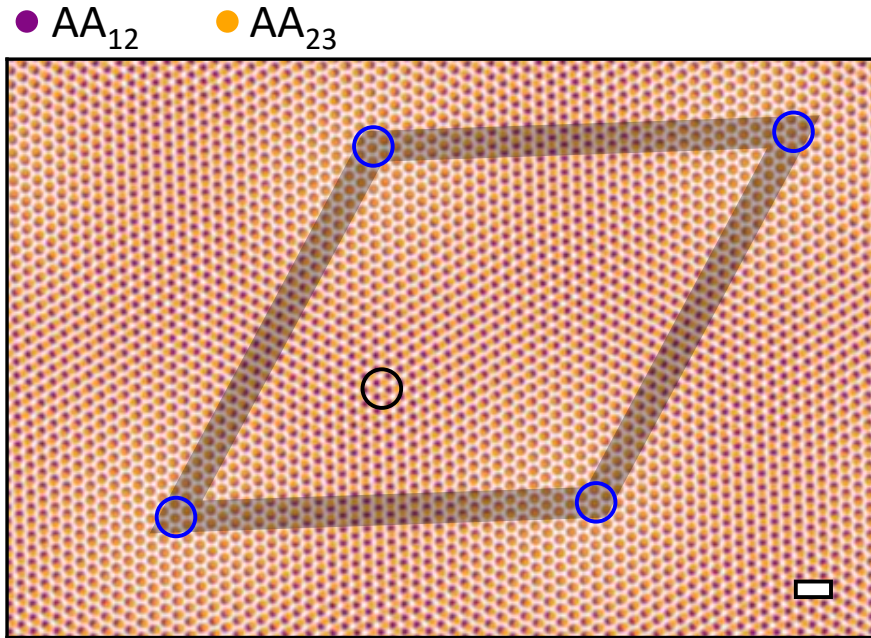
Yang, et al. PRB (2024)

Nakatsuji, et al. PRX (2023)

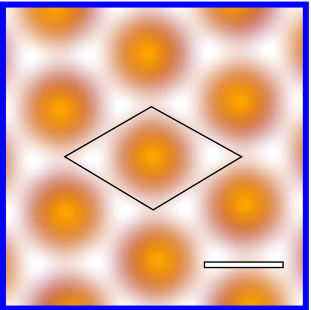
Foo, et al. PRR (2024)

Bistritzer and MacDonald, PNAS (2011)

Shift dependent electronic bands



$d = 0, C_6$



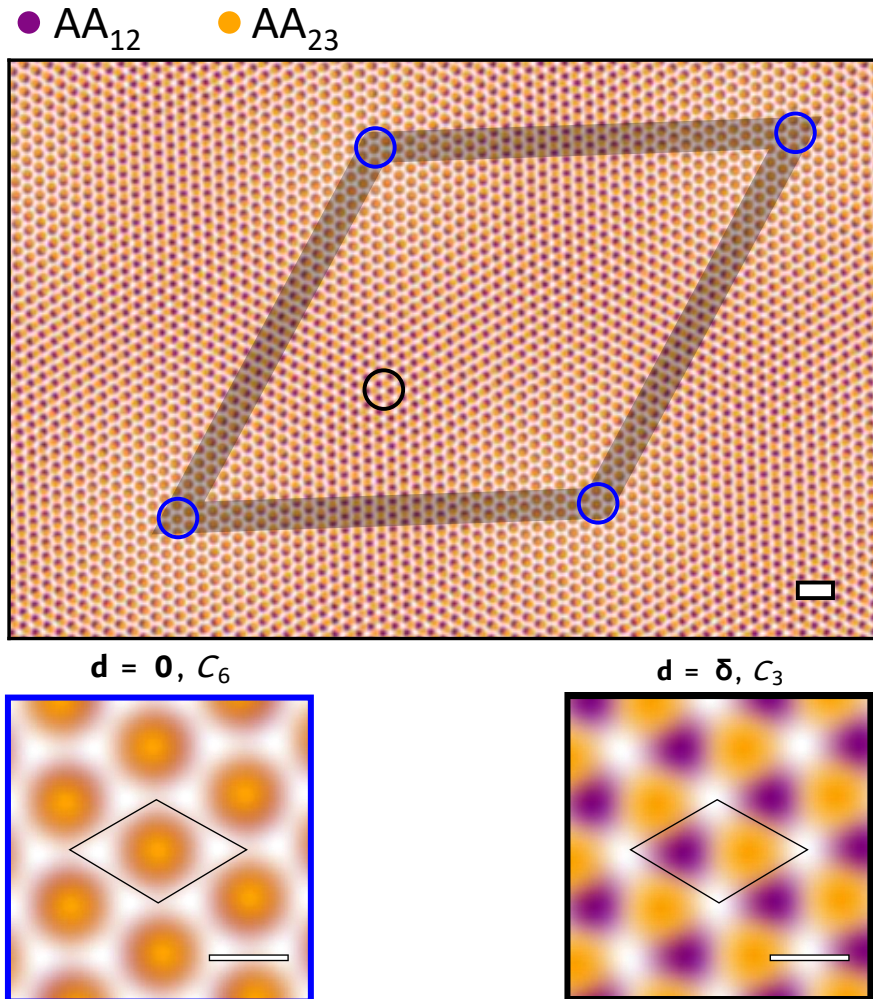
Yang, et al. PRB (2024)

Nakatsuji, et al. PRX (2023)

Foo, et al. PRR (2024)

Bistritzer and MacDonald, PNAS (2011)

Shift dependent electronic bands



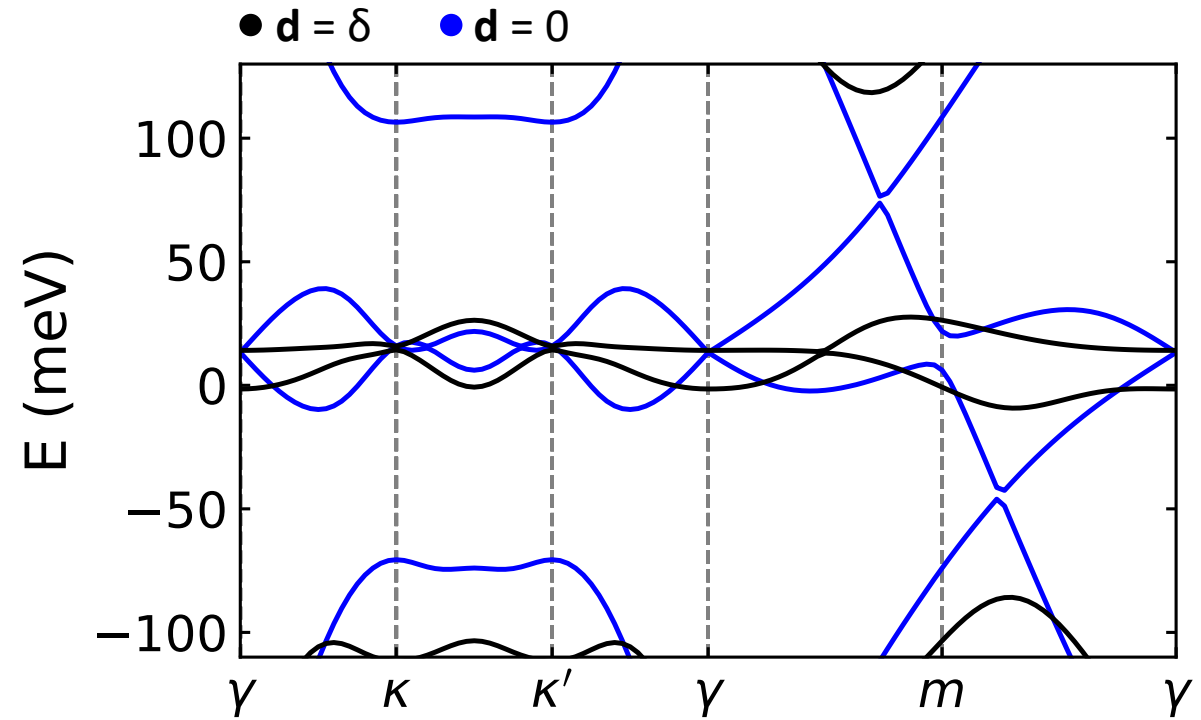
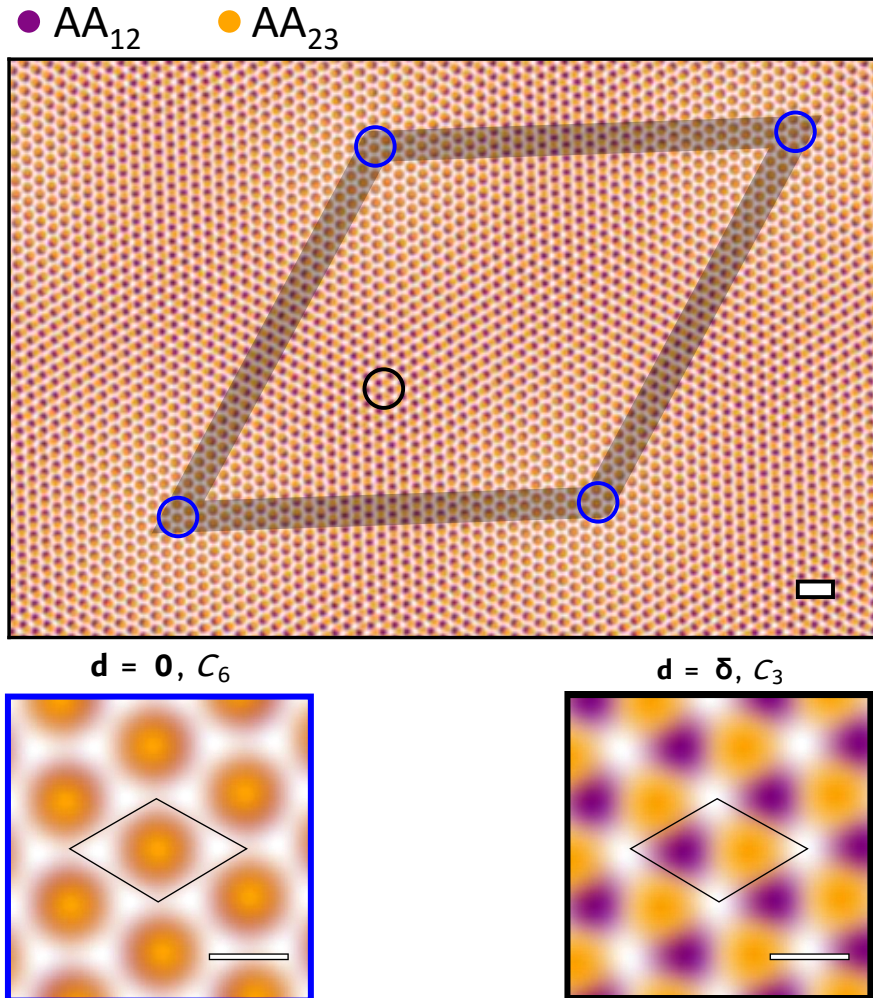
Yang, et al. PRB (2024)

Nakatsuji, et al. PRX (2023)

Foo, et al. PRR (2024)

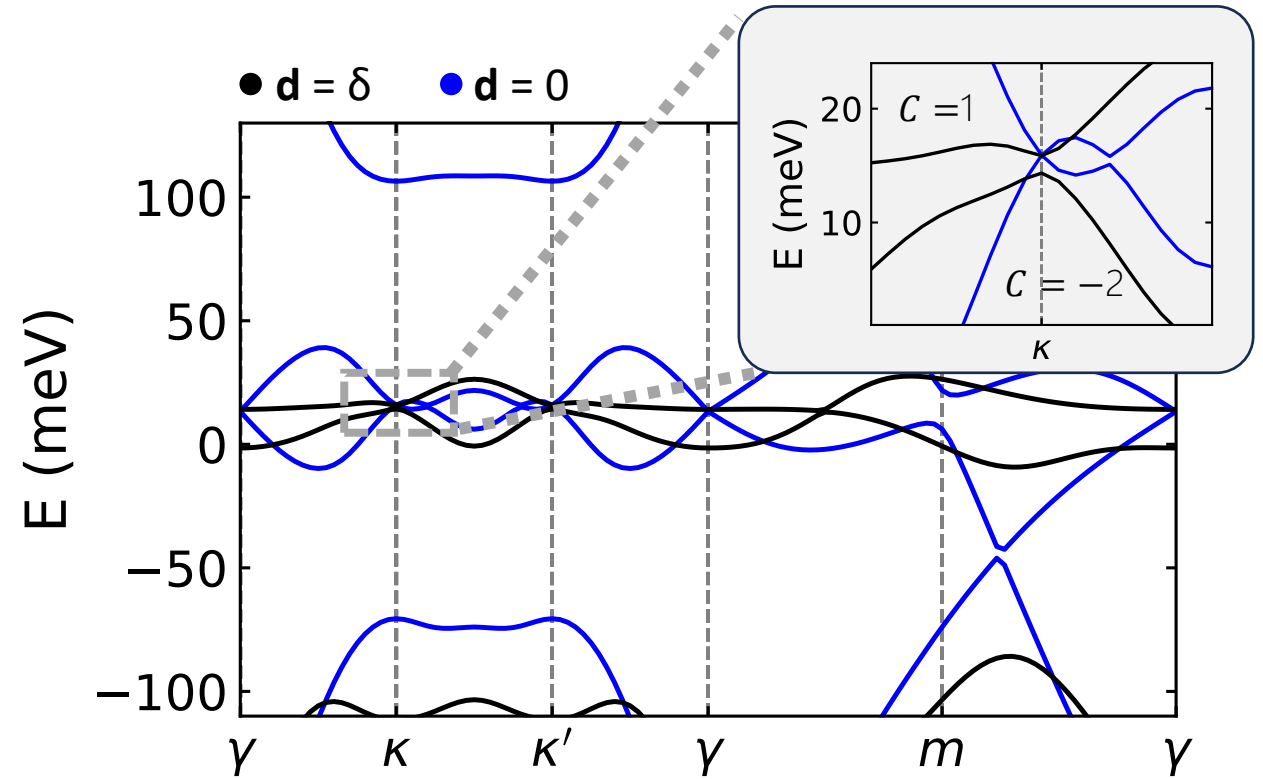
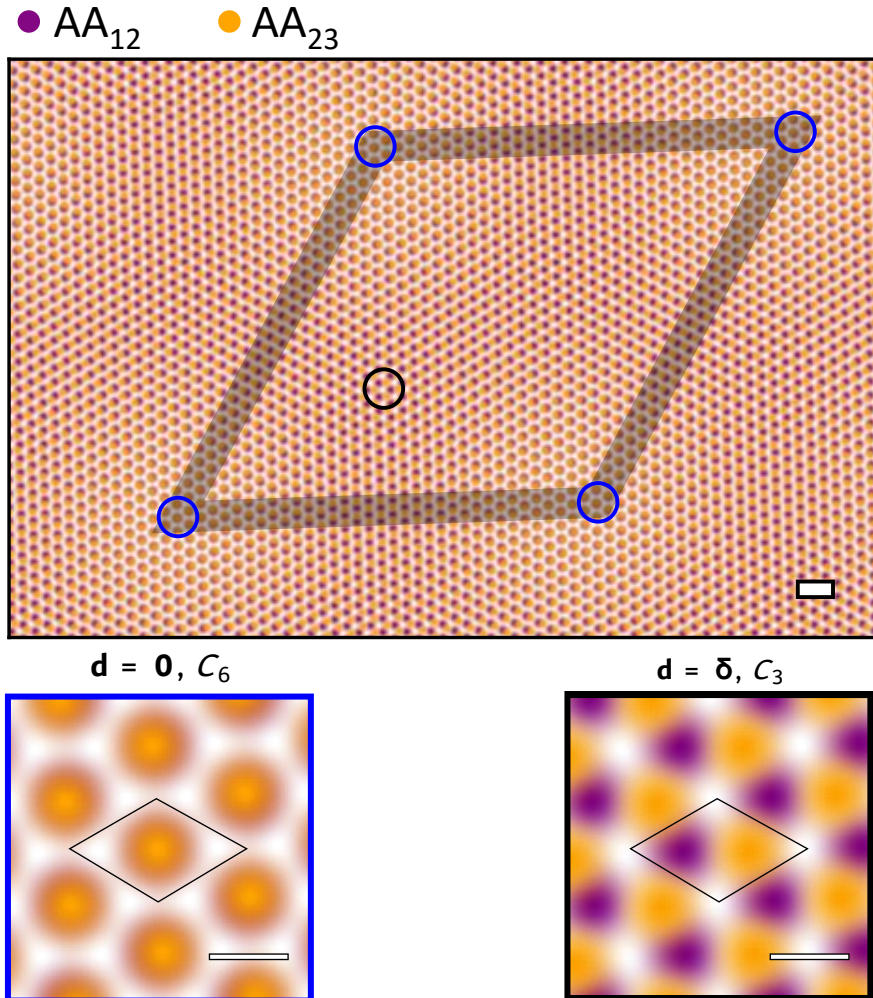
Bistritzer and MacDonald, PNAS (2011)

Shift dependent electronic bands



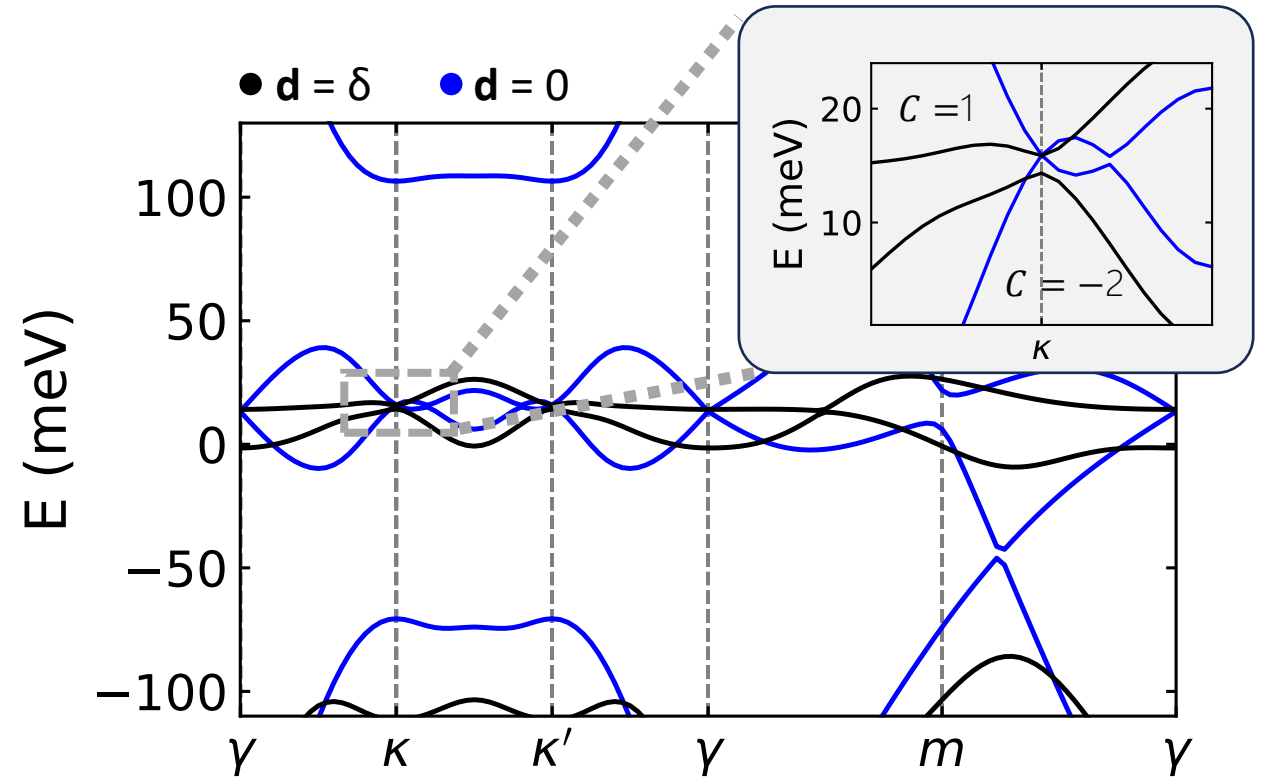
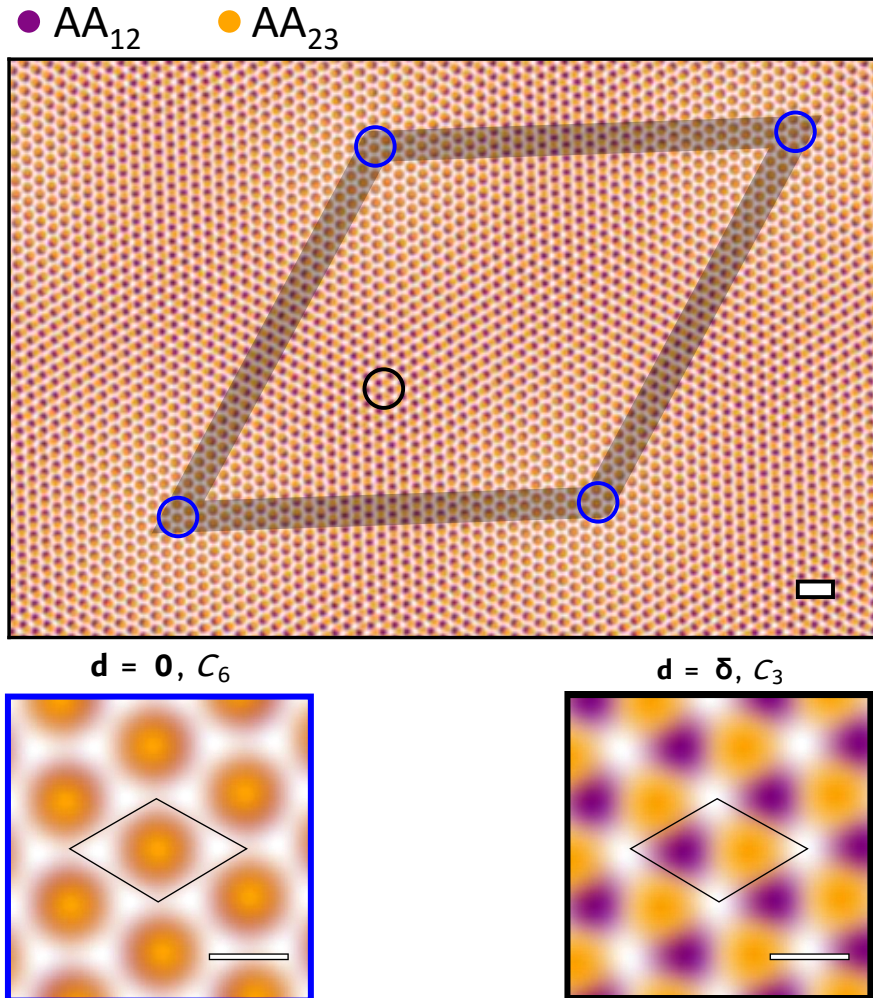
Yang, et al. PRB (2024)
 Nakatsuji, et al. PRX (2023)
 Foo, et al. PRR (2024)
 Bistritzer and MacDonald, PNAS (2011)

Shift dependent electronic bands



Yang, et al. PRB (2024)
 Nakatsuji, et al. PRX (2023)
 Foo, et al. PRR (2024)
 Bistritzer and MacDonald, PNAS (2011)

Shift dependent electronic bands

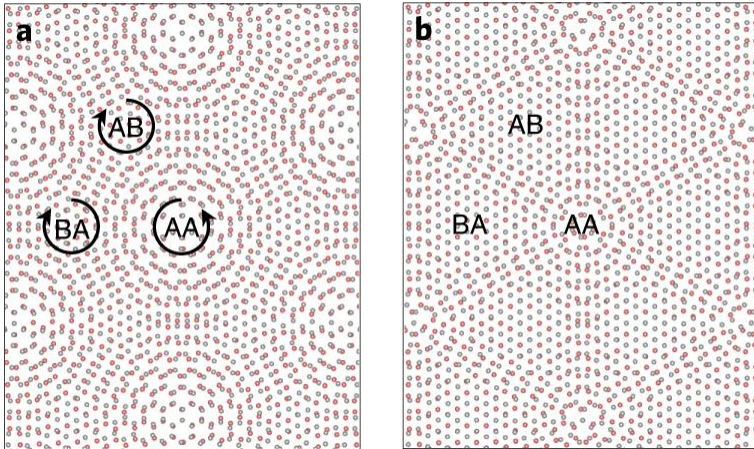


How do we weight intermoiré shifts?

Yang, et al. PRB (2024)
 Nakatsuji, et al. PRX (2023)
 Foo, et al. PRR (2024)
 Bistritzer and MacDonald, PNAS (2011)

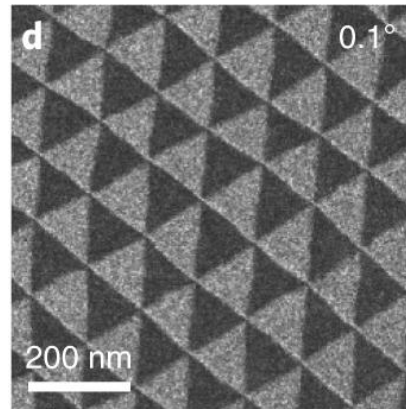
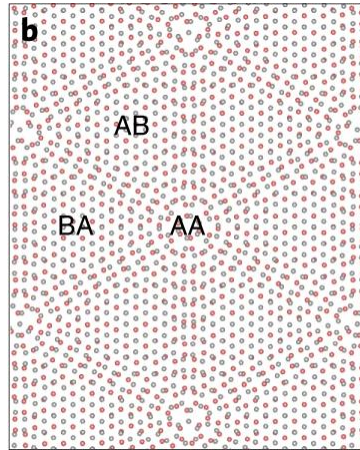
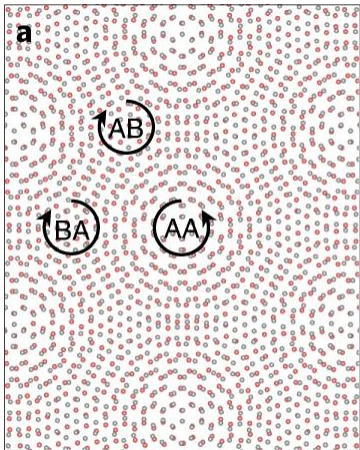
Lattice relaxation on the supermoiré scale

Relaxation at the moiré scale



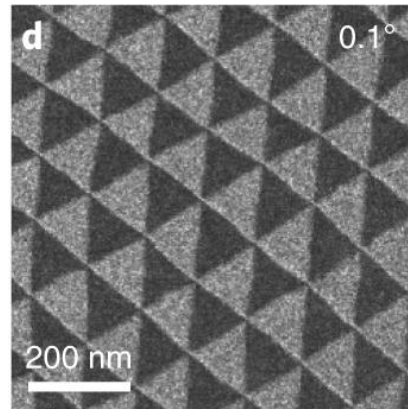
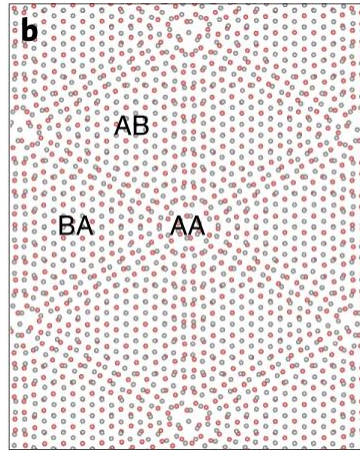
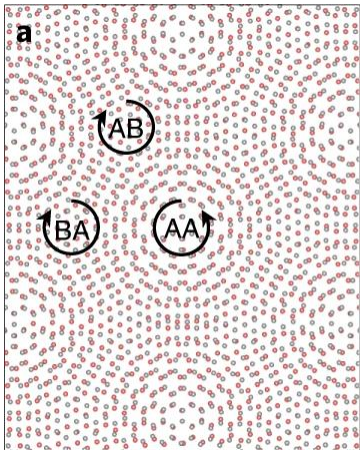
Lattice relaxation on the supermoiré scale

Relaxation at the moiré scale



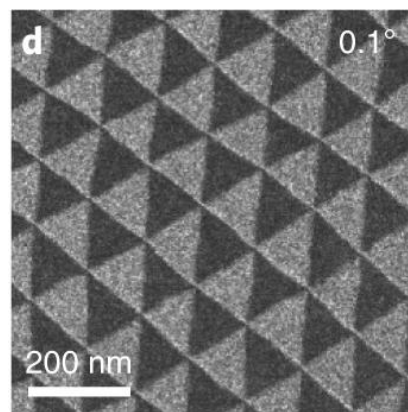
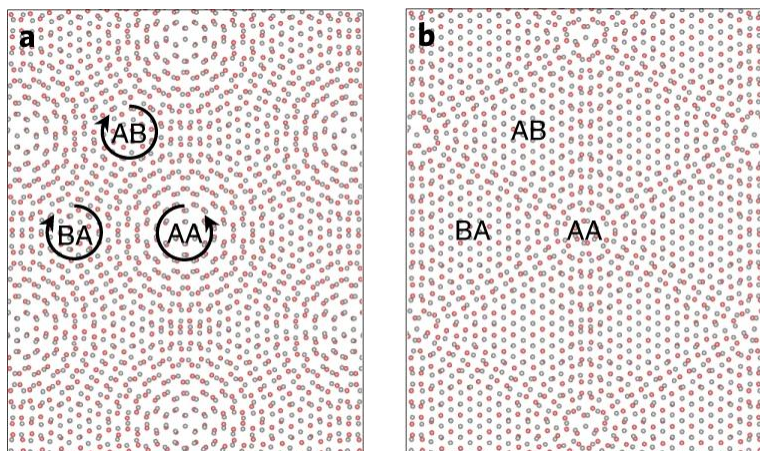
Lattice relaxation on the supermoiré scale

Relaxation at the moiré scale



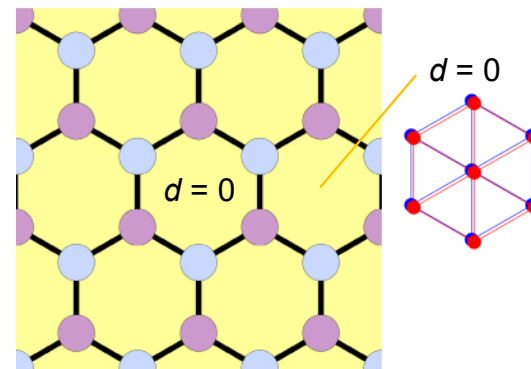
Lattice relaxation on the supermoiré scale

Relaxation at the moiré scale

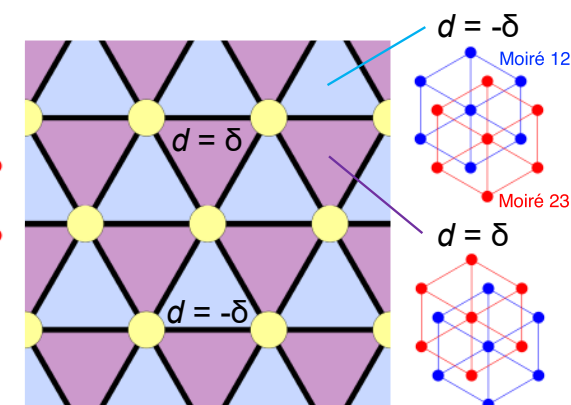


Relaxation at the supermoiré scale

Alternating

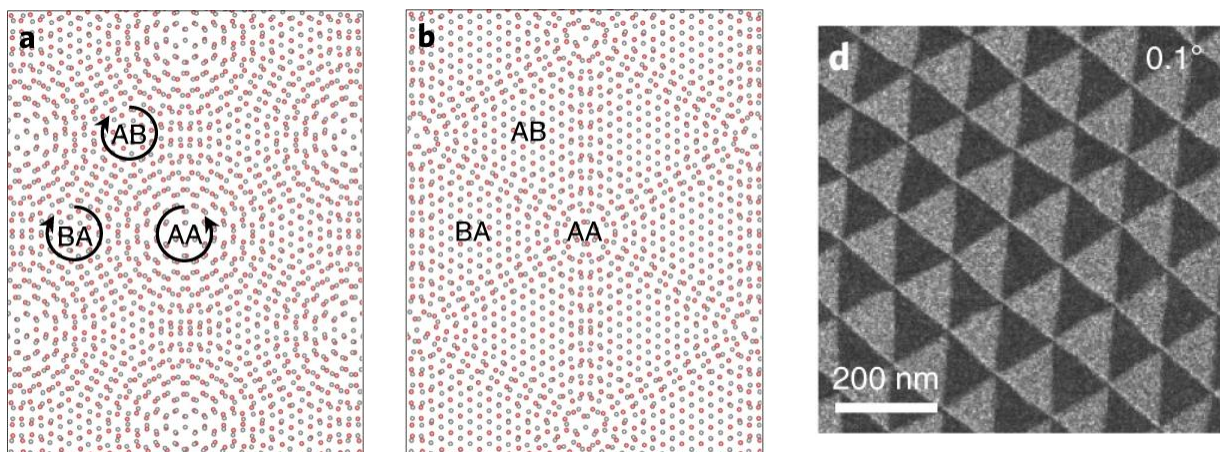


Helical



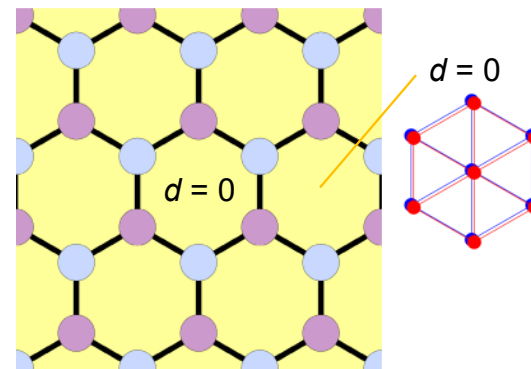
Lattice relaxation on the supermoiré scale

Relaxation at the moiré scale

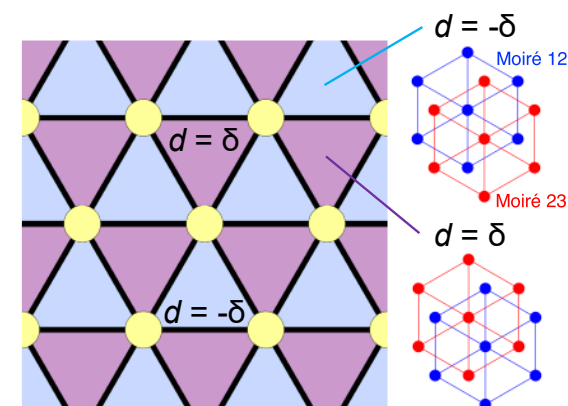


Relaxation at the supermoiré scale

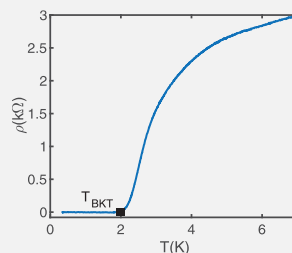
Alternating



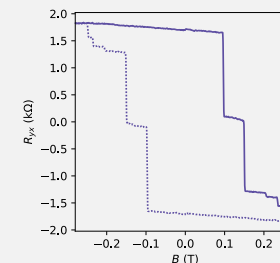
Helical



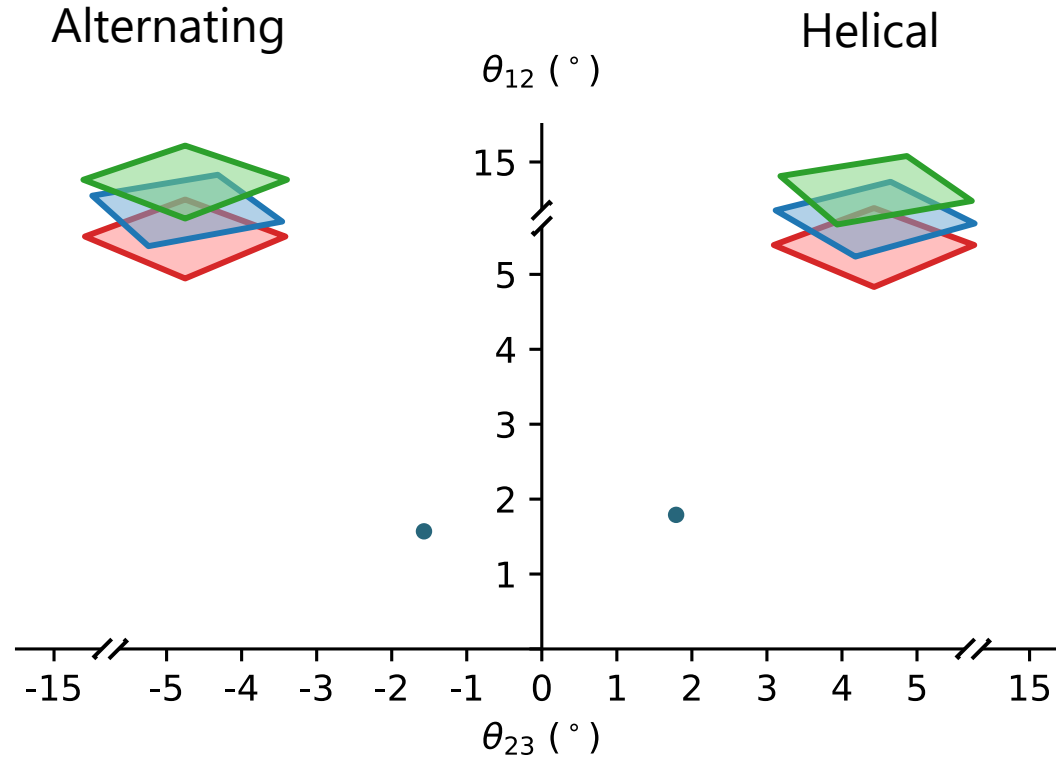
Yoo et al. Nat. Nano. (2019)
Nakatsuji, et al. PRX (2023)



MATTG: $d=0$ domains
MAHTG: xy -inversion
broken domains!

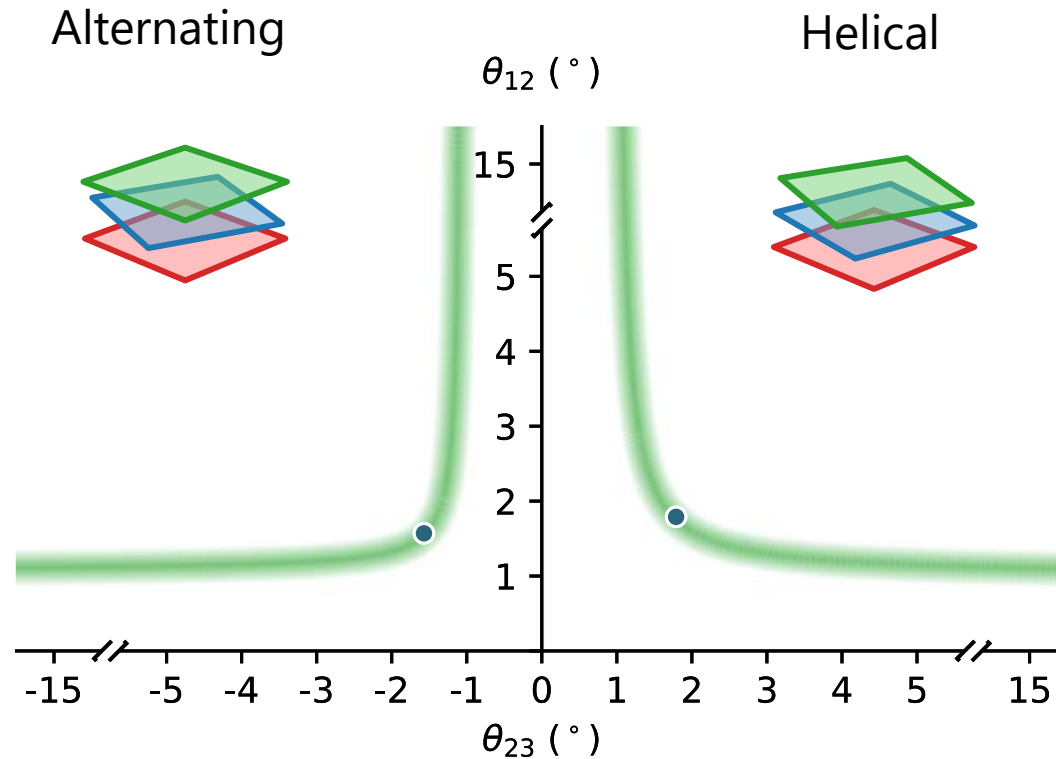


Expectations for the angle-angle phase diagram



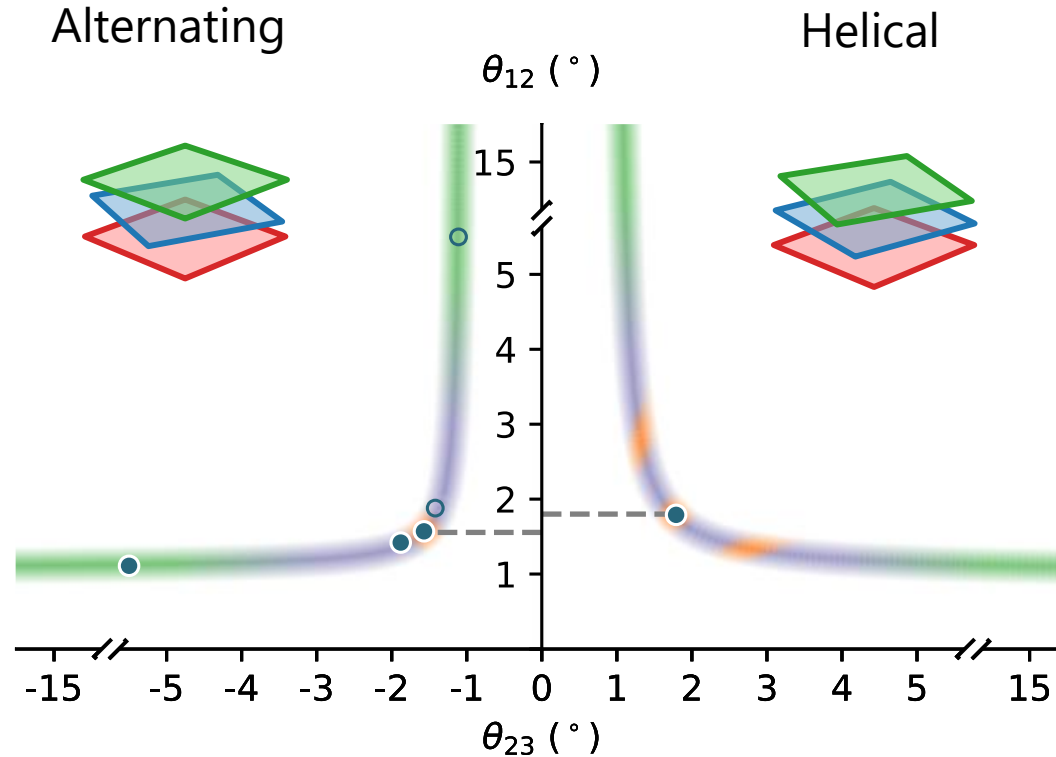
1. Start with a TTG structure that is nearly commensurate $\Rightarrow (\theta_{12}, \theta_{23}) = (p\theta, q\theta)$
2. Calculate bandstructure for all intermoiré shifts
3. Expectation for correlations: θ that maximizes single particle density of states
4. Strongly reconstruct?
Yes for $(\theta, \pm\theta)$ and $(\theta, 2\theta)$

Expectations for the angle-angle phase diagram



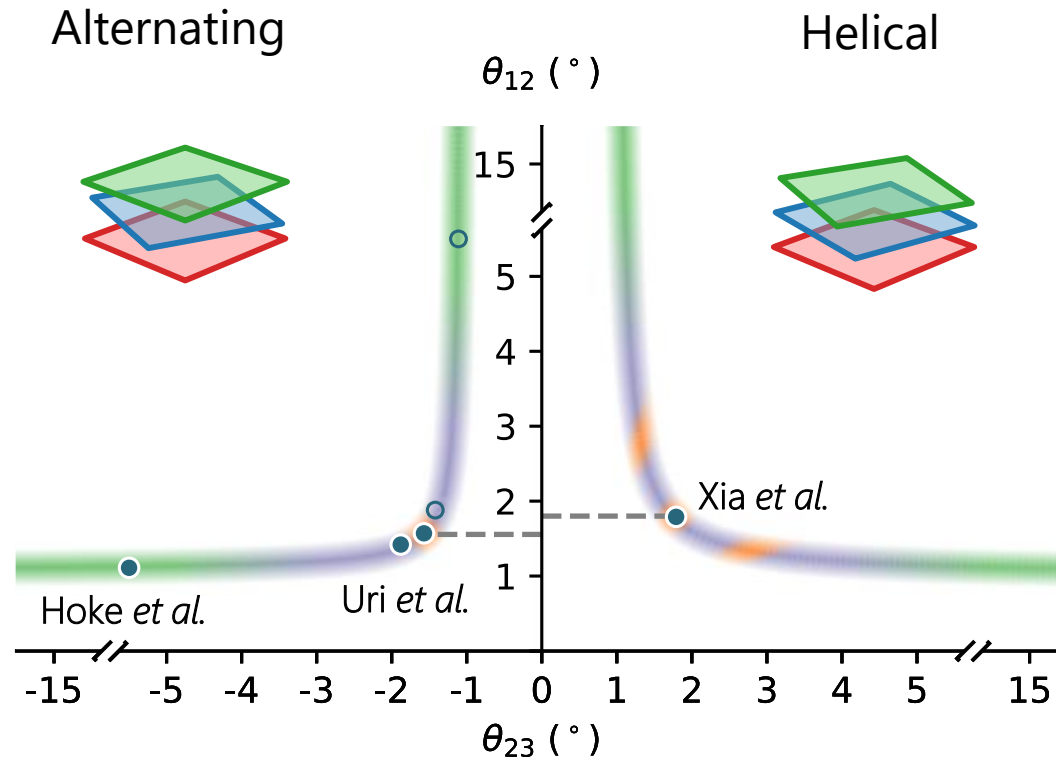
1. Start with a TTG structure that is nearly commensurate $\Rightarrow (\theta_{12}, \theta_{23}) = (p\theta, q\theta)$
2. Calculate bandstructure for all intermoiré shifts
3. Expectation for correlations: θ that maximizes single particle density of states
4. Strongly reconstruct?
Yes for $(\theta, \pm\theta)$ and $(\theta, 2\theta)$

Expectations for the angle-angle phase diagram



1. Start with a TTG structure that is nearly commensurate $\Rightarrow (\theta_{12}, \theta_{23}) = (p\theta, q\theta)$
2. Calculate bandstructure for all intermoiré shifts
3. Expectation for correlations: θ that maximizes single particle density of states
4. Strongly reconstruct?
Yes for $(\theta, \pm\theta)$ and $(\theta, 2\theta)$

Expectations for the angle-angle phase diagram



1. Start with a TTG structure that is nearly commensurate $\Rightarrow (\theta_{12}, \theta_{23}) = (p\theta, q\theta)$
2. Calculate bandstructure for all intermoiré shifts
3. Expectation for correlations: θ that maximizes single particle density of states
4. Strongly reconstruct?
Yes for $(\theta, \pm\theta)$ and $(\theta, 2\theta)$

Uri, et al. Nature (2023)
Hoke, et al. Nat. Comms. (2024)
Xia, et al. Nat. Phys. (2025)
Yang, et al. arXiv:2310.12961

Weakly Reconstructed
Strongly Reconstructed
MATBG + decoupled monolayer

Outline

Introduction:

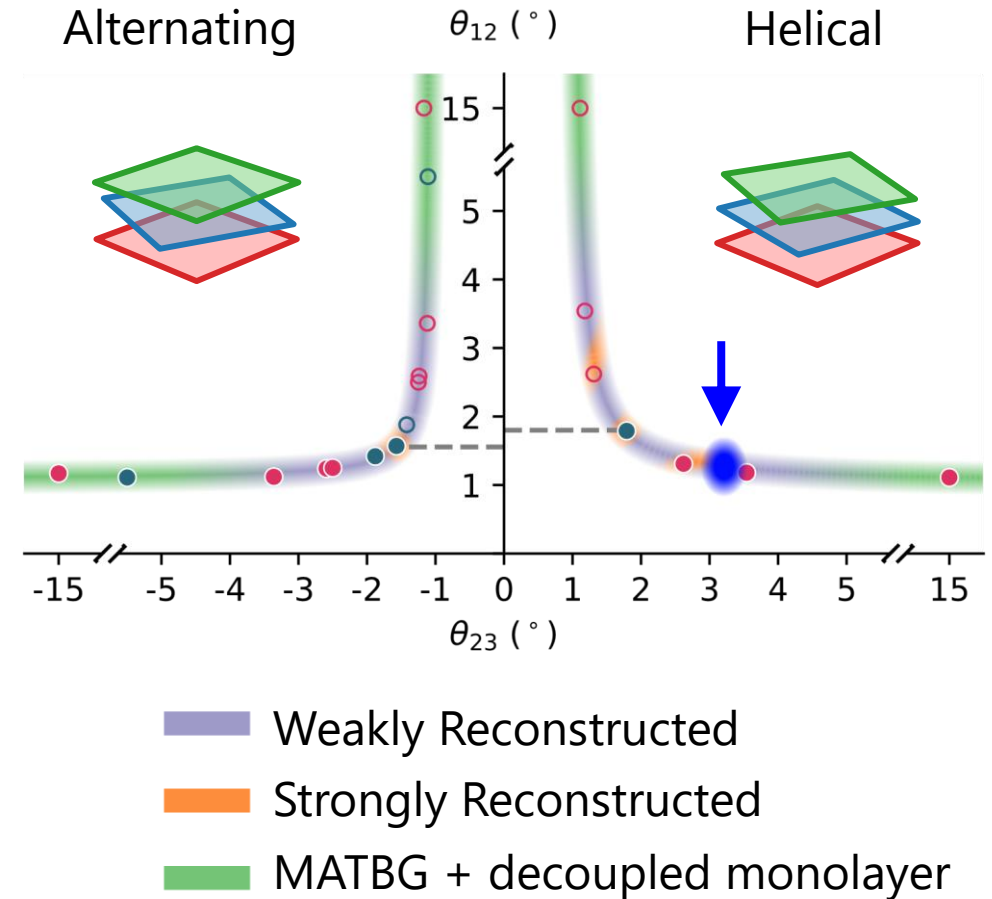
- From moiré to supermoiré
- Helical vs. alternating stacking
- Lattice relaxation

Linking thermodynamic probes and transport:

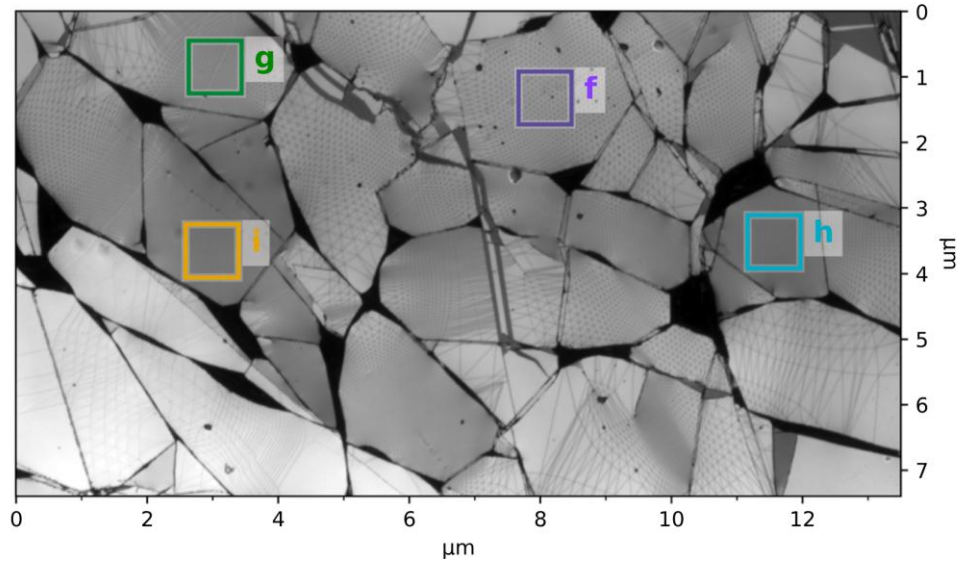
- Fine-grained evolution of correlations
- Cross comparison of distinct experimental probes

Exploring the “magic continuum”:

- Broad comparison across many devices in angle-angle space



These samples vary wildly



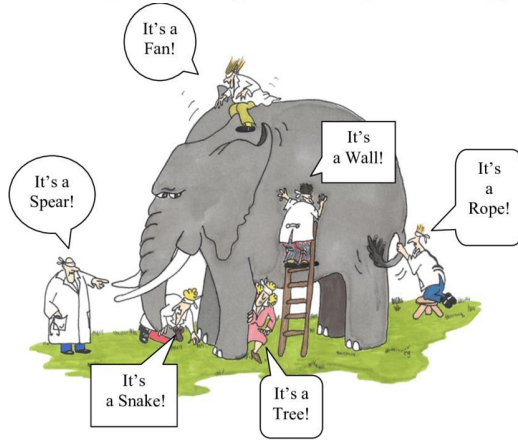
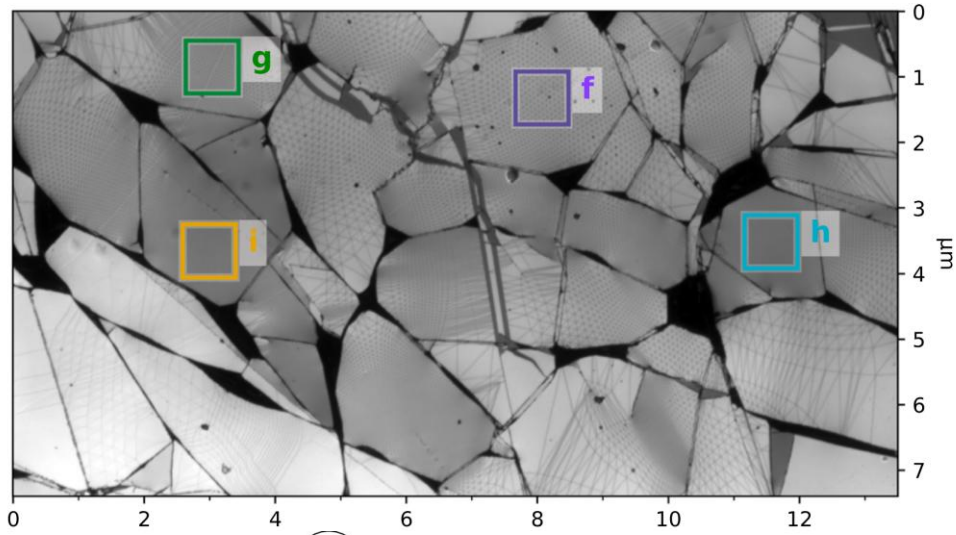
de Jong, et al. Nat. Comms. (2022)

Pendarkar, et al. PNAS (2024)

Xia*, Uri*, Yan*, Sharpe* et al. arXiv:2509.03583

Hoke,..., Sharpe, Feldman arXiv:2509.07977

These samples vary wildly



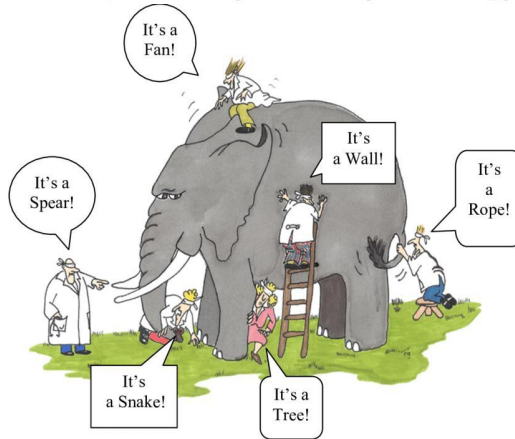
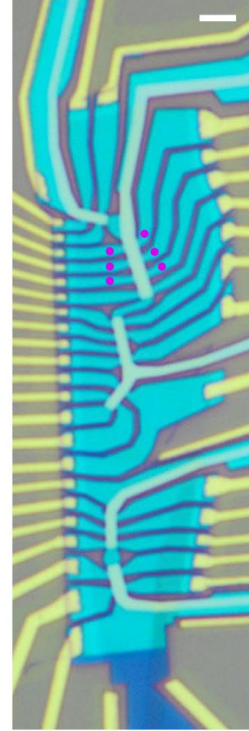
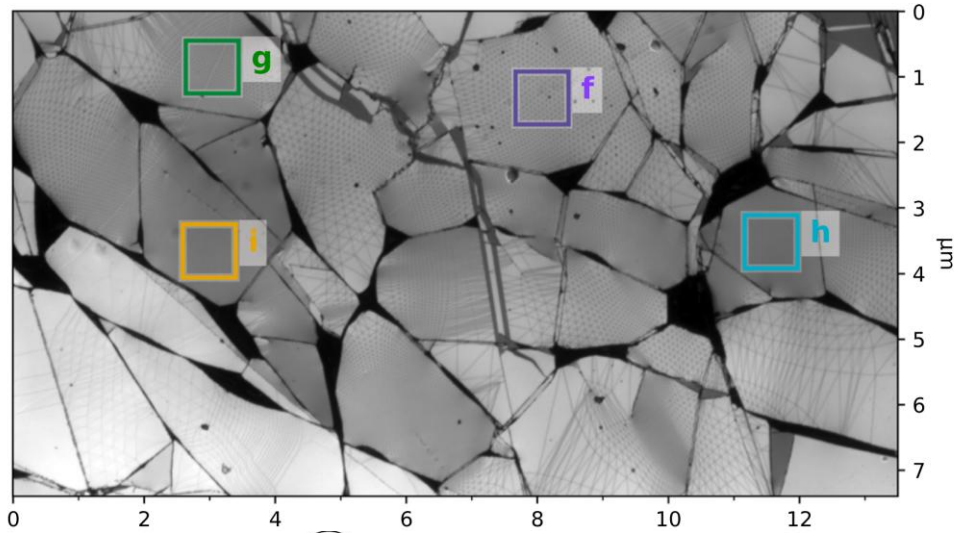
de Jong, et al. Nat. Comms. (2022)

Pendarkar, et al. PNAS (2024)

Xia*, Uri*, Yan*, Sharpe* et al. arXiv:2509.03583

Hoke,..., Sharpe, Feldman arXiv:2509.07977

These samples vary wildly

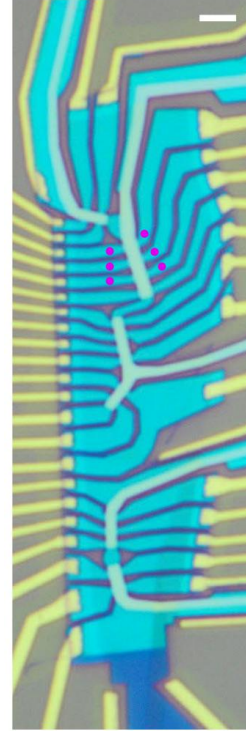
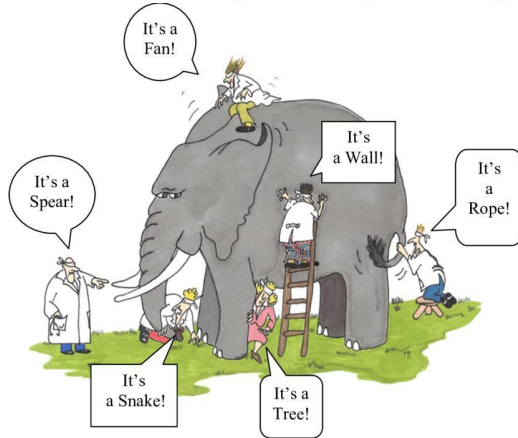
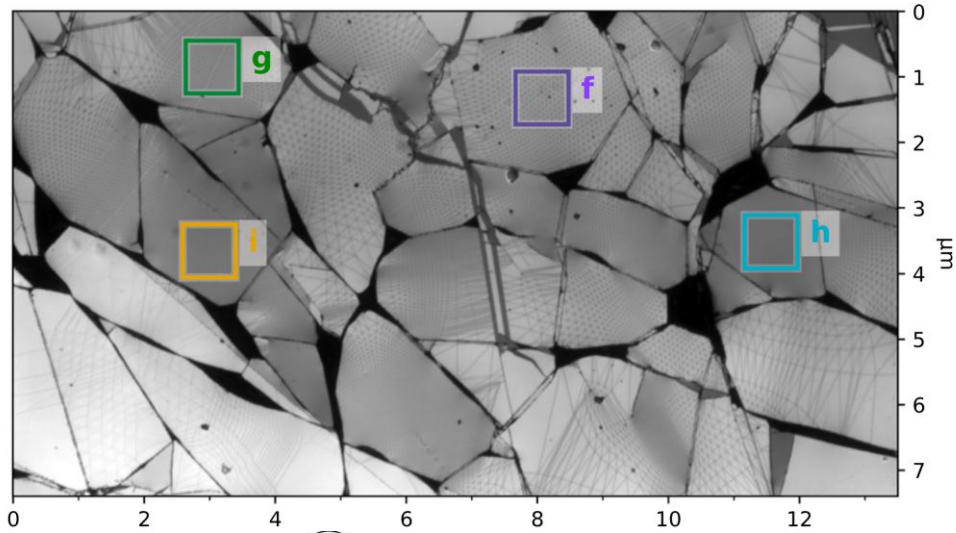


Usual approach: probe as much of the heterostructure as possible

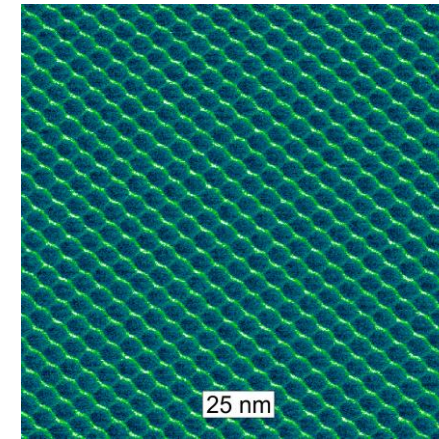
Can we take a better approach?

de Jong, et al. Nat. Comms. (2022)
Pendarkar, et al. PNAS (2024)
Xia*, Uri*, Yan*, Sharpe* et al. arXiv:2509.03583
Hoke,..., Sharpe, Feldman arXiv:2509.07977

These samples vary wildly



TFM

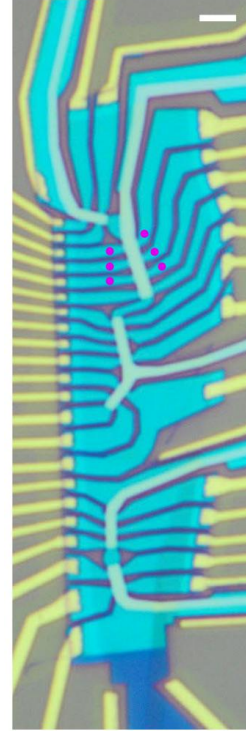
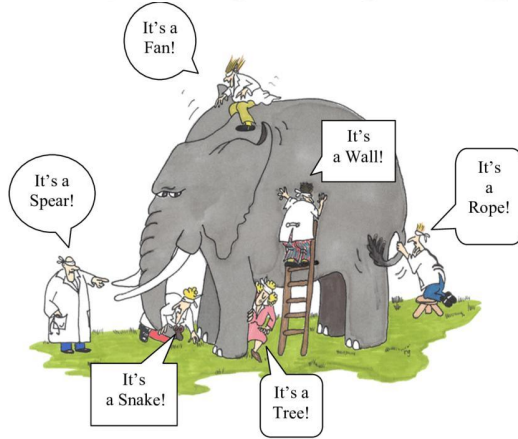
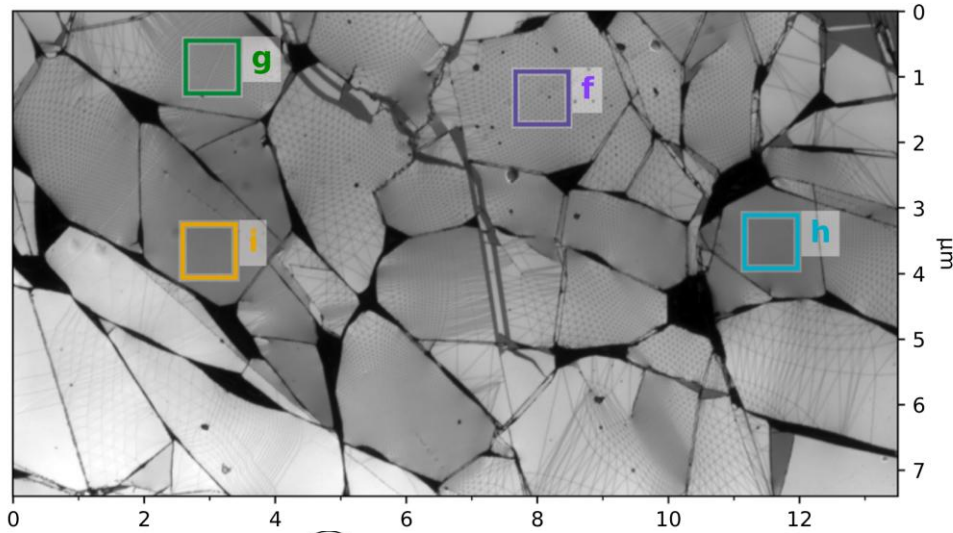


Usual approach: probe as much of the heterostructure as possible

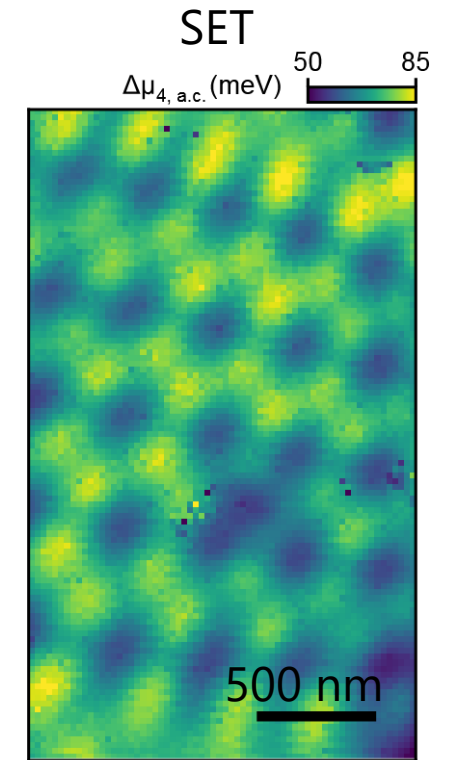
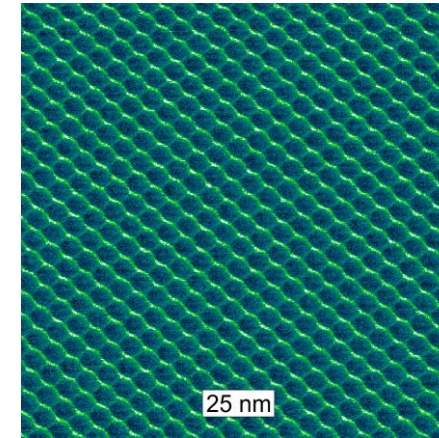
Can we take a better approach?

de Jong, et al. Nat. Comms. (2022)
Pendarkar, et al. PNAS (2024)
Xia*, Uri*, Yan*, Sharpe* et al. arXiv:2509.03583
Hoke,..., Sharpe, Feldman arXiv:2509.07977

These samples vary wildly



TFM

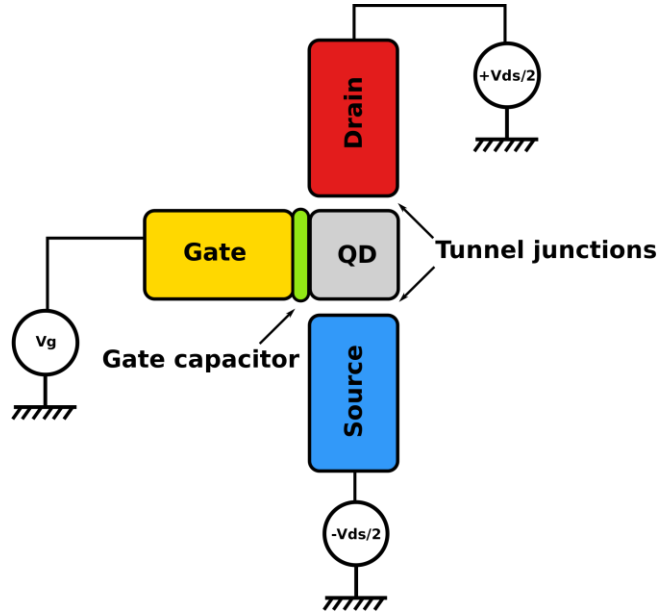


Usual approach: probe as much of the heterostructure as possible

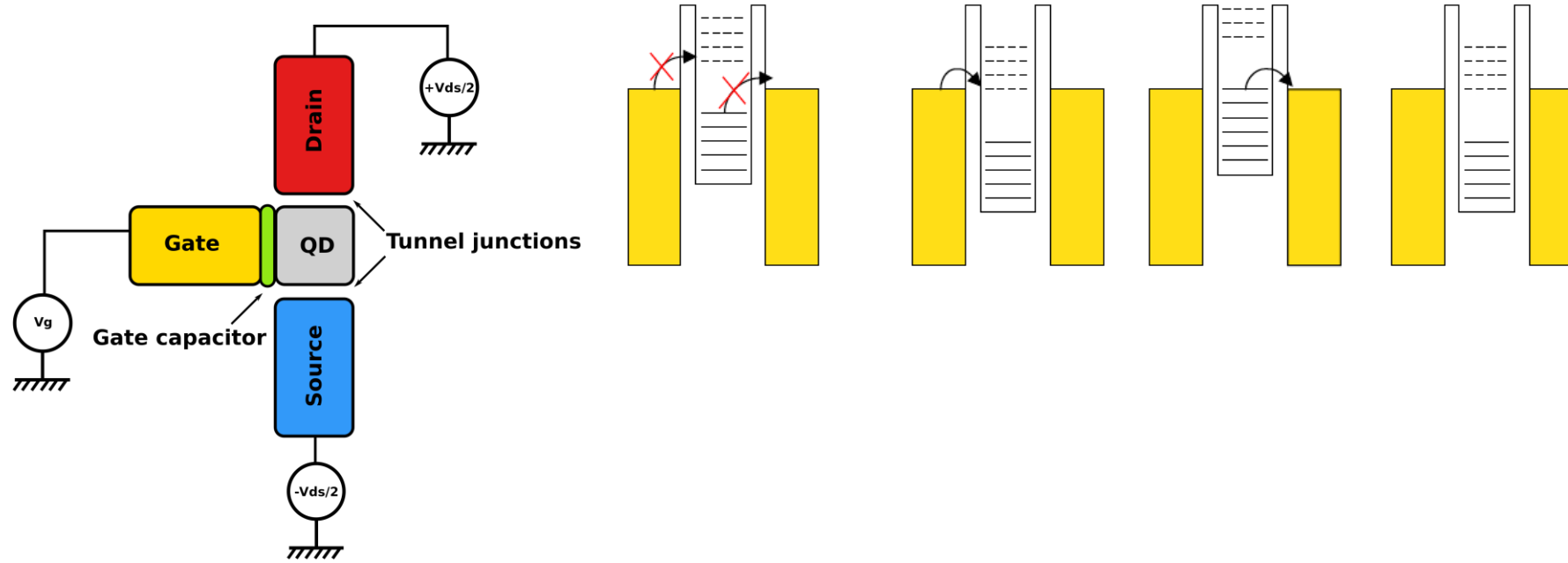
Can we take a better approach?

de Jong, et al. Nat. Comms. (2022)
Pendarkar, et al. PNAS (2024)
Xia*, Uri*, Yan*, Sharpe* et al. arXiv:2509.03583
Hoke,..., Sharpe, Feldman arXiv:2509.07977

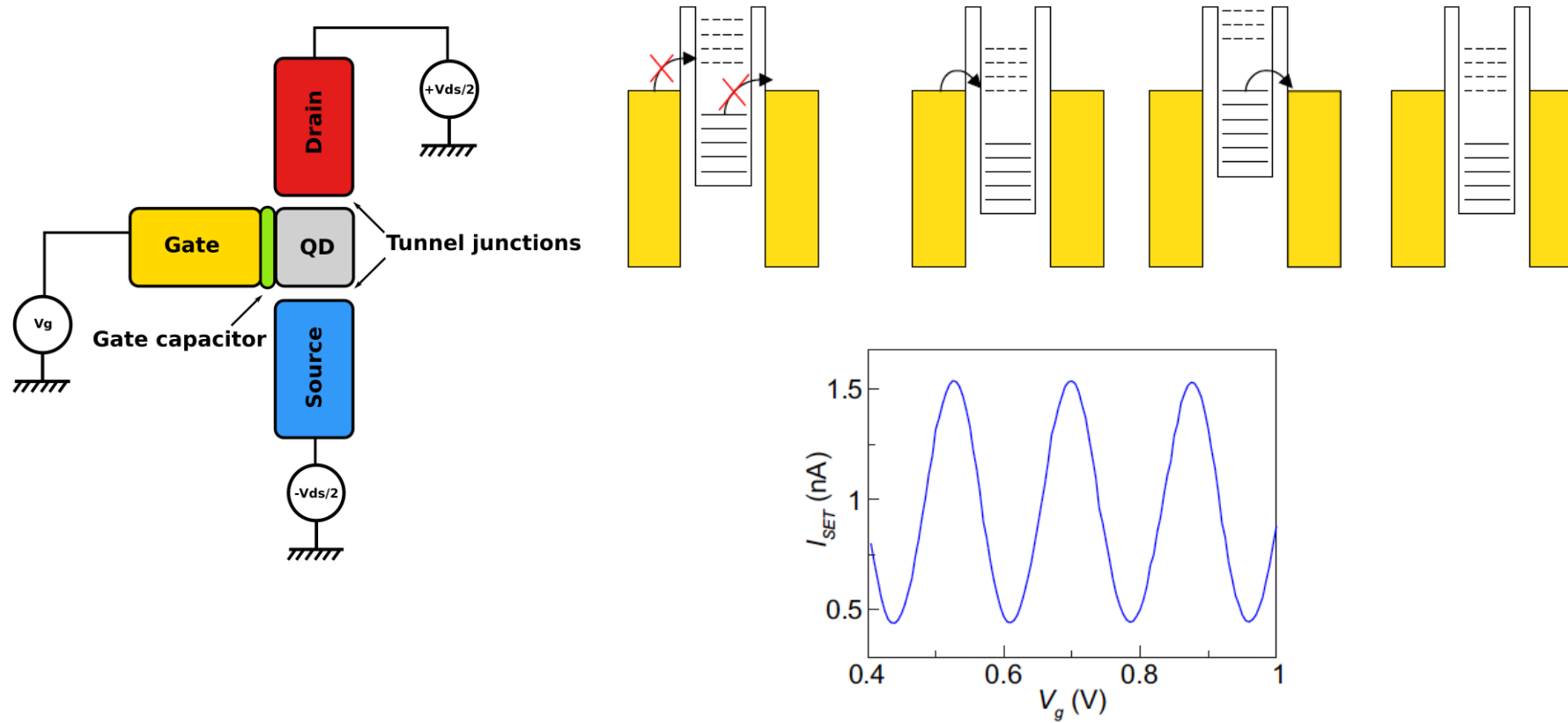
Scanning single-electron transistor



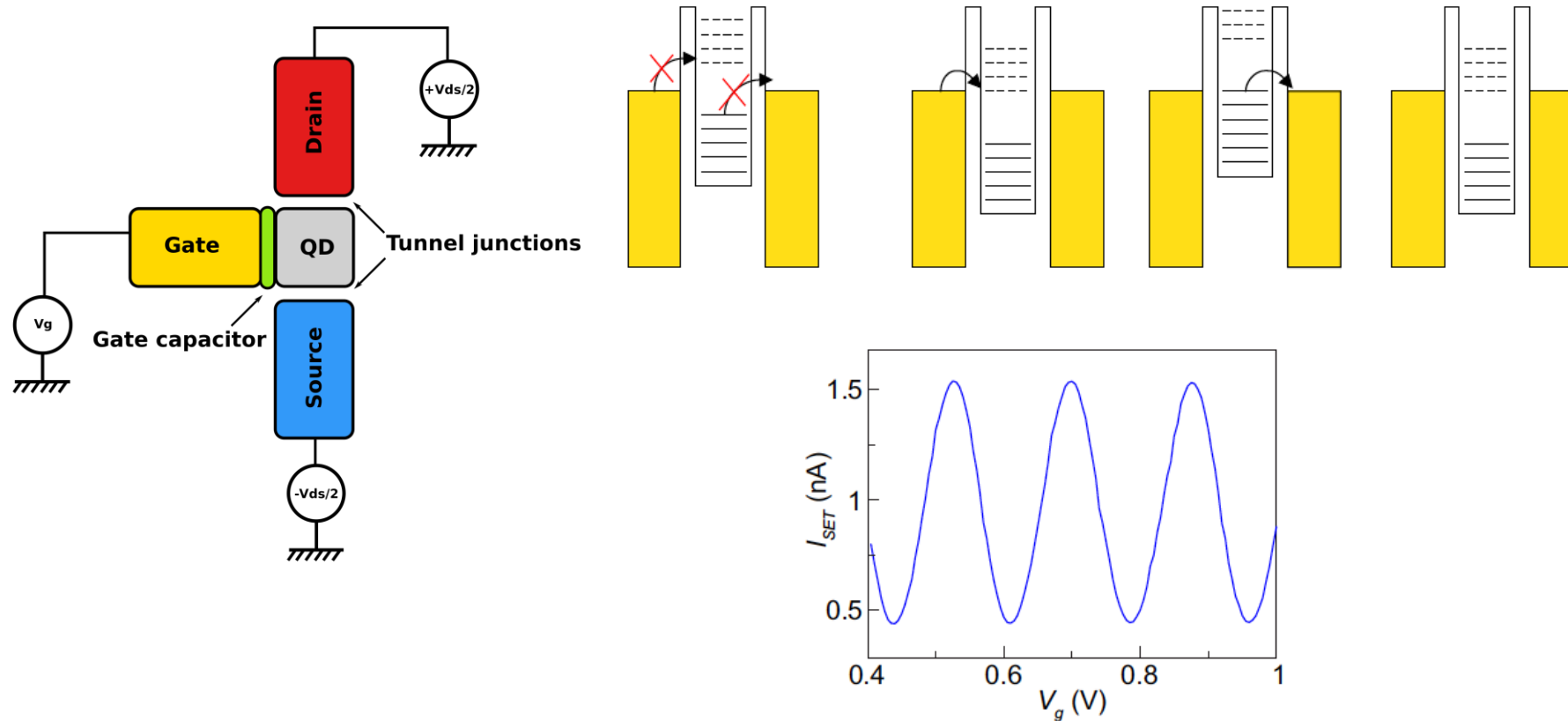
Scanning single-electron transistor



Scanning single-electron transistor

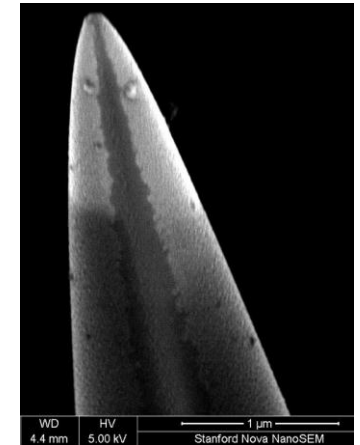
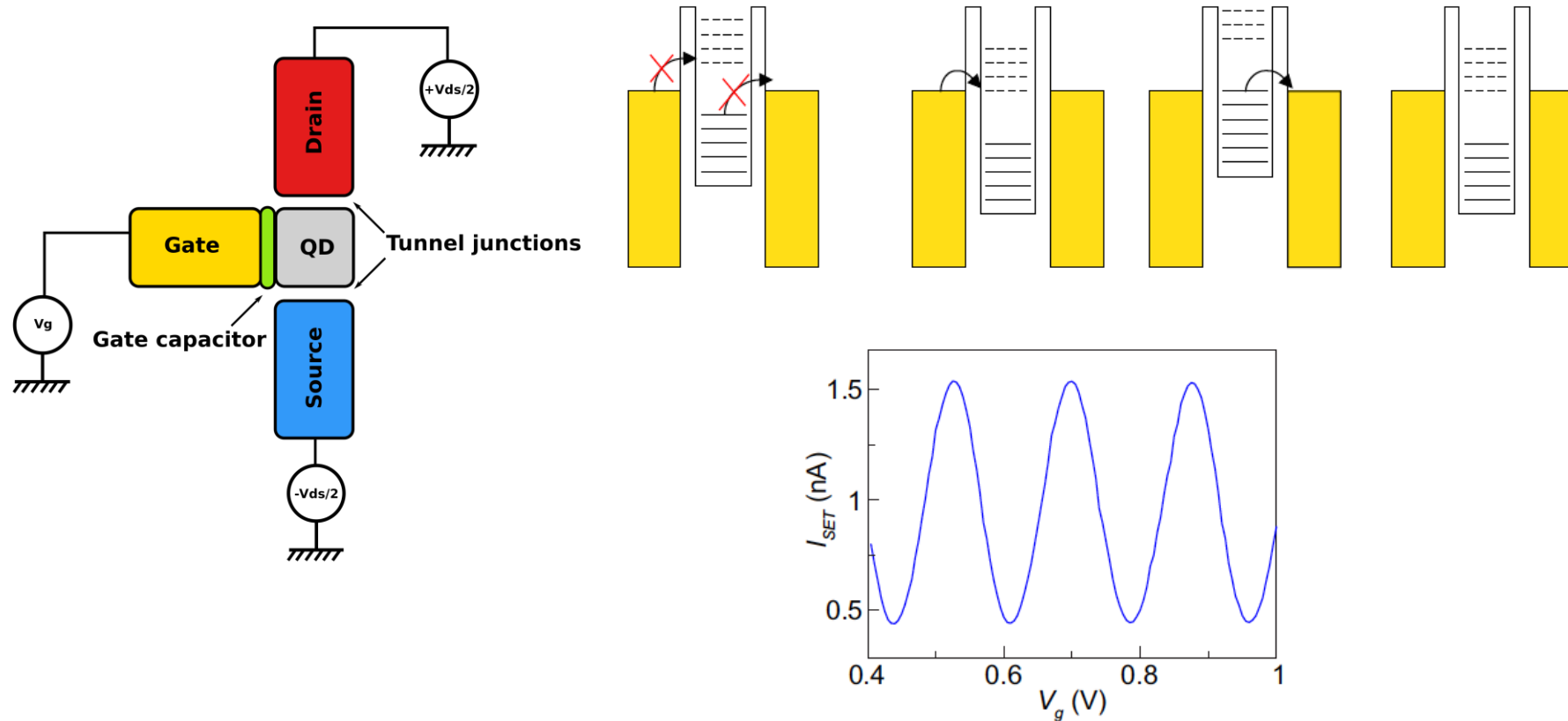


Scanning single-electron transistor



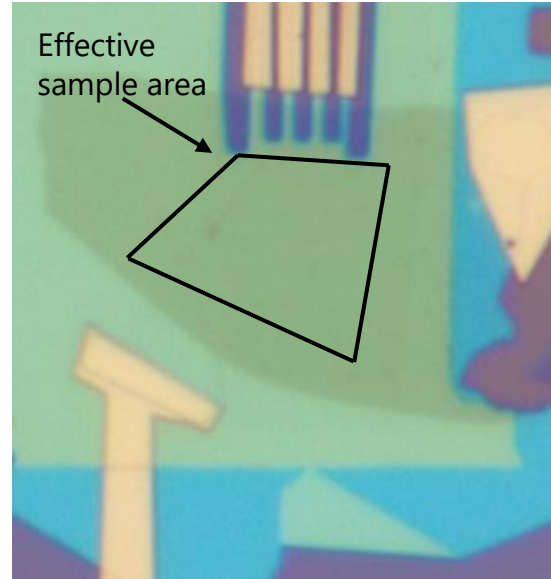
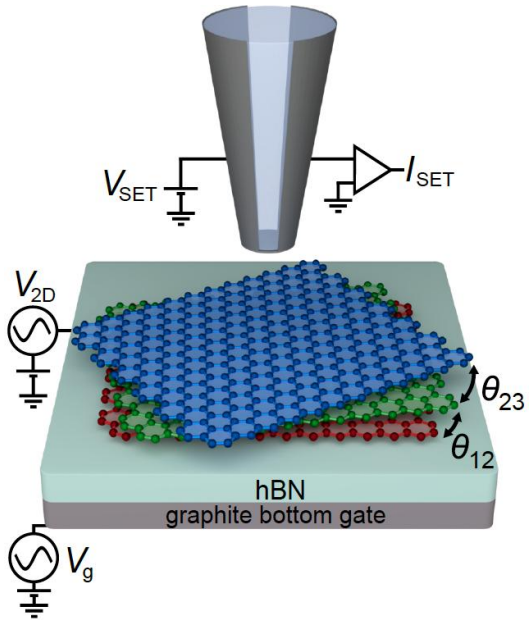
Coulomb blockade: makes SET exquisitely sensitive to electrostatic potential

Scanning single-electron transistor



Coulomb blockade: makes SET exquisitely sensitive to electrostatic potential

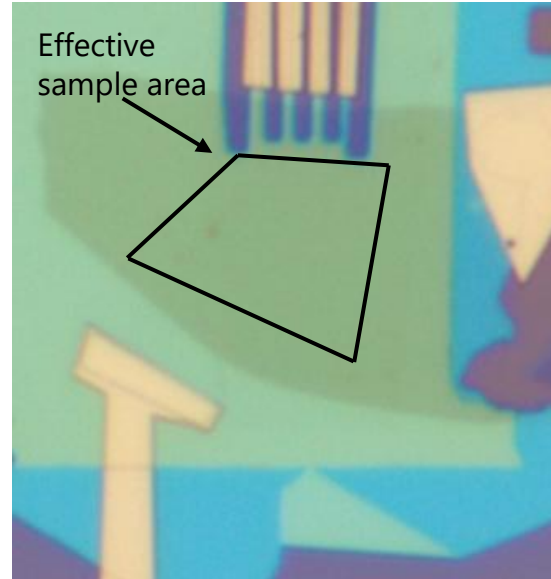
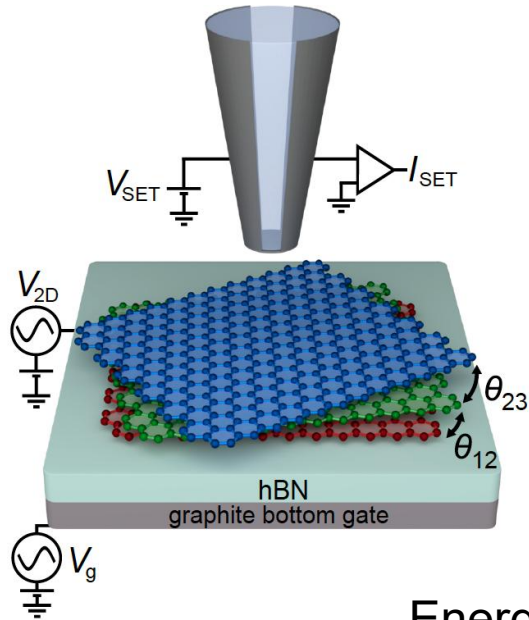
Scanning single-electron transistor



Sample held at fixed
electrochemical potential
$$eV = \mu + e\phi$$

Spatial resolution: ~ 100 nm
Sensitivity: $\sim 10 \mu\text{V}/\text{VHz}$
 $T_{\text{base}} = 330$ mK

Scanning single-electron transistor

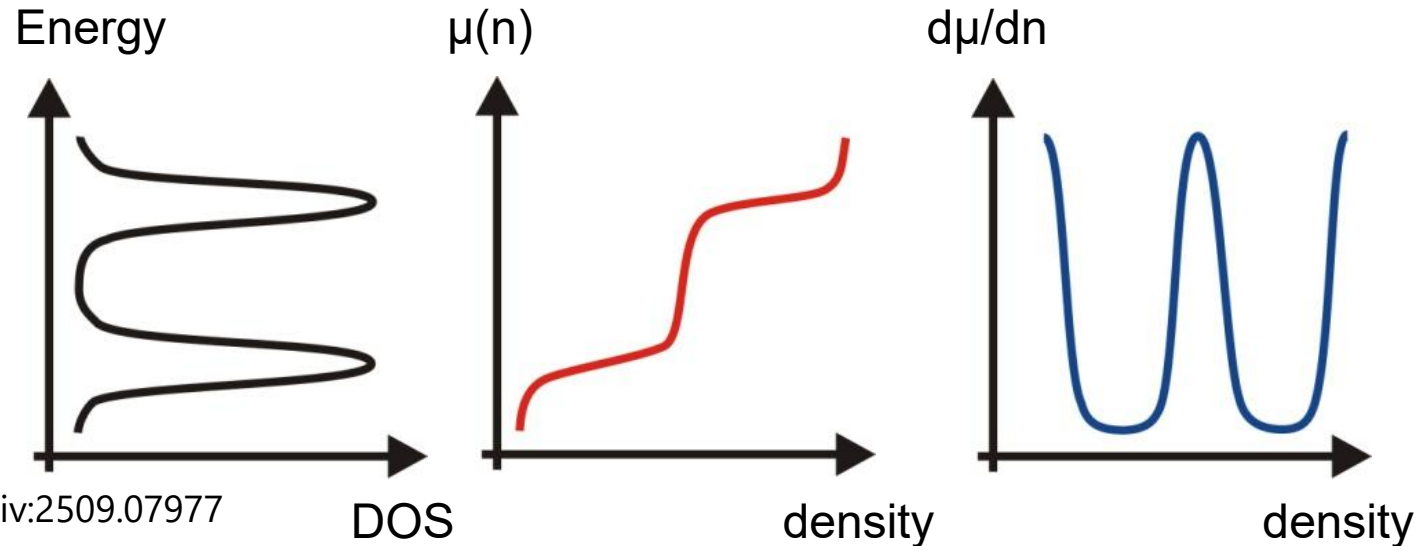


Sample held at fixed
electrochemical potential
 $eV = \mu + e\phi$

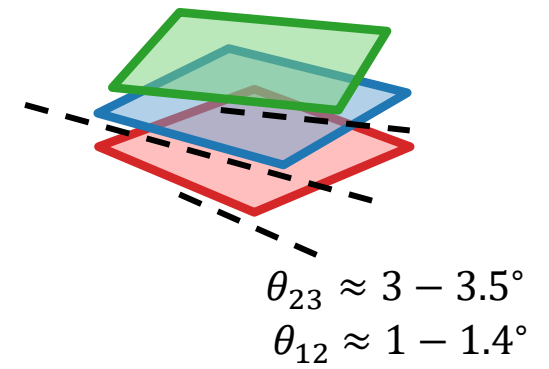
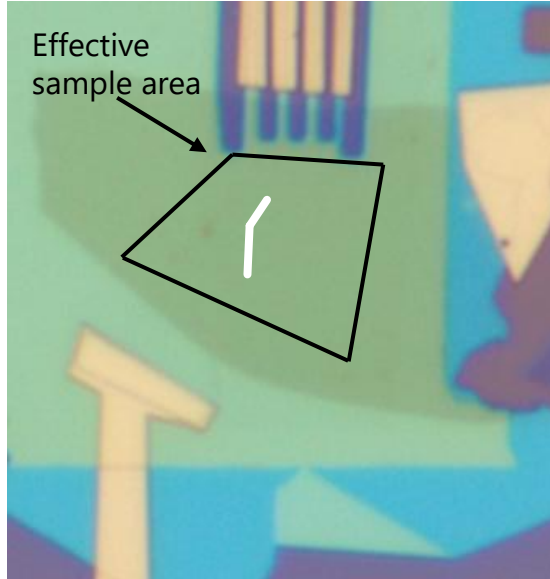
Spatial resolution: ~ 100 nm

Sensitivity: $\sim 10 \mu\text{V}/\text{VHz}$

$T_{\text{base}} = 330$ mK

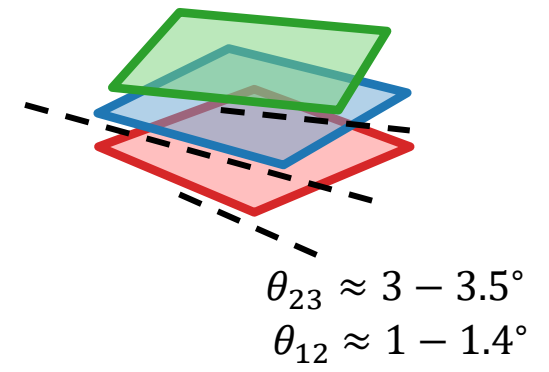
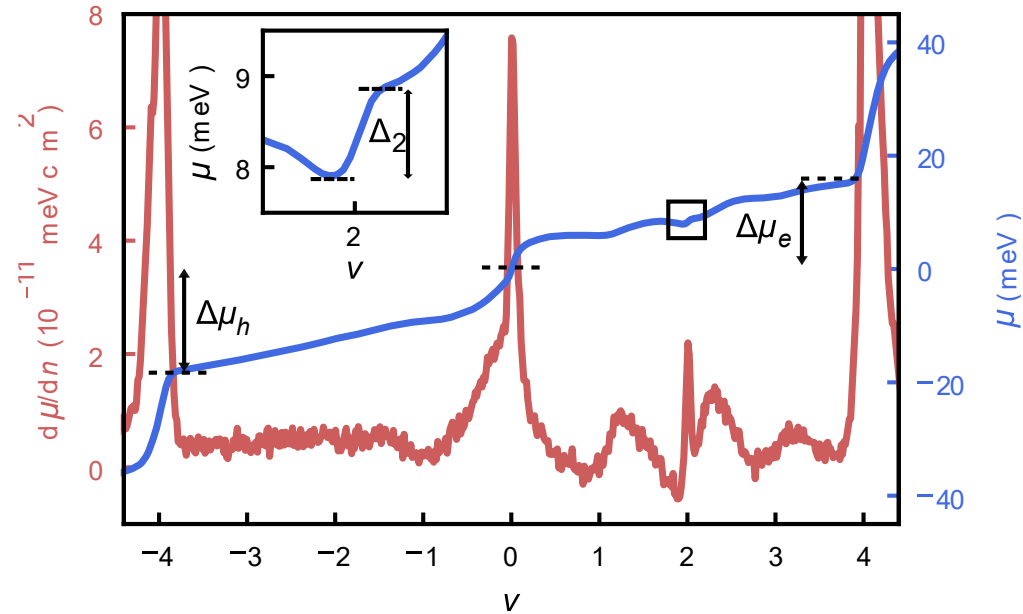
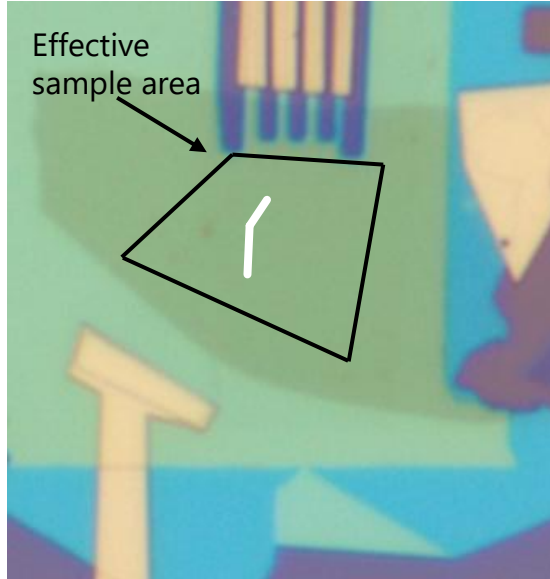


Imaging twisted trilayer compressibility⁻¹



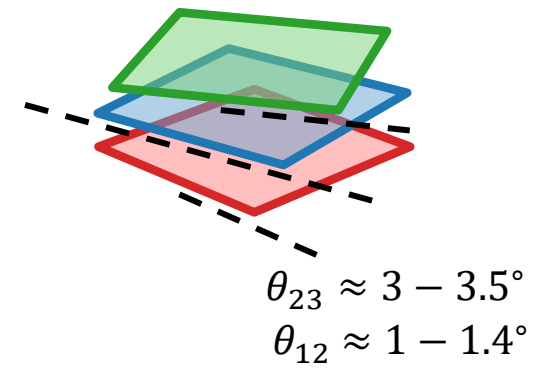
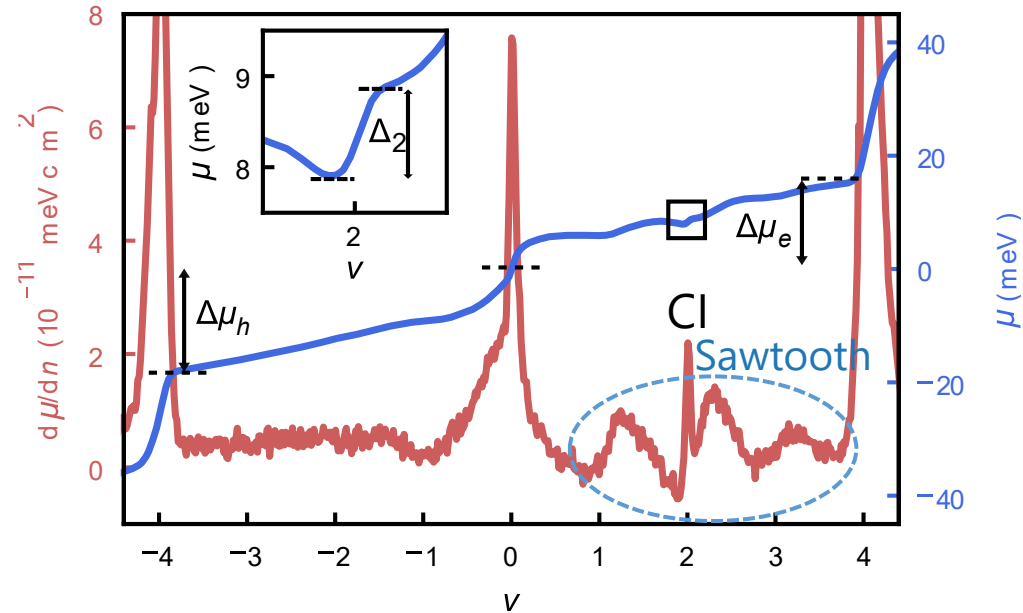
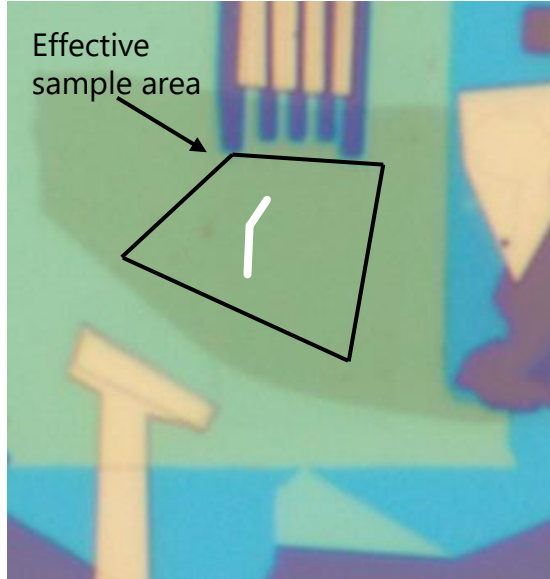
Zondiner, et al. Nature (2020)
Wong, et al. Nature (2020)
Liu, et al. Nat. Phys (2022)
Turkel et al. Science (2022)
Hoke,..., Sharpe, Feldman arXiv:2509.07977

Imaging twisted trilayer compressibility⁻¹



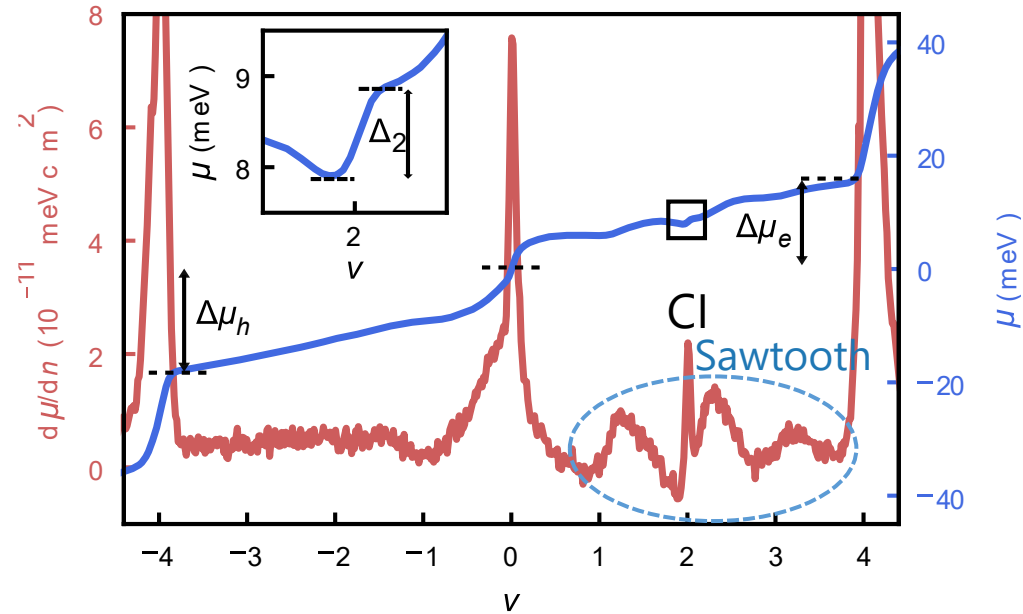
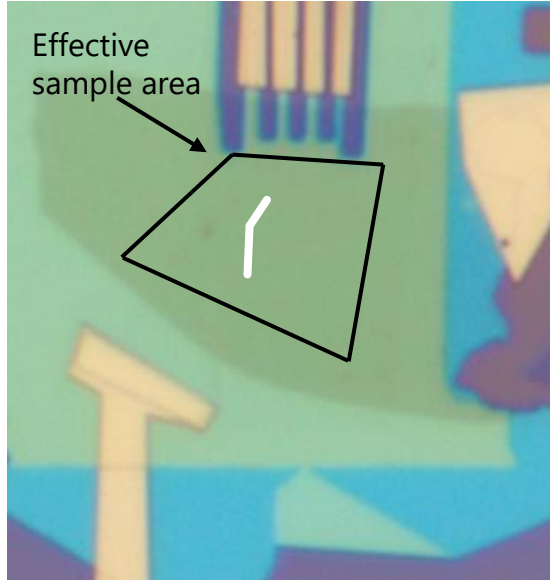
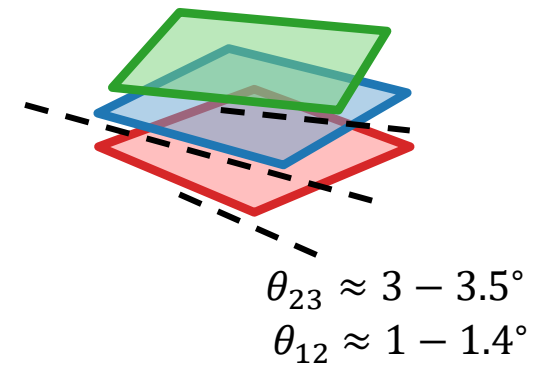
Zondiner, et al. Nature (2020)
Wong, et al. Nature (2020)
Liu, et al. Nat. Phys (2022)
Turek et al. Science (2022)
Hoke,..., Sharpe, Feldman arXiv:2509.07977

Imaging twisted trilayer compressibility⁻¹

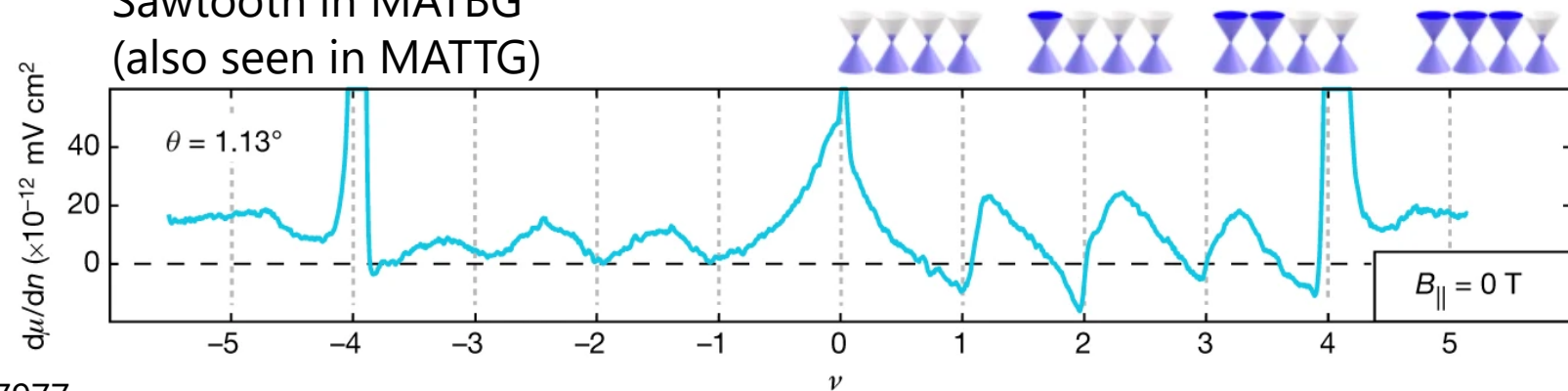


Zondiner, et al. Nature (2020)
Wong, et al. Nature (2020)
Liu, et al. Nat. Phys (2022)
Turler et al. Science (2022)
Hoke,..., Sharpe, Feldman arXiv:2509.07977

Imaging twisted trilayer compressibility⁻¹

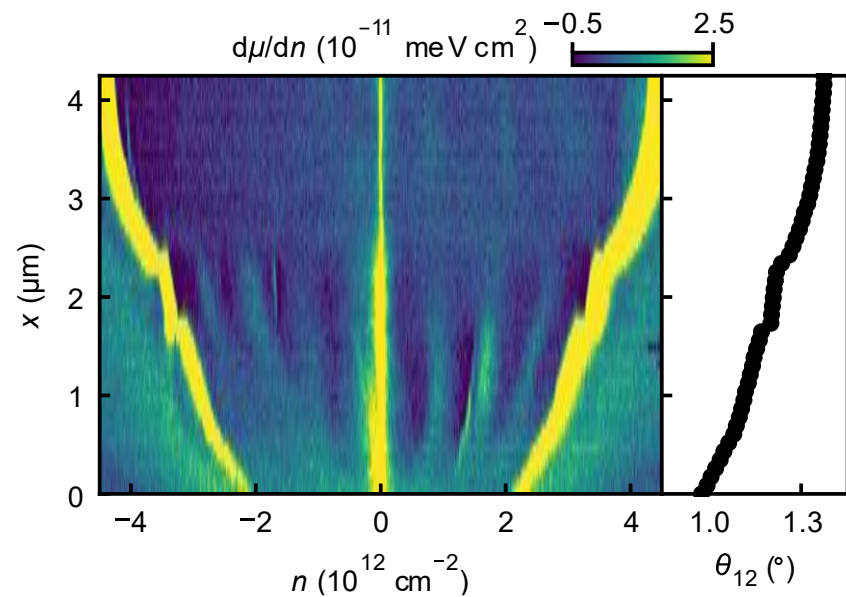


Sawtooth in MATBG
 (also seen in MATTG)

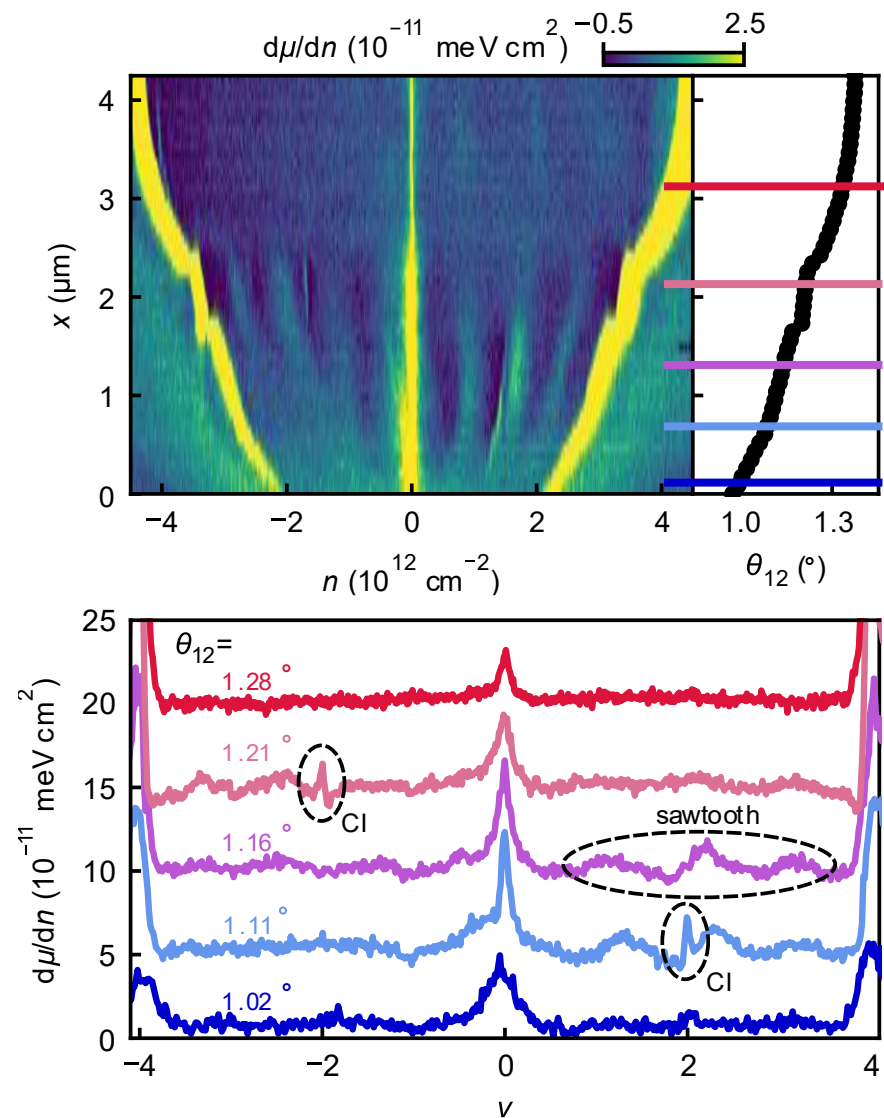


Zondiner, et al. Nature (2020)
 Wong, et al. Nature (2020)
 Liu, et al. Nat. Phys (2022)
 Turkel et al. Science (2022)
 Hoke,..., Sharpe, Feldman arXiv:2509.07977

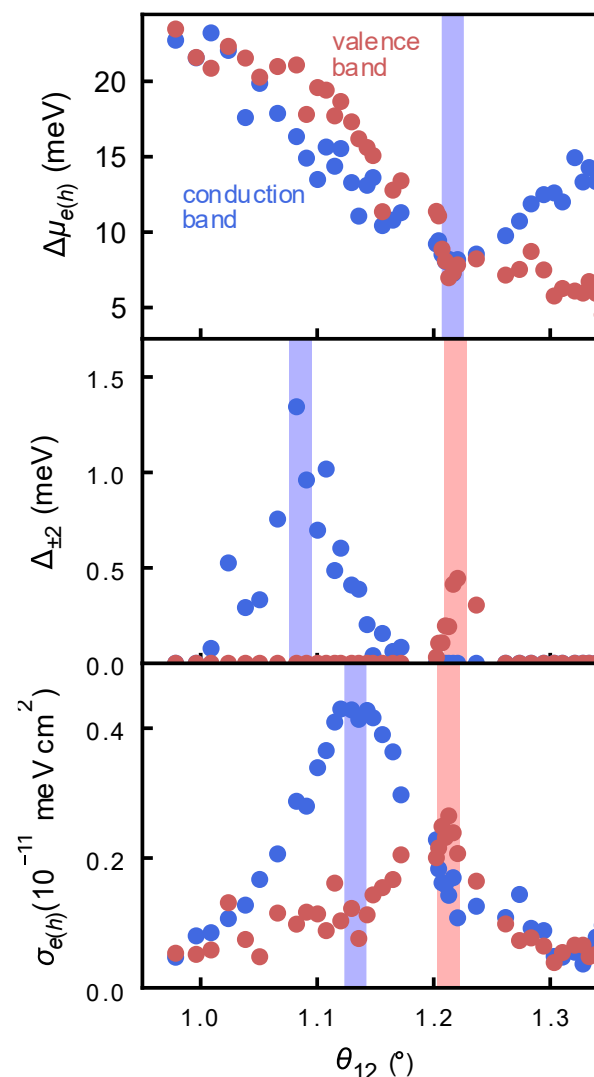
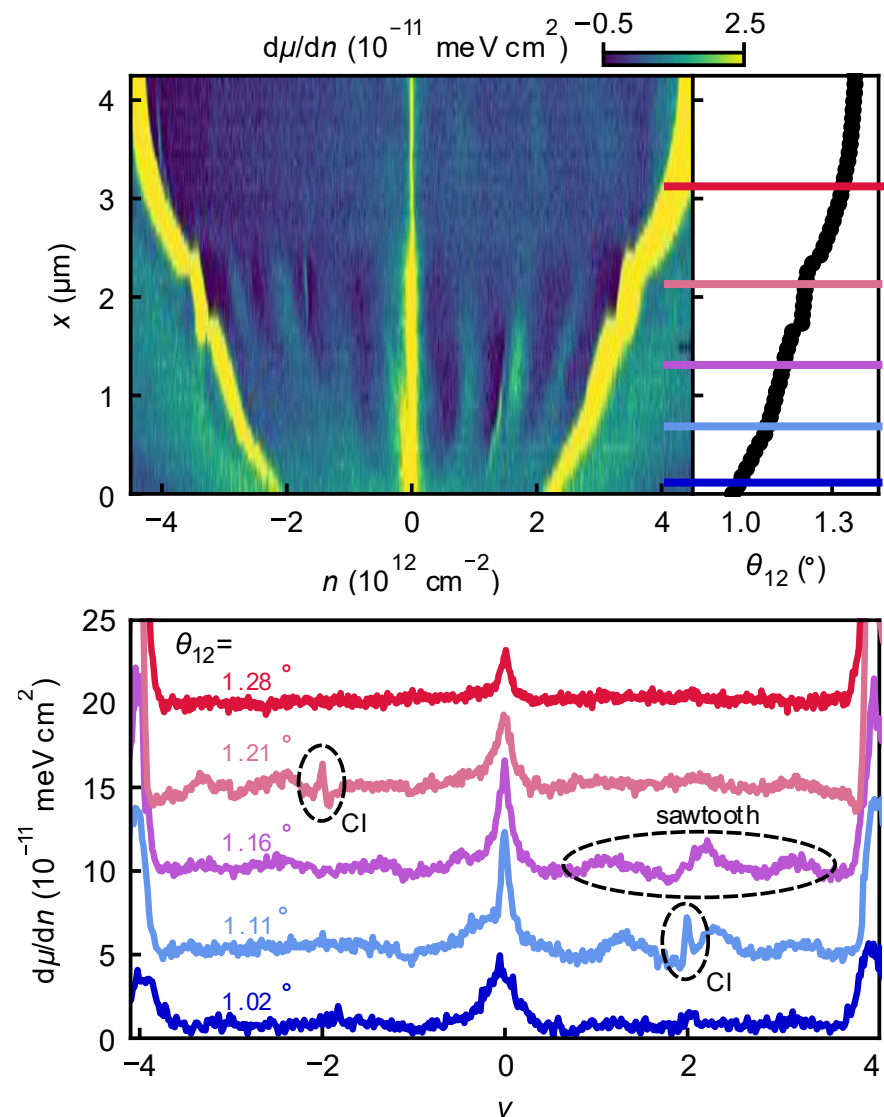
Twist angle dependent correlations



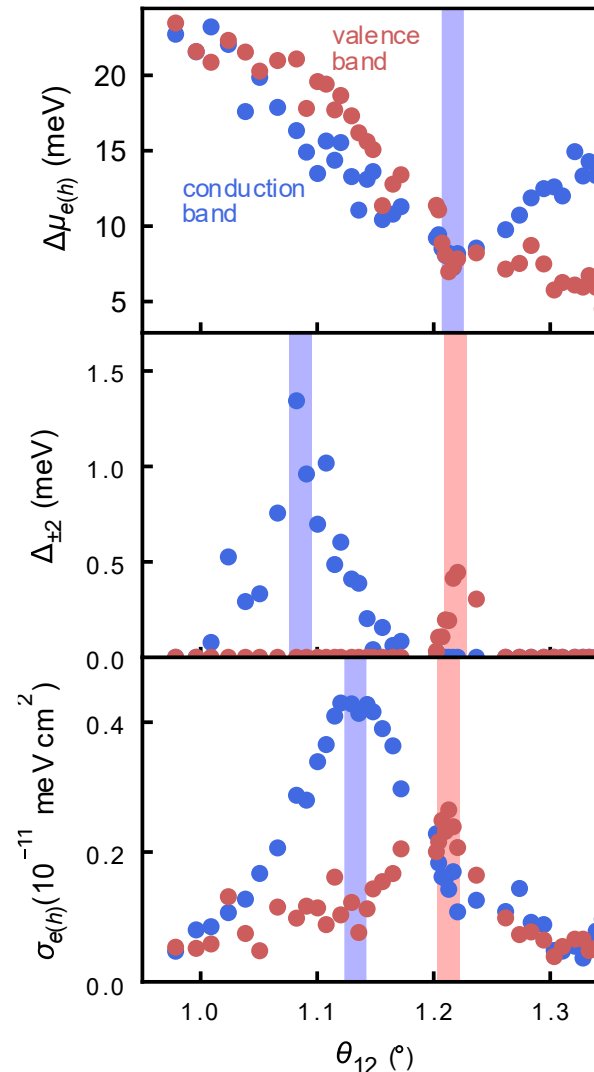
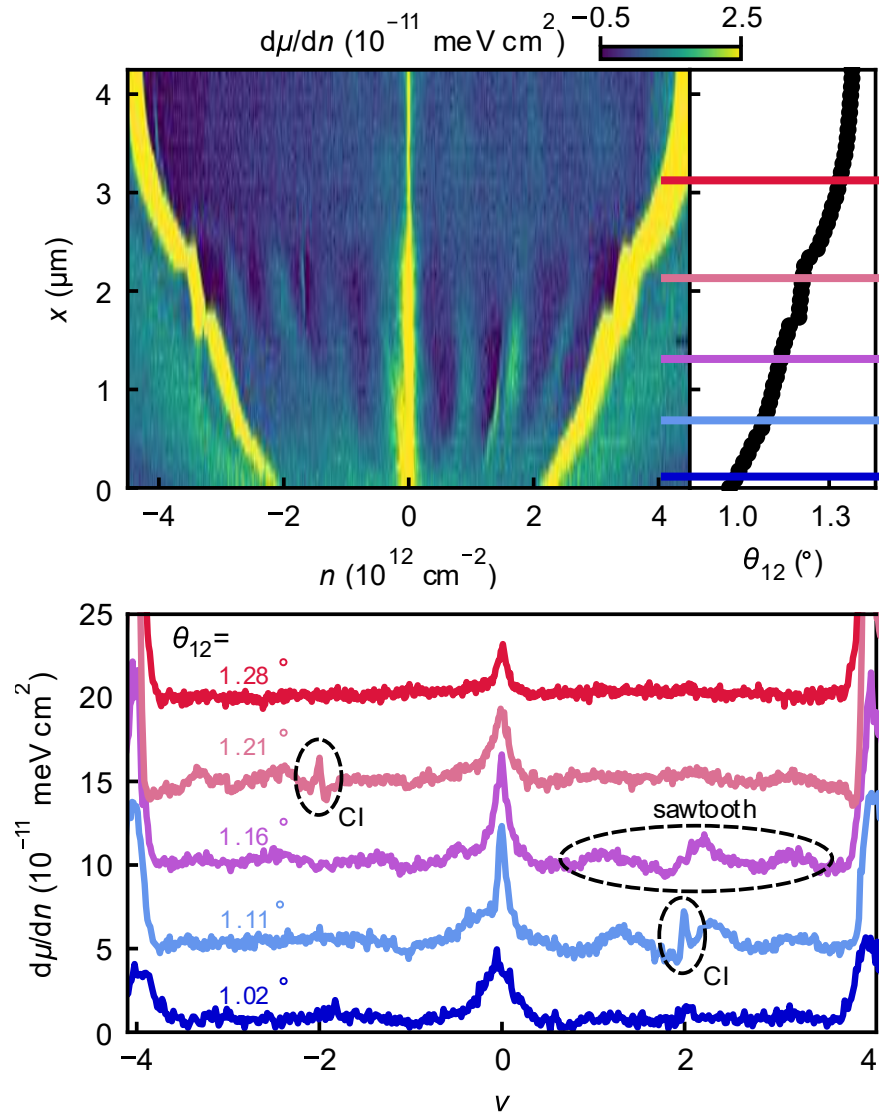
Twist angle dependent correlations



Twist angle dependent correlations

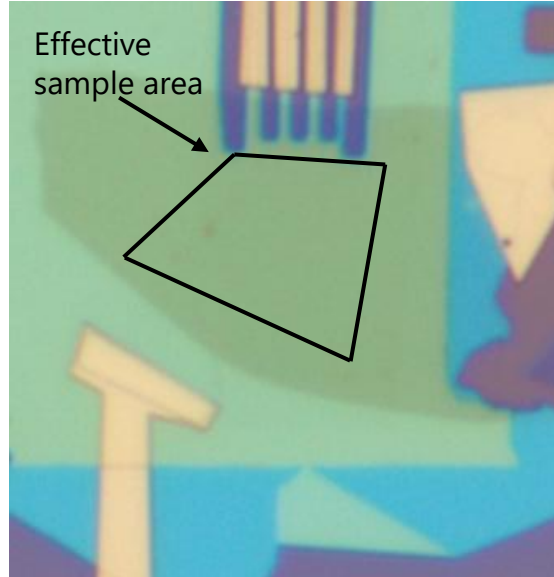


Twist angle dependent correlations

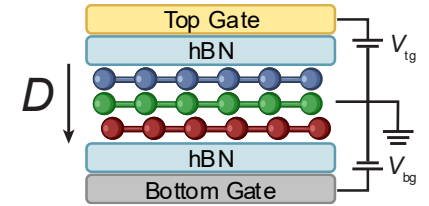
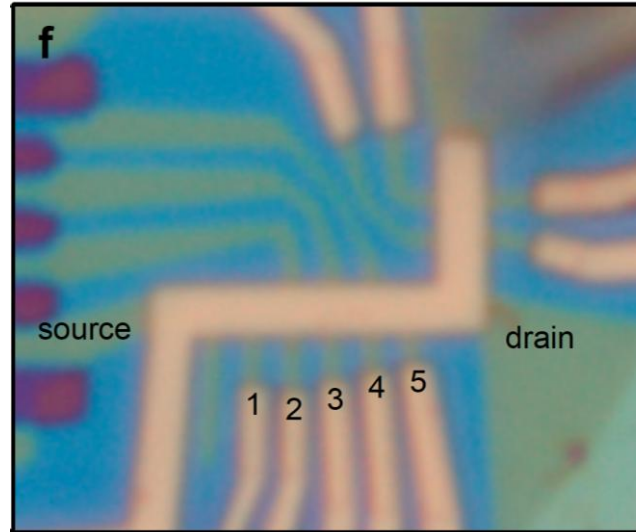


1. Distinct "magic angles" for electrons and holes
2. Correlated insulators like MATBG and TTG
3. "Sawtooth" in compressibility⁻¹ also like MATBG/MATTG

Compare to Transport

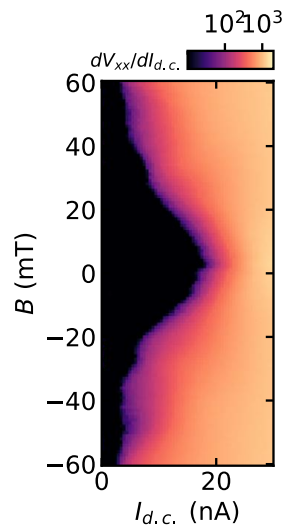


→
Fab

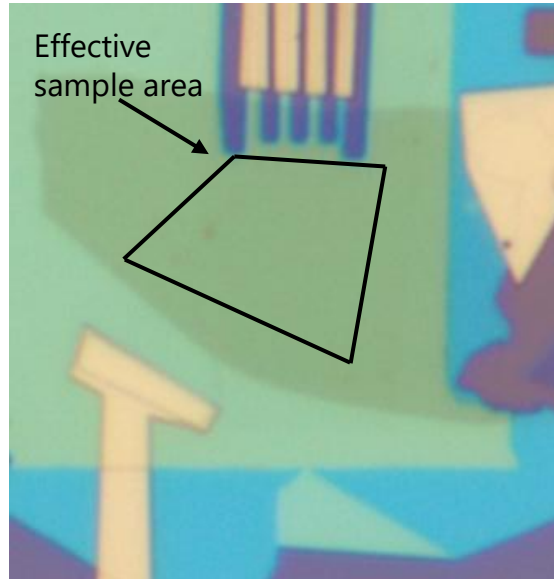


Etched a 'magic' region of the device and added a top gate

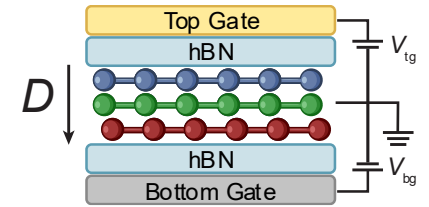
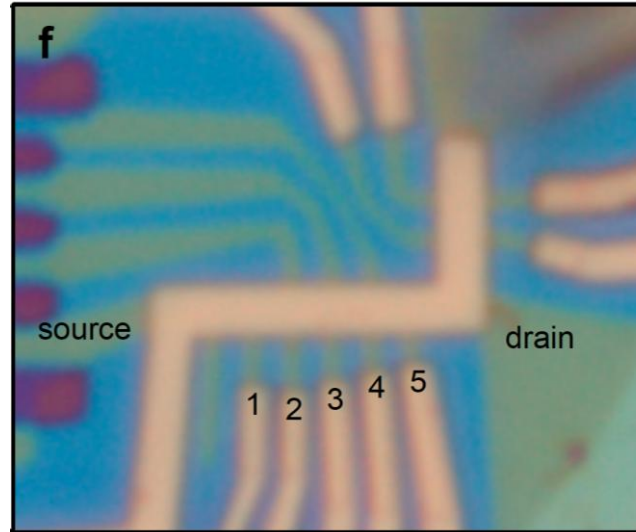
Direct comparison of compressibility⁻¹ and transport in the same device!



Compare to Transport

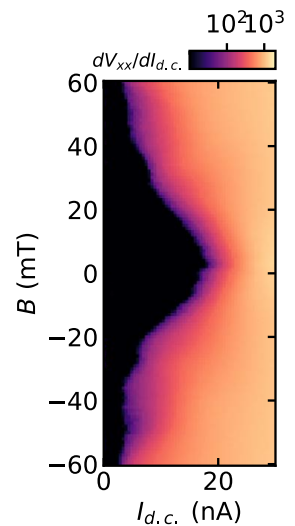
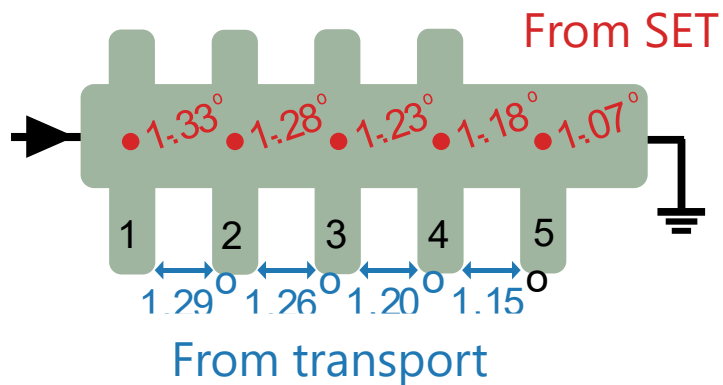


→
Fab

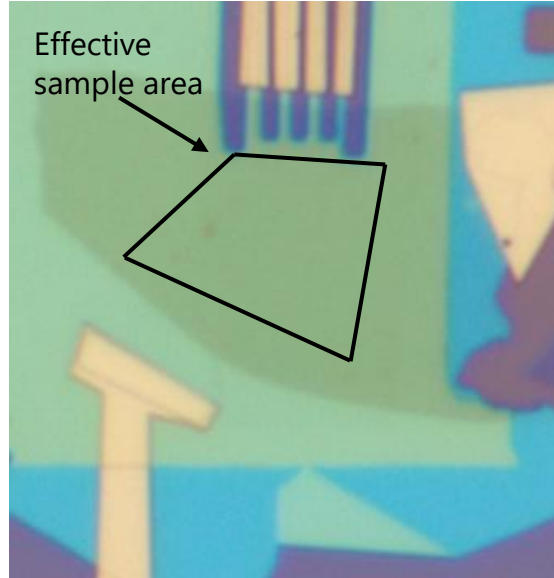


Etched a 'magic' region of the device and added a top gate

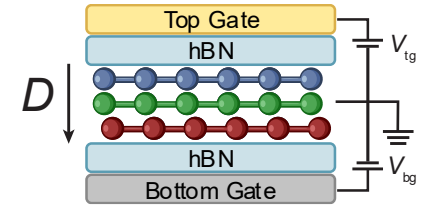
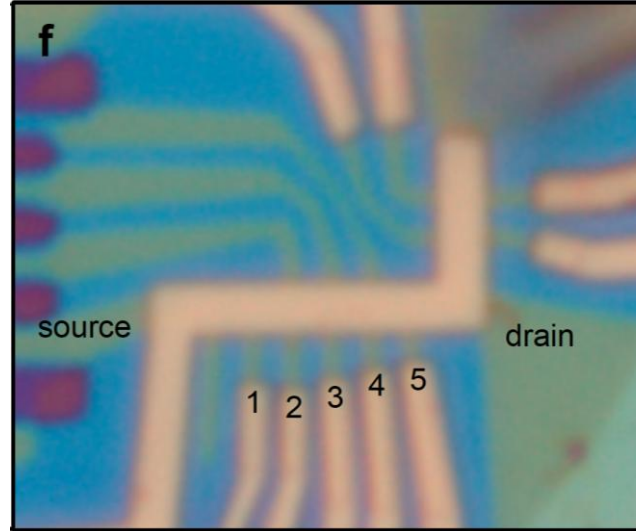
Direct comparison of compressibility⁻¹ and transport in the same device!



Compare to Transport

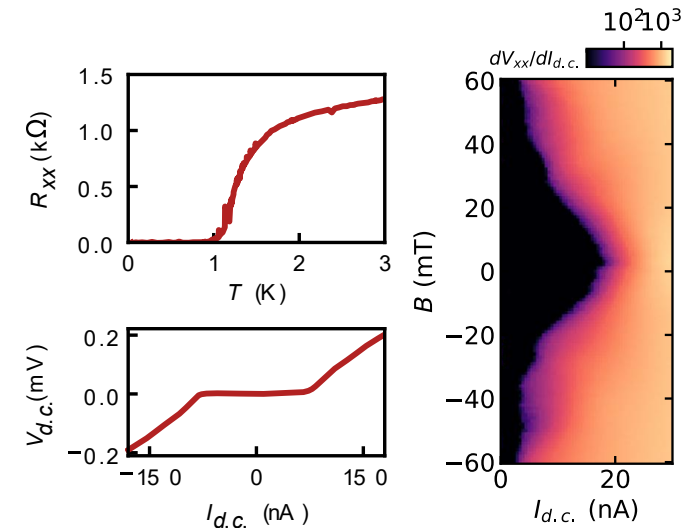
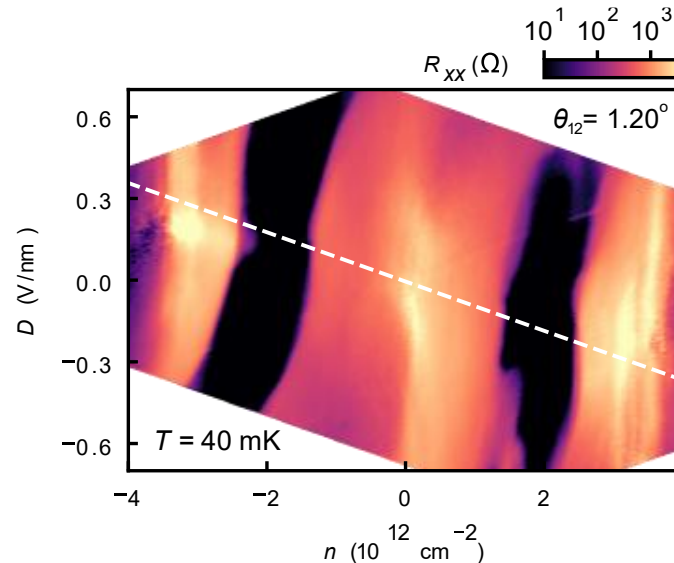
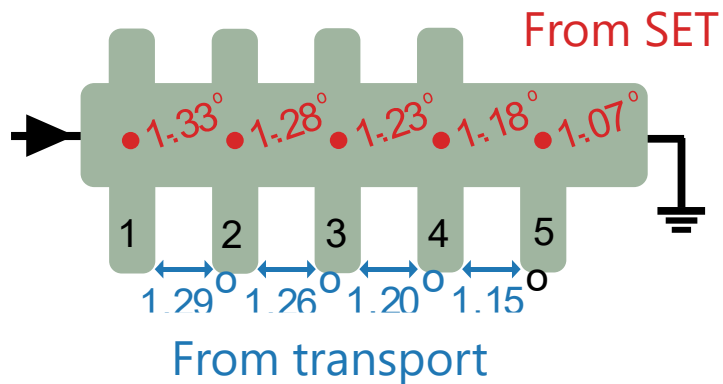


→
Fab

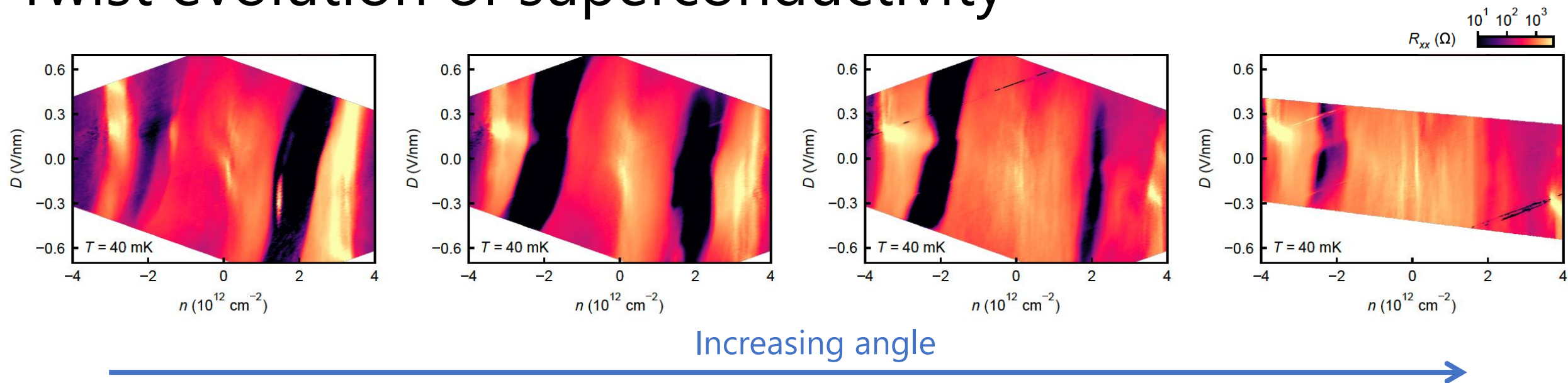


Etched a 'magic' region of the device and added a top gate

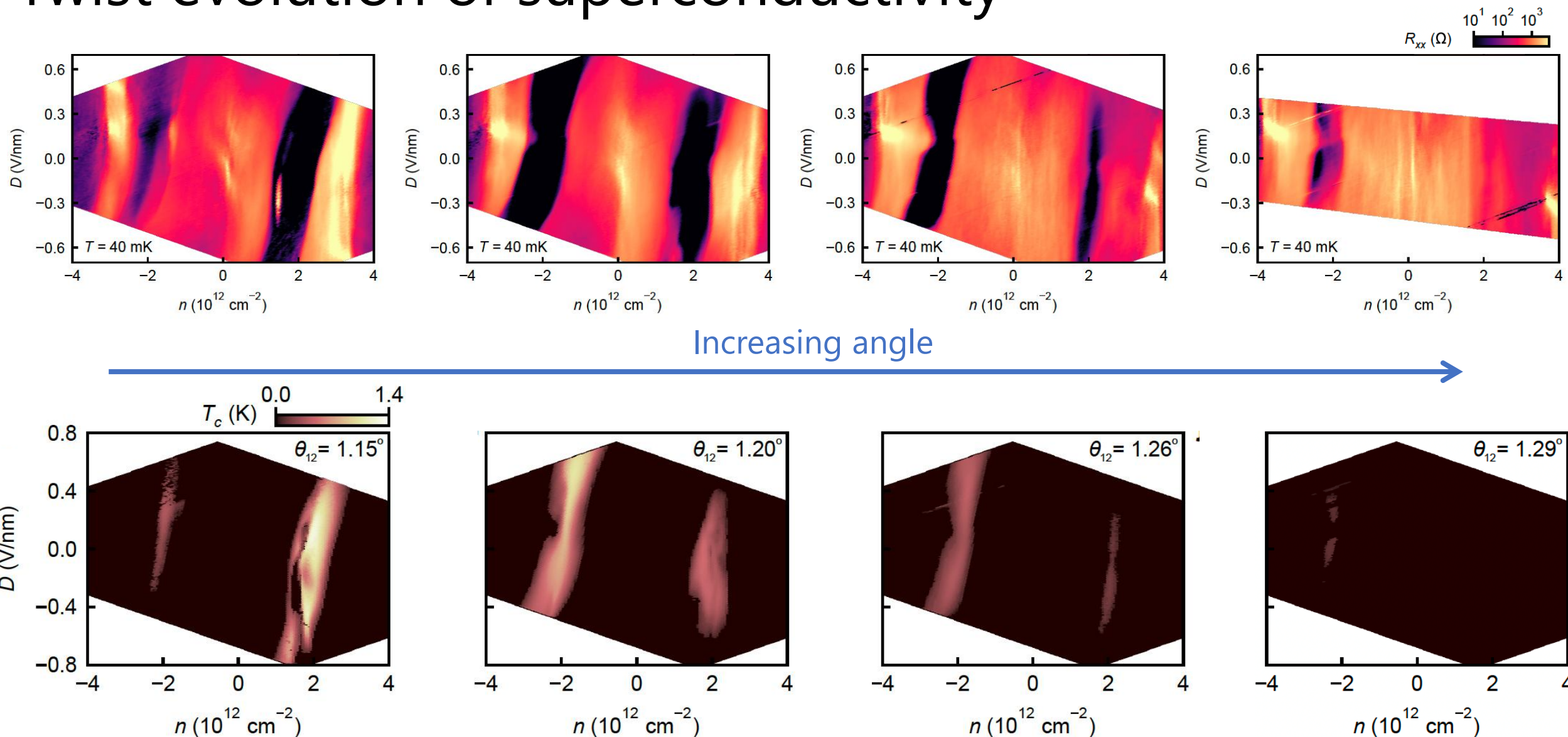
Direct comparison of compressibility⁻¹ and transport in the same device!



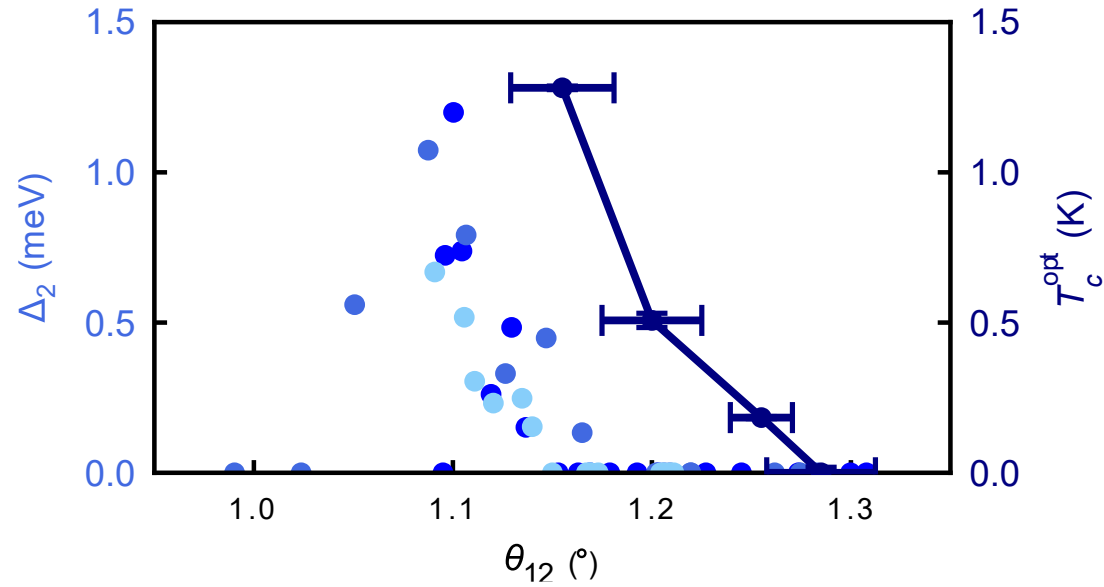
Twist evolution of superconductivity



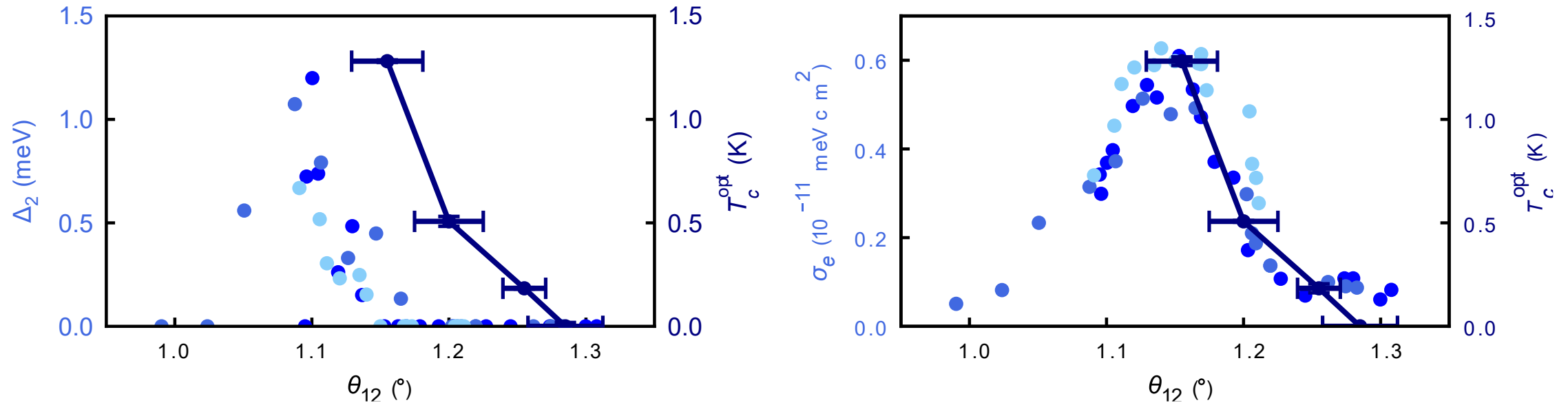
Twist evolution of superconductivity



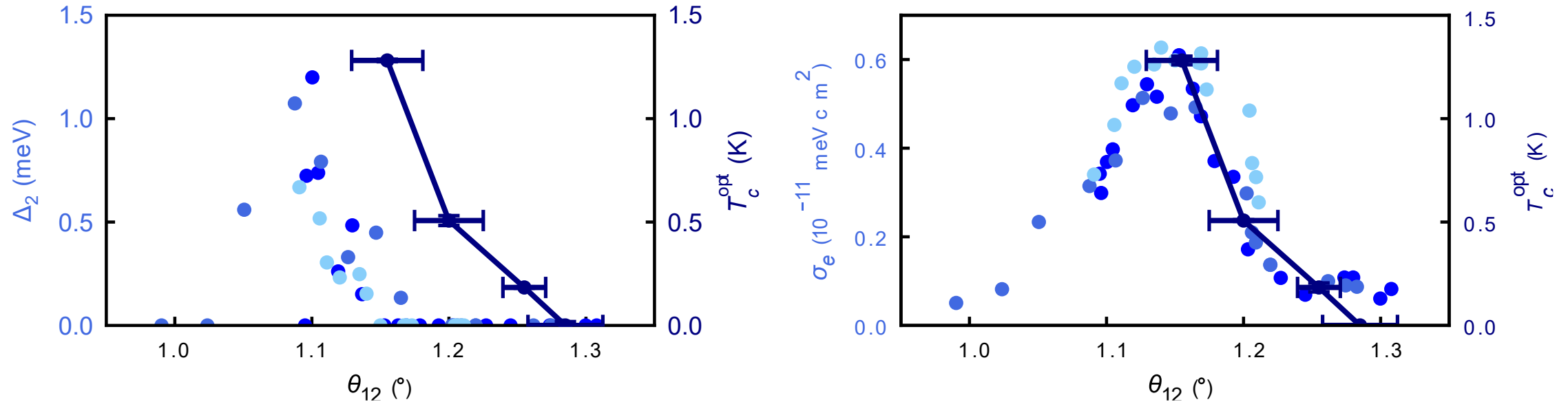
Comparing superconductivity and sawtooth strength



Comparing superconductivity and sawtooth strength



Comparing superconductivity and sawtooth strength



- Strength (T_c) of superconductivity scales similarly with sawtooth
- No clear correspondence to correlated insulators
- For now: just an observation (there is correlation), but no crisp conclusion about origin of superconductivity

Outline

Introduction:

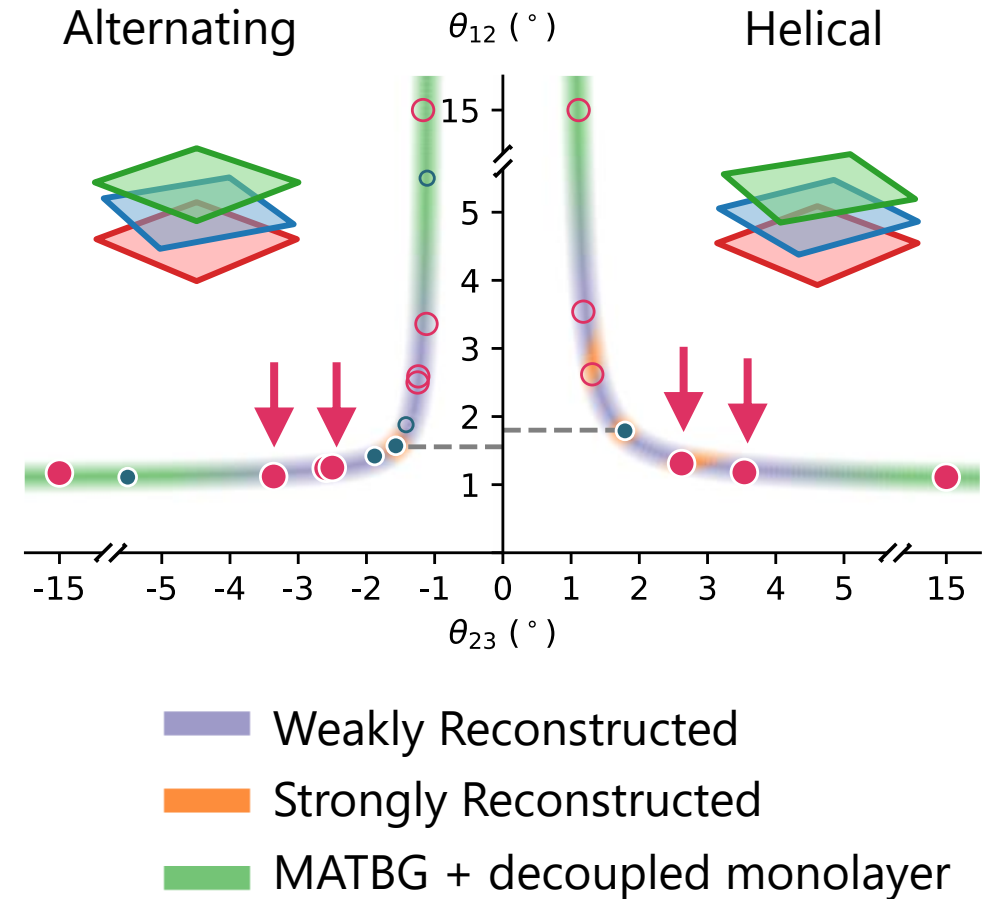
- From moiré to supermoiré
- Helical vs. alternating stacking
- Lattice relaxation

Linking thermodynamic probes and transport:

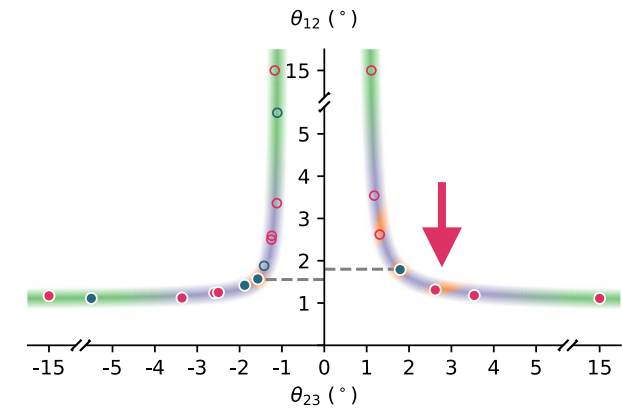
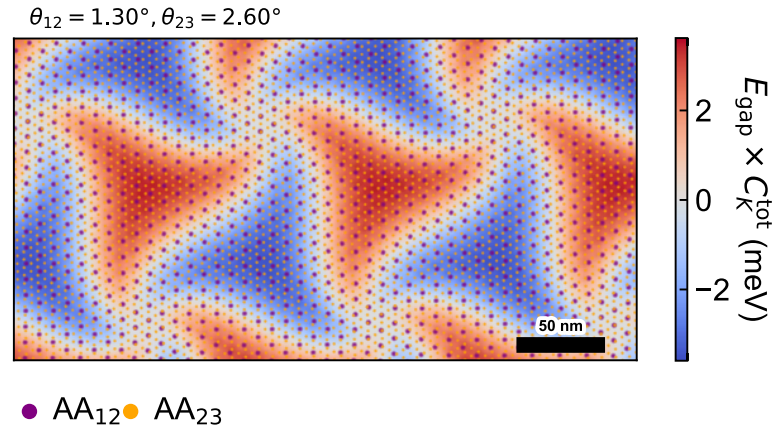
- Fine-grained evolution of correlations
- Cross comparison of distinct experimental probes

Exploring the “magic continuum”:

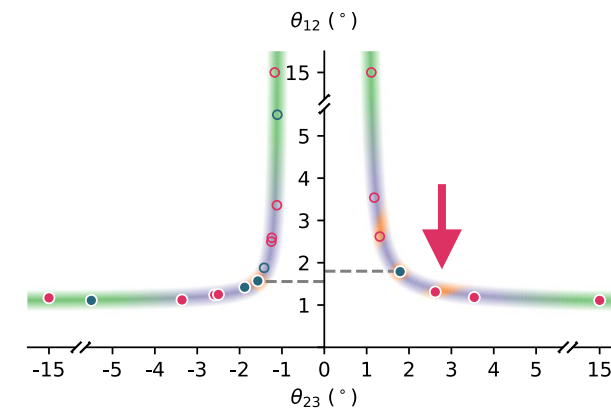
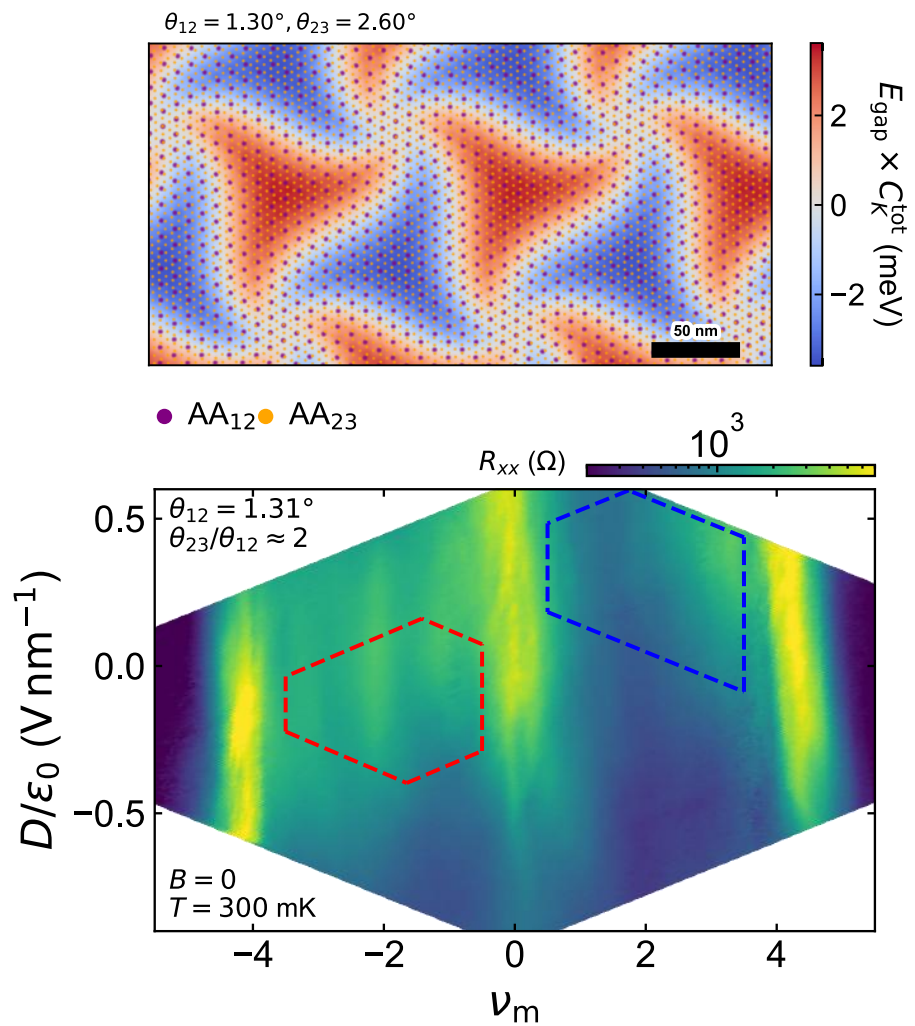
- Broad comparison across many devices in angle-angle space



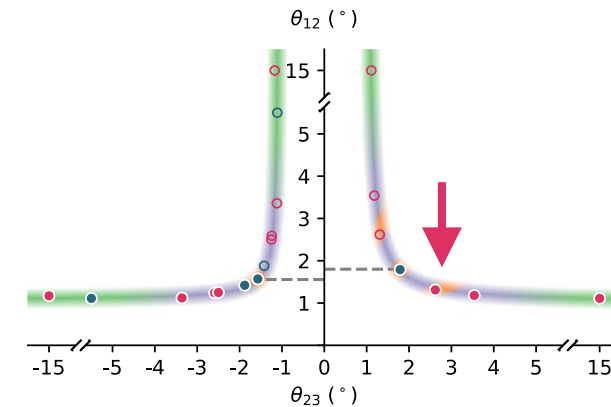
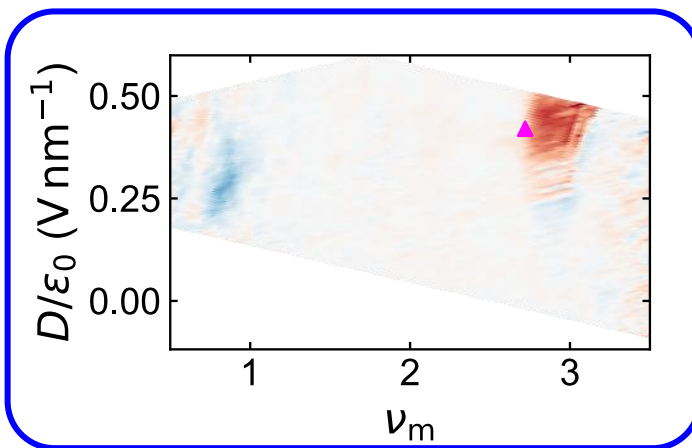
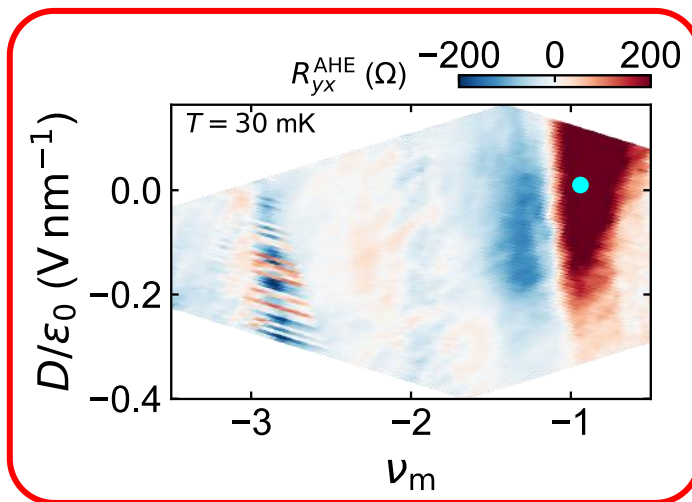
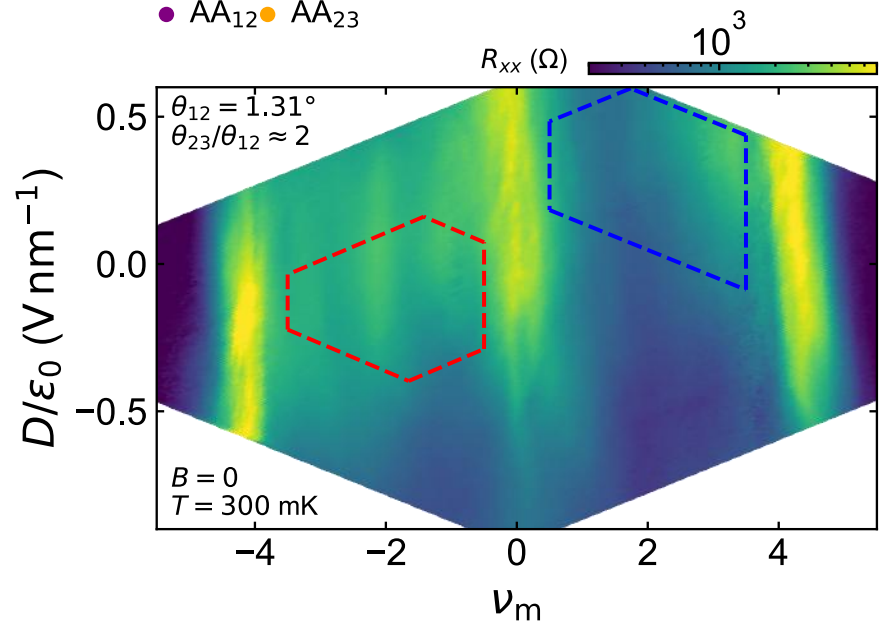
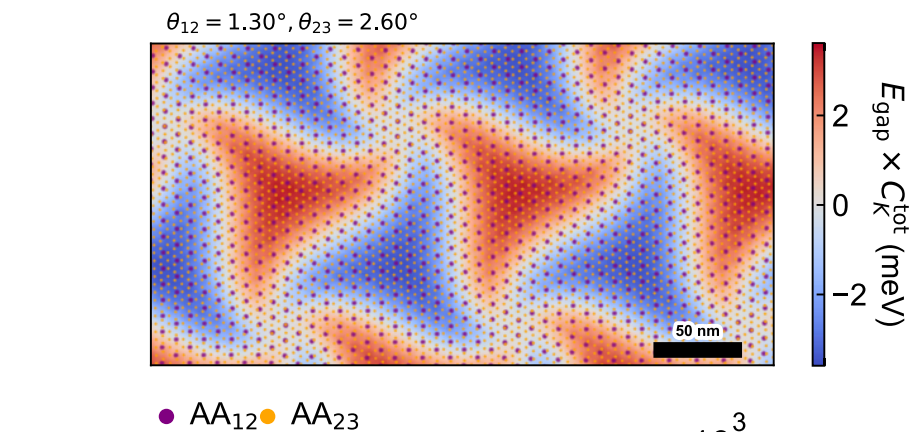
$(2\theta, \theta)$ – Helical Trilayer Graphene



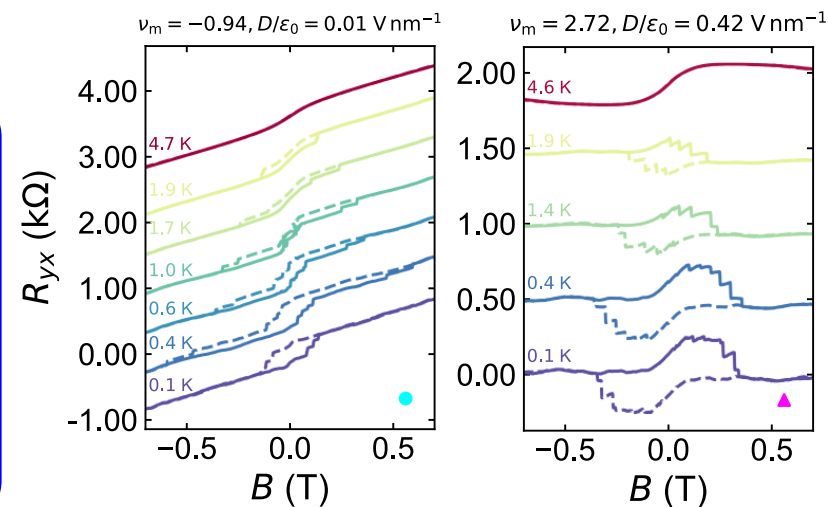
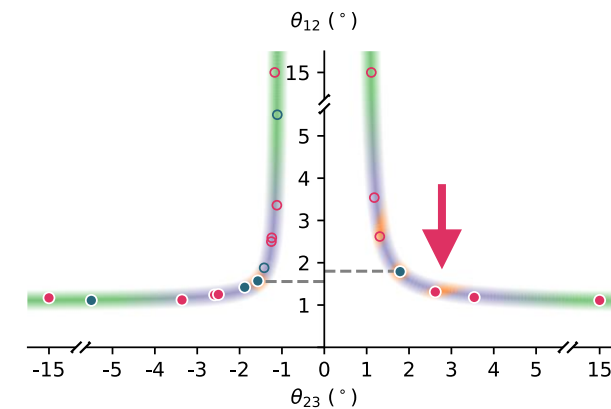
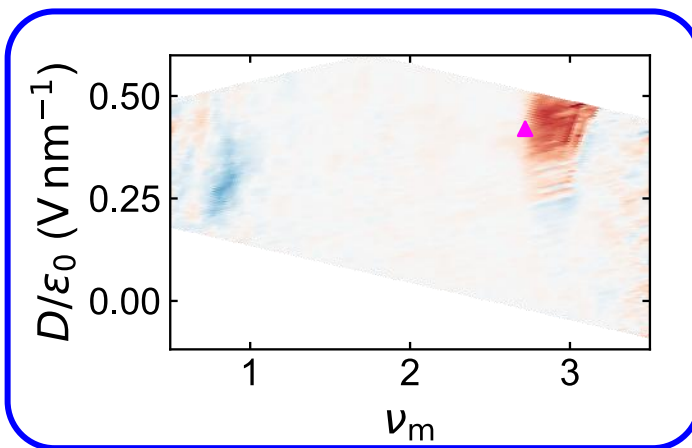
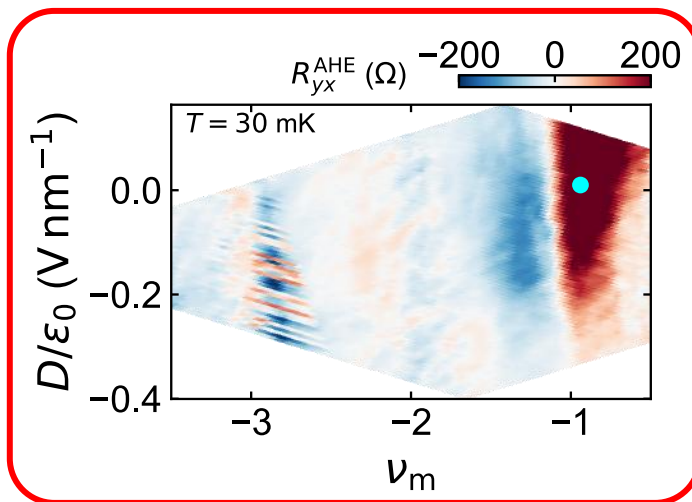
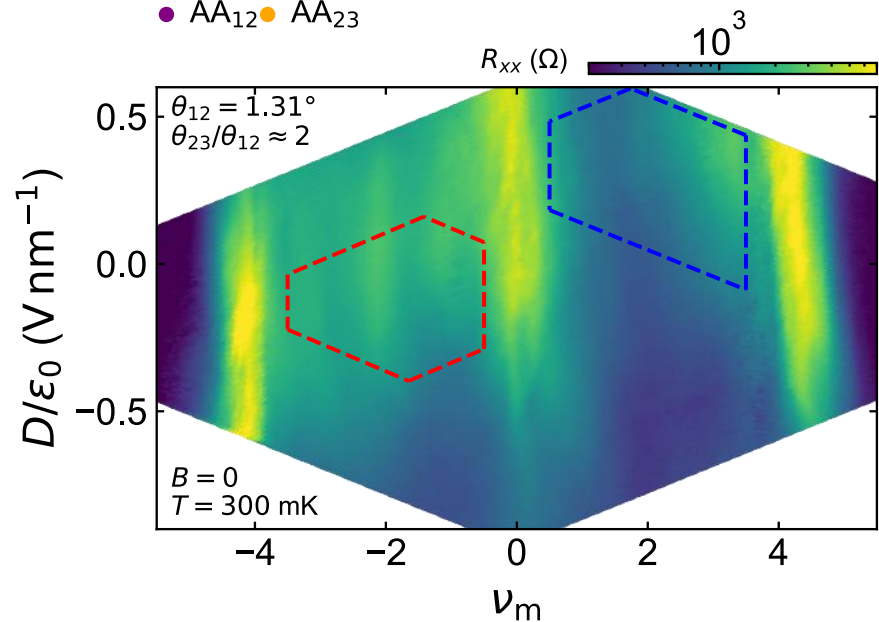
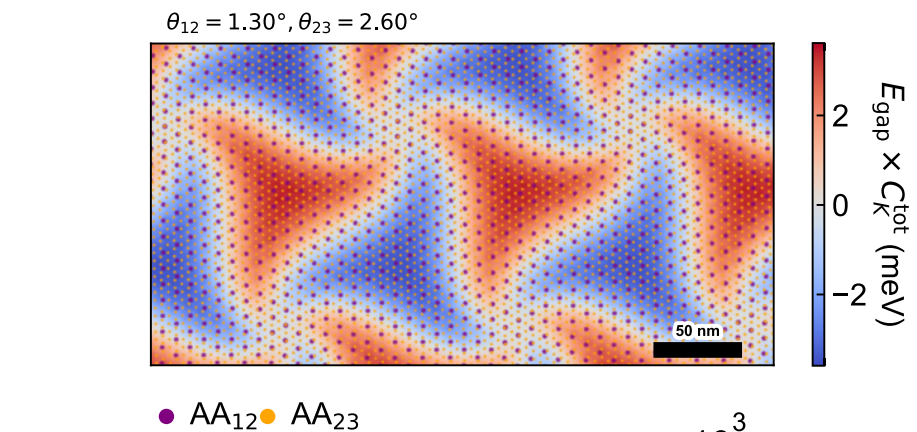
$(2\theta, \theta)$ – Helical Trilayer Graphene



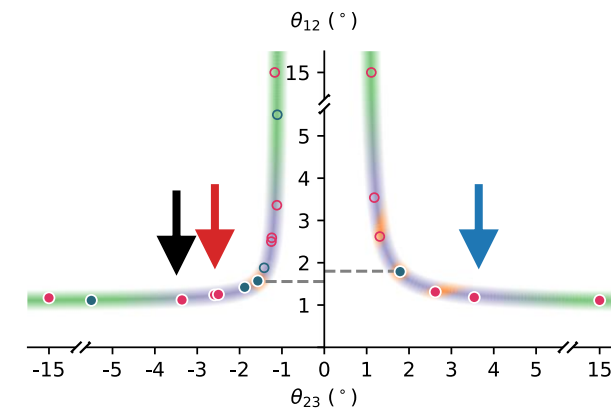
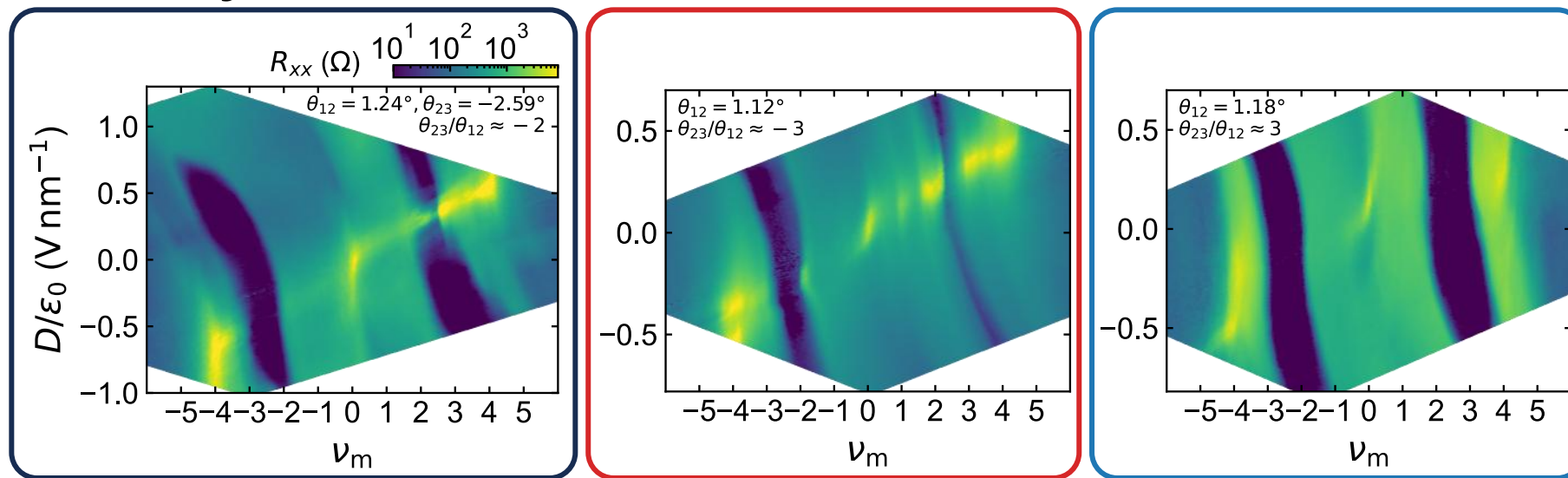
$(2\theta, \theta)$ – Helical Trilayer Graphene



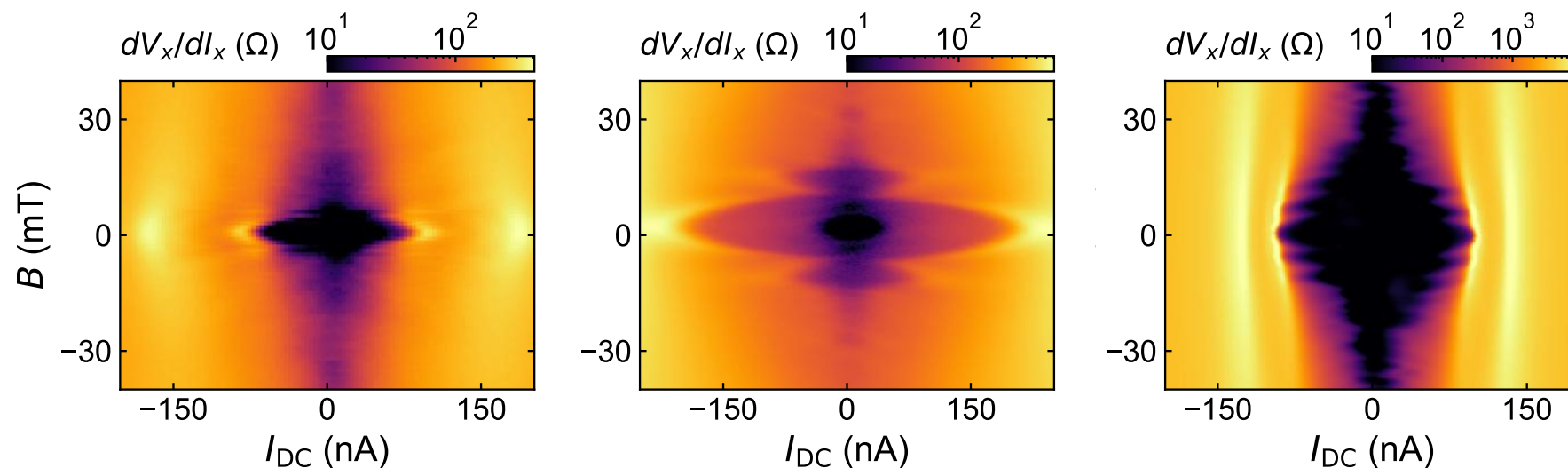
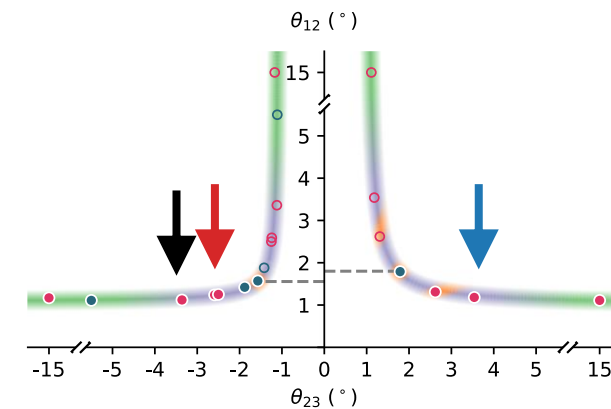
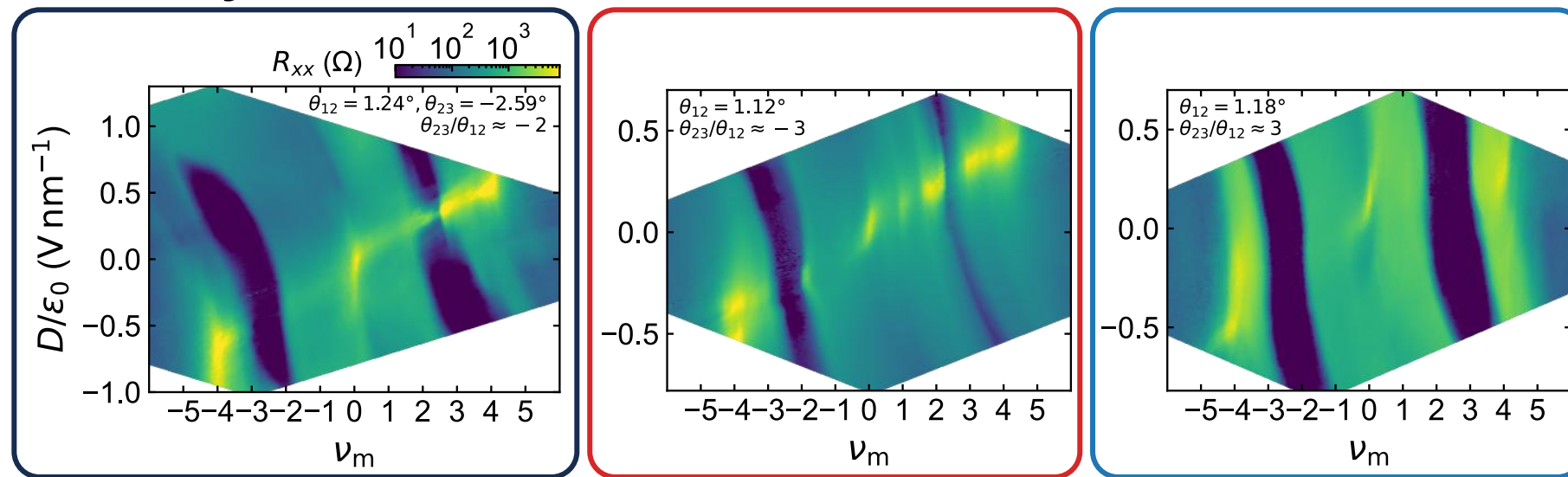
$(2\theta, \theta)$ – Helical Trilayer Graphene



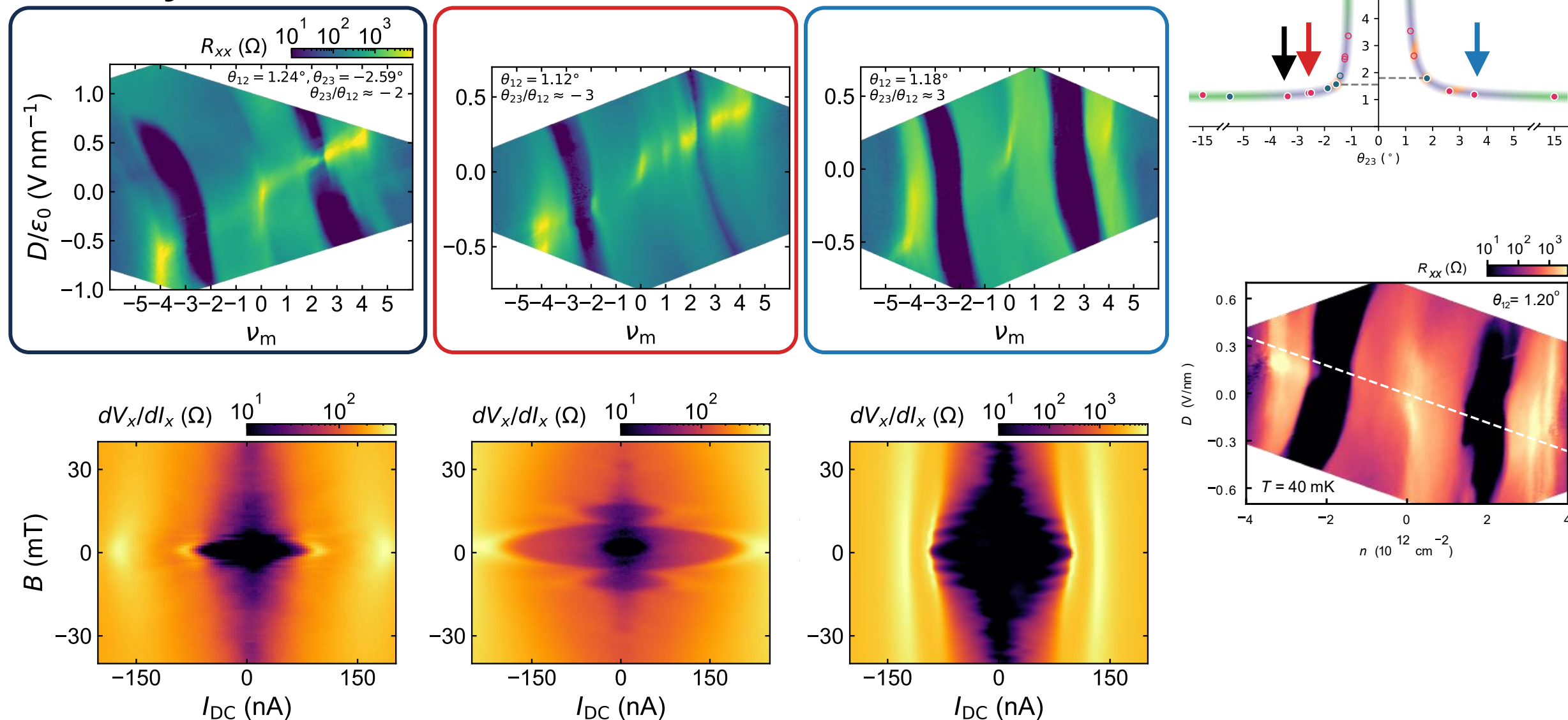
Weakly-Reconstructed TTG



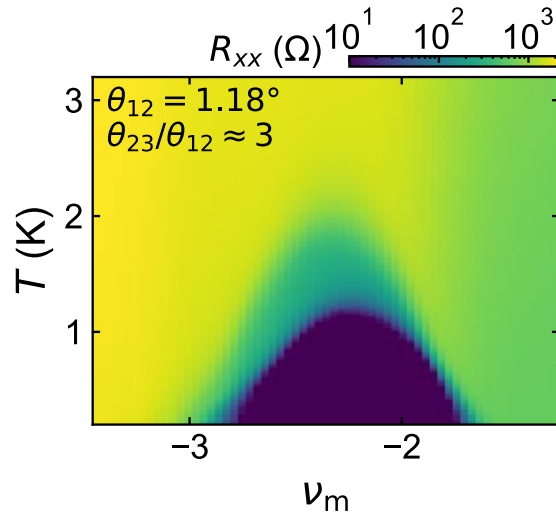
Weakly-Reconstructed TTG



Weakly-Reconstructed TTG



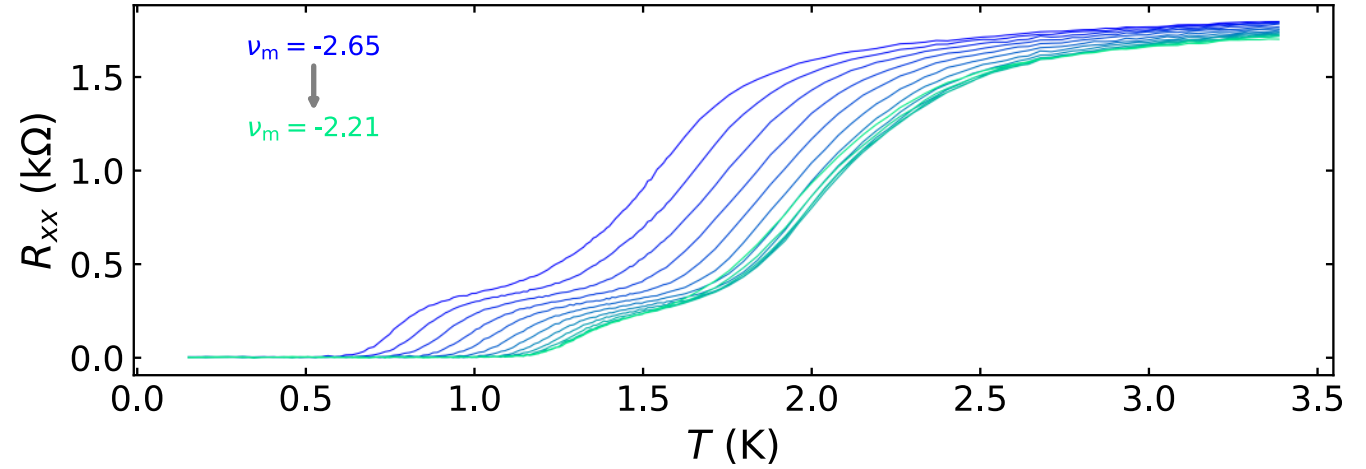
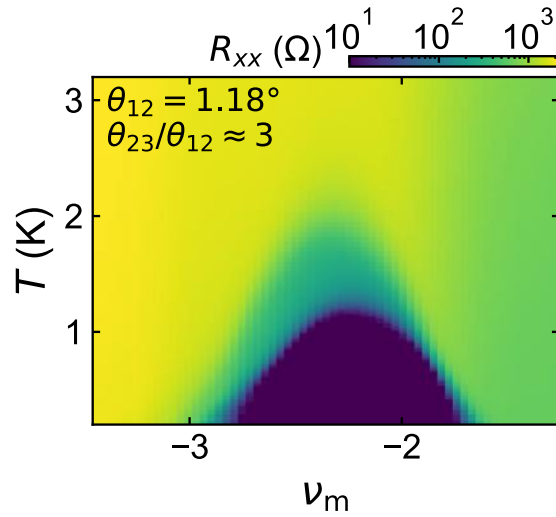
Supermoiré-modulation of superconductivity?



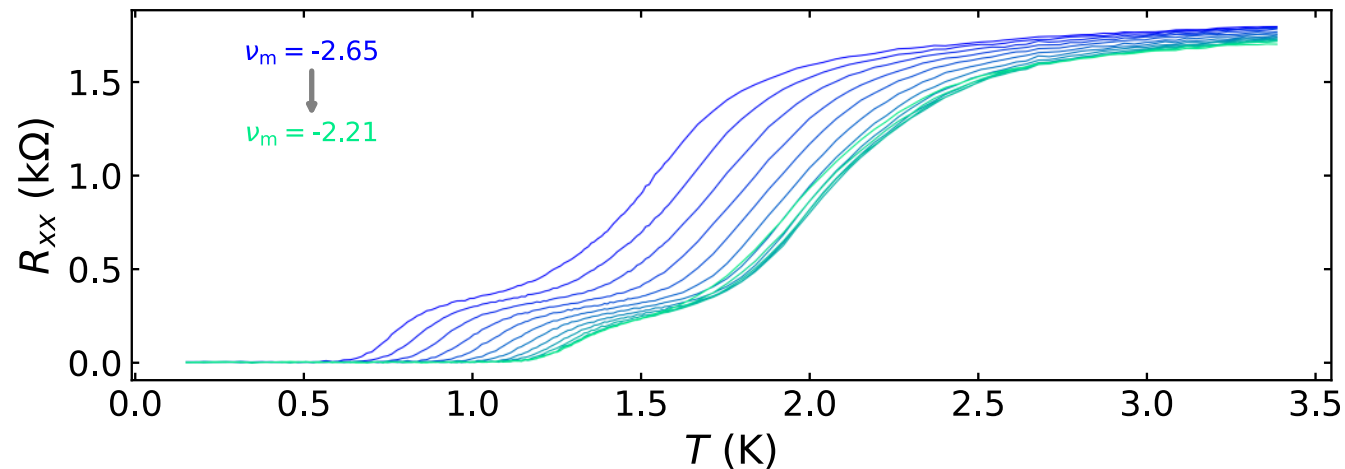
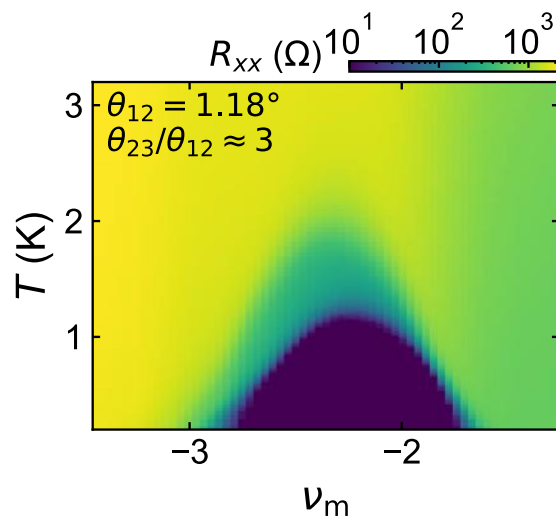
Eley, et al. Nat. Phys (2011)

Xia*, Uri*, Yan*, Sharpe* et al. arXiv:2509.03583

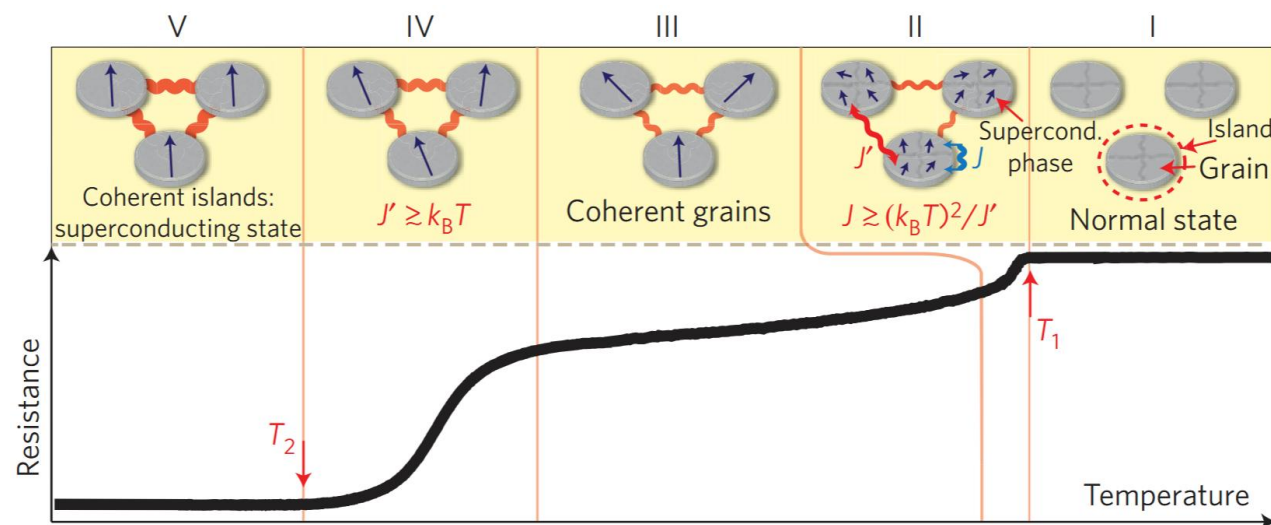
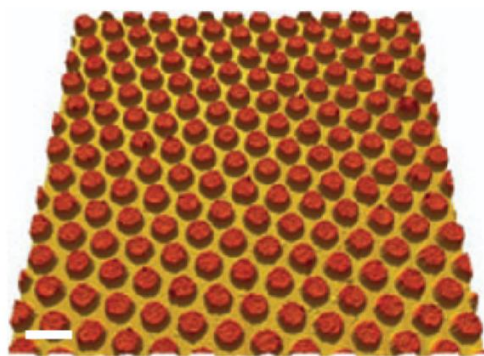
Supermoiré-modulation of superconductivity?



Supermoiré-modulation of superconductivity?

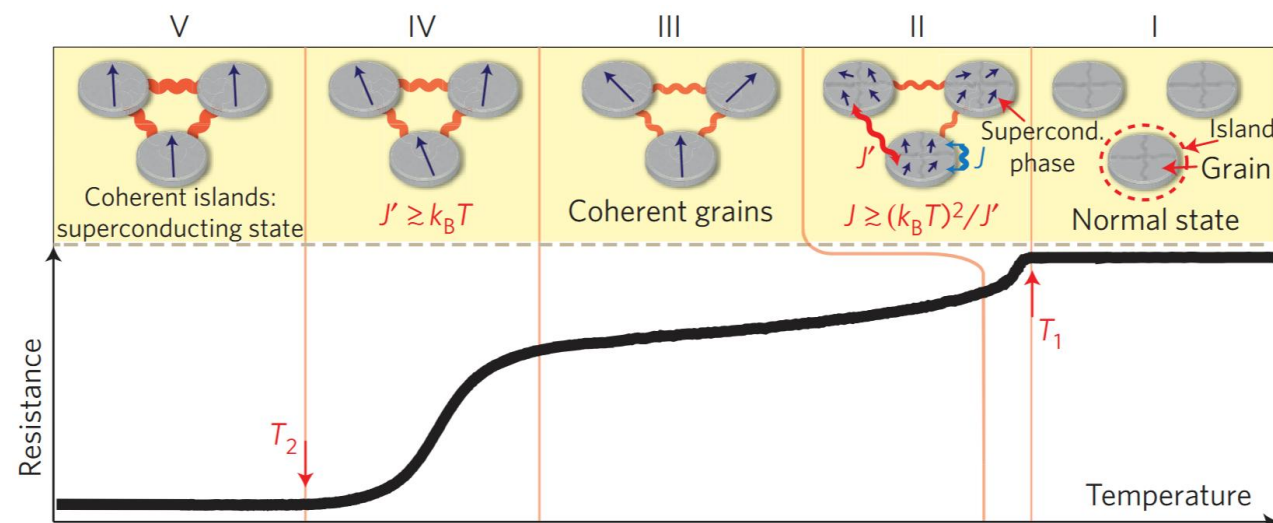
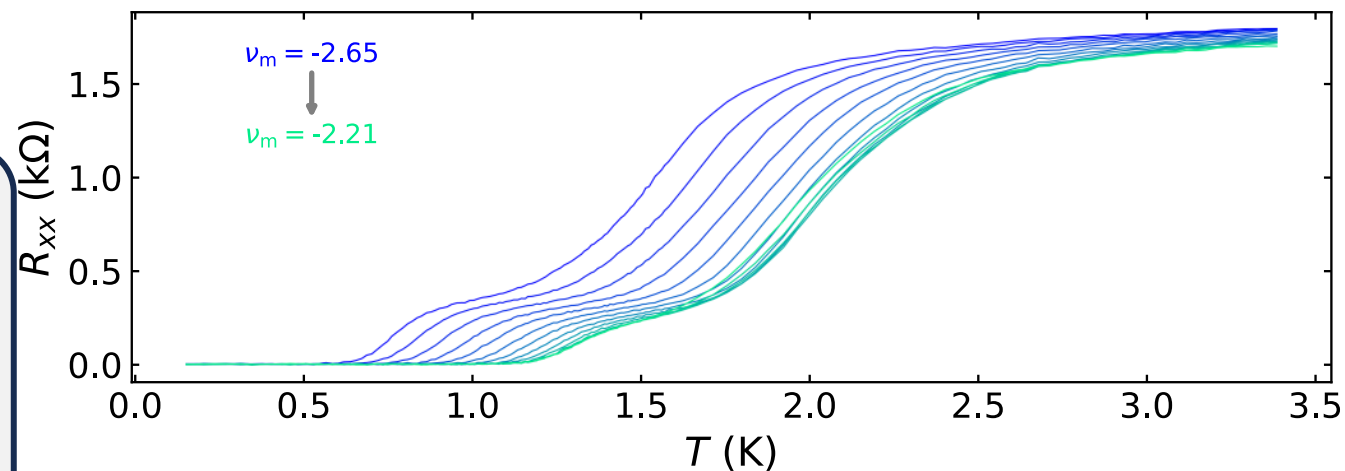
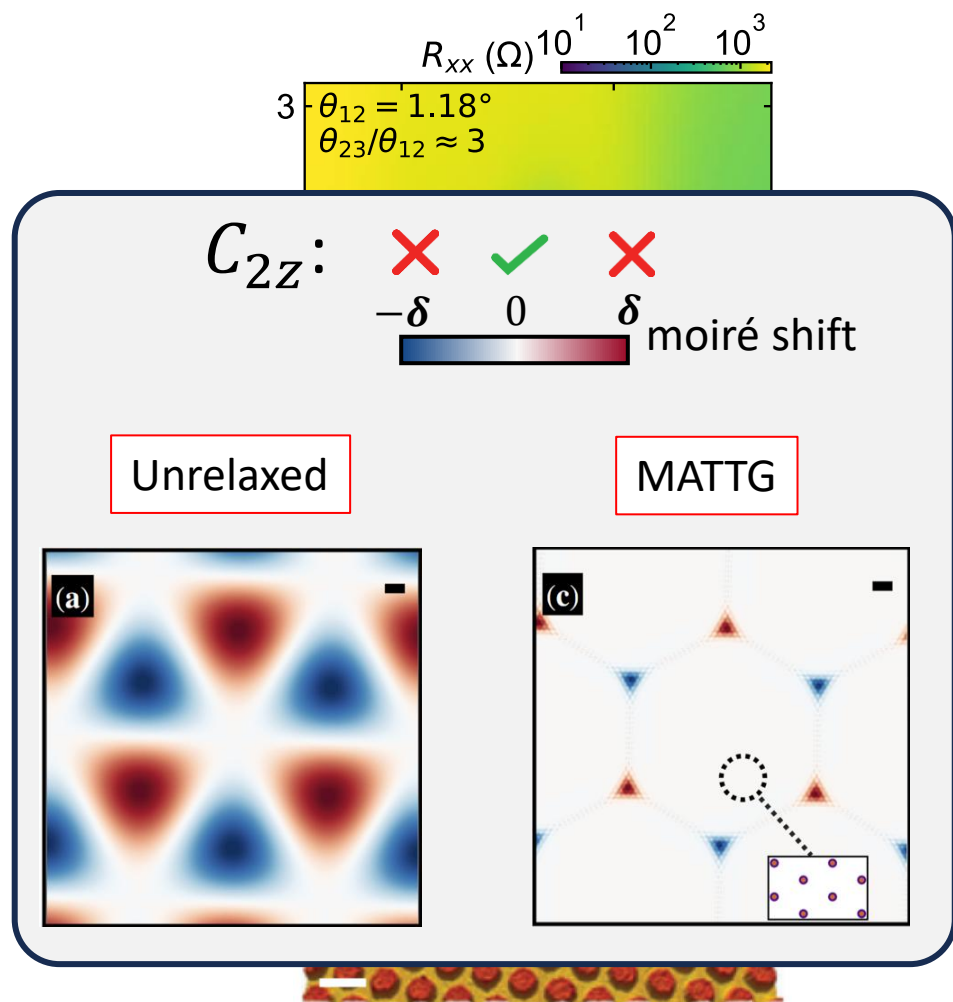


Nb islands on Cu



Eley, et al. Nat. Phys (2011)
 Xia*, Uri*, Yan*, Sharpe* et al. arXiv:2509.03583

Supermoiré-modulation of superconductivity?



Acknowledgments



Jesse Hoke



Yifan Li



Yuwen Hu



Julian
May-Mann



Liqiao Xia



Aviram Uri



Jiaoji Yan



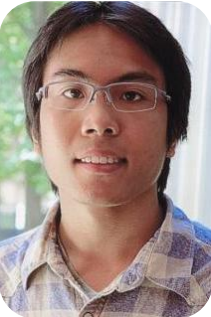
Filippo Gaggioli



Nicole Ticea



Ben Feldman



Trithep Devakul



Jarillo-Herrero



Liang Fu



Jurgen Smet

Thanks

Takashi Taniguchi and Kenji Watanabe

Charles Yang, Yves Kwan, Zoe Zhu, B. Andrei Bernevig,
Xi Dai, Patrick Ledwith, Marc A. Kastner

National High Magnetic Field Laboratory



nano @
stanford



MIT.nano



Center for
Nanoscale
Systems
Harvard University
FAS • SEAS

Main takeaways:

1. VdW systems are an exciting platform for exploring the interplay between **electronic correlations** and **topology**
2. Experimentally, **distinct correlated phenomena** at different points in **angle-angle space**.
3. Driver of phenomenology is more subtle than **helical vs alternating stacking**. Our modeling suggests **lattice relaxations** are paramount.
4. Through **local thermodynamic probes and transport**, we study evolution of correlations in a narrow region of this space. We find a **link between compressibility and superconductivity**.

